EXPLICIT AND ASYMPTOTIC ESTIMATES FOR THE ZERO-COUNTING FUNCTION OF THE RIEMANN ZETA-FUNCTION

VÍCTOR CASAÑA-GINER

ABSTRACT. Let $N(T)$ be the number of non-trivial zeros of the Riemann zeta-function with imaginary part in $(0,T]$, counted with multiplicity, and let $S(T)$ be the argument of ζ on the critical line, normalised so that $N(T)=\frac{1}{\pi}\vartheta(T)+1+S(T)$. Two estimates are proved by the method of Turing type that Amberger introduced for Dedekind zeta-functions and that Bellotti and Wong extended to Artin L-functions. The first is explicit: for every $T\geq 1$,

$$\left|N(T)-\frac{T}{2\pi}\log\frac{T}{2\pi e}\right|\leq 0.08382\log T+5.326,\qquad |S(T)|\leq 0.08382\log T+4.444,$$

and a piecewise linear bound, formed by six lines, lies below the explicit bounds of Amberger and of Bellotti and Wong for every $T\geq 1$. The kernels of the method are found by convex optimisation instead of by hand, the constant coming from the Euler product is computed exactly up to 10^8 and bounded beyond by a prime-number estimate of Rosser and Schoenfeld, and the elementary part is evaluated as one function of T. The second estimate is asymptotic: $\limsup_{T\to\infty}|S(T)|/\log T<0.0748596142711$. It comes from a hybrid of the kernel majorant with Jensen's formula on discs to the right of the critical line, where the growth of ζ is controlled by the subconvexity bounds of Bourgain, of Trudgian and Yang, of Tao, Trudgian and Yang, and of Heath-Brown. The constant is smaller than the value $0.0786252\ldots$ that the method of Backlund gives for discs centred near $s=1$ even under the Lindelöf hypothesis. Every inequality between elementary functions that enters a proof is certified in ball arithmetic, and no numerical information on the zeros is used.

CONTENTS

1. Introduction 2
2. Results 4
3. The identity of Turing type 5
4. The explicit estimate 8
4.1. The decomposition 8
4.2. The Euler product 10
4.3. The elementary part 11
4.4. The kernel inequality 13
4.5. Proof of Theorem 2.1 and of Corollary 2.2 14
5. Jensen discs and admissible profiles 15
6. The asymptotic estimate 17
7. Methods and remarks 19
7.1. Where the kernel inequality binds 19
7.2. Optimisation 19
7.3. Certification 20
7.4. Comparison of the two methods 20
Data availability 20
AI declaration 20
Mathematical software 21

Date: 1 November 2026.
ORCID 0000-0002-8252-2608.

References 21
Appendix A. Data tables 22
Appendix B. Programs 25
Appendix C. Outputs of the certification programs 41
Appendix D. Exact data 45

1. Introduction

Let ζ be the Riemann zeta-function and

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\tfrac{s}{2})\zeta(s) \tag{1.1}$$

its completed form, an entire function that satisfies $\xi(s) = \xi(1-s)$ and is real and positive on the real axis [27, §2.1, (2.1.12)–(2.1.13)]. By the Euler product, ζ has no zeros in $\operatorname{Re} s > 1$, by the theorem of Hadamard and de la Vallée Poussin none on $\operatorname{Re} s = 1$ [9, Ch. 13], and by the functional equation none in $\operatorname{Re} s \le 0$ other than the trivial zeros $-2, -4, \ldots$, which are cancelled in ξ by the poles of $\Gamma(s/2)$. The zeros of ξ are therefore the non-trivial zeros of ζ, all in the strip $0 < \operatorname{Re} s < 1$. Throughout, $\rho = \beta + i\gamma$ runs over these zeros, counted with multiplicity, and

$$N(T) = \#\{\rho :\ 0 < \gamma \le T\} \qquad (T > 0), \tag{1.2}$$

so that $N(T)$ is right-continuous and $N(T) = N(T^+)$ at an ordinate T. Since $\xi(\bar{s}) = \overline{\xi(s)}$, the zeros are symmetric with respect to the real axis, and $N(T)$ counts half of the zeros with $|\gamma| \le T$. Riemann stated and von Mangoldt [32] proved that

$$N(T) = \mathcal{M}(T) + O(\log T), \qquad \mathcal{M}(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}. \tag{1.3}$$

The function

$$\vartheta(T) = \operatorname{Im}\operatorname{Log}\Gamma(\tfrac{1}{4} + \tfrac{iT}{2}) - \frac{T}{2}\log\pi \tag{1.4}$$

is the Riemann–Siegel theta function, with the branch of $\operatorname{Log}\Gamma$ that is real on $(0,\infty)$, and the argument of ζ on the critical line is the function S defined for $T > 0$ by

$$N(T) = \frac{1}{\pi}\vartheta(T) + 1 + S(T) \tag{1.5}$$

[27, §9.3]; with the convention (1.2), $S(T) = S(T^+)$ at the ordinates. Since $\vartheta(T)/\pi + 1 = \mathcal{M}(T) + \frac{7}{8} + O(1/T)$ (Lemma 4.8 below), the estimate (1.3) is equivalent to $S(T) = O(\log T)$.

The problem of making (1.3) explicit, in the form

$$\bigl|N(T) - \mathcal{M}(T)\bigr| \le C_1\log T + C_2\log\log T + C_3 \qquad (T \ge T_0), \tag{1.6}$$

goes back to Backlund [3], after von Mangoldt [32] and Grossmann [13], and was taken up by Rosser [23], Trudgian [28, 29], Platt and Trudgian [20], Hasanalizade, Shen and Wong [14] and Bellotti and Wong [5], all by the method of Backlund: the variation of $\arg\zeta$ on the horizontal segment from $\frac{1}{2} + iT$ to the right is controlled by Jensen's formula on a disc centred at a point $c + iT$ with $c > 1$, and the growth of ζ on that disc by convexity or subconvexity bounds. Table 1 lists the constants obtained in this way, in the form (1.6) and with the first height T_0 from which each bound is proved. The leading constant obtainable by this method is limited from below: for the family of discs used in [8], centred near $s = 1$ with radius near 1, the leading constant cannot be smaller than

$$\Lambda(0) = \frac{3\sqrt{3} - \pi}{12\pi\log 2} = 0.0786252\ldots \tag{1.7}$$

for any admissible growth profile of ζ, even under the Lindelöf hypothesis [8, Remark 5.2]; whether other discs could do better is not settled there.

In 2025 Amberger [1] introduced a different method, inspired by Turing's [31, Lemma 1], for Dedekind zeta-functions, and Bellotti and Wong [6] extended it to Artin L-functions and sharpened it. Instead of a disc, it uses an exact identity: for $d > \frac{1}{2}$, $N(T)$ equals a combination of the arguments of ξ at the two points $\frac{1}{2}+d+iT$ and $\frac{1}{2}+2d+iT$, which lie in the zero-free half-plane $\operatorname{Re} s > 1$, plus a sum over all the zeros of an explicit function $f(\frac{1}{2}-\beta, T-\gamma)$ (Lemma 3.1). The sum over the zeros is then estimated from above and from below by sums of functions of the form $\operatorname{Re}(x+it)^{-j}$ and $\operatorname{Im}(x+it)^{-j}$, whose sums over the zeros are the derivatives of ξ'/ξ at $\frac{1}{2}+d+iT$, again in the zero-free half-plane. For the Riemann zeta-function the method gives $|N(T) - \mathcal{M}(T)| \le 0.097 \log T + 4.807$ for all $T \ge 1$ in [1, Cor. 2], and $\min(0.0946 \log T + 5.746,\ 0.097 \log T + 4.954)$ in [6, Cor. 1.3]. Both articles choose the coefficients of the majorant by trial, with three or four of them, and Bellotti and Wong [6, §3] ask what the method can give when more coefficients are allowed and chosen by convex optimisation, and whether the infimum of the leading constant over all such majorants is zero.

This article contains two contributions to that method. The first is quantitative. The majorant is taken from a family with nineteen coefficients and the coefficients are chosen by a second-order cone programme, with the inequality that defines the majorant imposed on a grid and refined by cutting planes. The constant that comes from the Euler product is computed exactly in ball arithmetic over all prime powers up to 10^8, taking, for each prime, the maximum over the common phase of all its powers (an observation of Amberger [1, Lemma 9]), and bounded beyond 10^8 by partial summation with the inequality $\psi(x) < 1.03883x$ of Rosser and Schoenfeld [24, Thm. 12]. The remaining part, which comes from Γ, from $s(s-1)$ and from the main term, is evaluated as a single function of T whose supremum over $T \ge 1$ is certified. The result (Theorem 2.1) is a piecewise linear bound, valid for all $T \ge 1$, that lies below the bounds of [1] and [6] at every height; its last line has leading constant 0.08382.

The second contribution is methodological. By the maximum principle, the constraints that determine the optimal majorant are those at hypothetical zeros with β near 0 or near 1, and the method of Turing type, which uses no information on ζ inside the critical strip, cannot exclude them. In the hybrid of Section 5 the majorant is allowed to fail near $\beta \in \{0, 1\}$, and the excess is paid by weights of the form $\log^+(R/|\rho^* - s_0|)$, where $s_0 = c + i(T+v)$ with $c > 1$ and ρ^* is the one of ρ, $1-\bar{\rho}$ with real part $\ge \frac{1}{2}$. The sum of such a weight over the zeros is bounded by Jensen's formula on the disc $|s - s_0| < R$, and the integral of $\log|\zeta|$ on its boundary by the subconvexity bounds for ζ in the strip. The coefficients of the majorant and the weights of the discs are chosen together by linear programming. This gives (Theorem 2.3)

$$\limsup_{T\to\infty} \frac{|S(T)|}{\log T} < 0.0748596142711,$$

below the value (1.7). The bound is asymptotic: the subconvexity estimates used have no explicit constants, and no height from which the inequality holds is computed.

The identity of Lemma 3.1, the family of kernel functions and the way their sums over the zeros are evaluated are due to Amberger [1] and to Bellotti and Wong [6], as is the maximum over the phase in the Euler product; they are proved here again, with the normalisation used in this article, so that the article is self-contained. The convex optimisation, the certification of every inequality in ball arithmetic [18], and the hybrid with Jensen's formula are, to the author's knowledge, new. Backlund's method and the method of Turing type are compared in Section 7, where the computations are also described. All numerical statements in the proofs are certified in Arb ball arithmetic [18] through `python-flint`; the programs and their outputs are reproduced after the bibliography (Appendices B–D), and no computation of zeros of ζ, in particular no verification of the Riemann hypothesis up to some height, is used anywhere.

2. Results

Write $L = \log T$ and, for a finite set of lines $\ell_i(L) = A_{1,i}L + A_{3,i}$,

$$\min_i \ell_i(L) = \min_i (A_{1,i}L + A_{3,i}). \tag{2.1}$$

Theorem 2.1 (Explicit estimate)**.** *For every $T \geq 1$, with $L = \log T$,*

$$\begin{aligned} |N(T) - \mathcal{M}(T)| \leq \min\{&0.20176L + 1.369,\ 0.10823L + 2.171,\ 0.09473L + 2.834,\\ &0.09349L + 2.987,\ 0.08399L + 5.202,\ 0.08382L + 5.326\}, \end{aligned} \tag{2.2}$$

$$\begin{aligned} |S(T)| \leq \min\{&0.20176L + 0.844,\ 0.10823L + 1.286,\ 0.09473L + 1.946,\\ &0.09349L + 2.099,\ 0.08399L + 4.318,\ 0.08382L + 4.444\}. \end{aligned} \tag{2.3}$$

The six lines are called $\ell_5, \ell_{20}, \ell_{60}, \ell_{150}, \ell_{400}, \ell_{6000}$, after the value of L for which each was optimised. In (2.2) they are the smallest, respectively, on $[0, L_1]$, $[L_1, L_2]$, $[L_2, L_3]$, $[L_3, L_4]$, $[L_4, L_5]$ and $[L_5, \infty)$, where the breakpoints, at which two adjacent lines agree, are

$$\begin{aligned} &L_1 = \tfrac{80200}{9353} = 8.574788\ldots, \qquad L_2 = \tfrac{442}{9} = 49.111111\ldots, \qquad L_3 = \tfrac{3825}{31} = 123.387096\ldots,\\ &L_4 = \tfrac{4430}{19} = 233.157894\ldots, \qquad L_5 = \tfrac{12400}{17} = 729.411764\ldots \end{aligned}$$

Each line is proved on its own for all $T \geq 1$; in particular

$$|N(T) - \mathcal{M}(T)| \leq 0.08382 \log T + 5.326, \qquad |S(T)| \leq 0.08382 \log T + 4.444 \qquad (T \geq 1). \tag{2.4}$$

Corollary 2.2 (Comparison)**.** *For every $T \geq 1$ the right side of* (2.2) *is smaller than* $\min(0.0946L + 5.746,\ 0.097L + 4.954)$ [6, Cor. 1.3] *by at least* 2.231*, and smaller than the minimum of the seven bounds* $0.097L + 4.807$*,* $0.1L + 4.140$*,* $0.12L + 2.830$*,* $0.14L + 2.306$*,* $0.16L + 2.027$*,* $0.18L + 1.847$*,* $0.2L + 1.722$ *of* [1, Cor. 2 and Table 6] *by at least* 0.291*. The single line* $0.08399L + 5.202$ *is smaller than* $0.0946L + 5.746$ *for every $T \geq 1$, and smaller than* $0.097L + 4.954$ *for $L > (5.202 - 4.954)/(0.097 - 0.08399) = 19.062\ldots$; the single line* $0.08382L + 5.326$ *is smaller than* $0.0946L + 5.746$ *for every $T \geq 1$, and smaller than* $0.097L + 4.954$ *for $L > (5.326 - 4.954)/(0.097 - 0.08382) = 28.224\ldots$.*

In the form (1.6) with $C_2 = 0$ and $T_0 = 1$, the last line of Theorem 2.1 has $(C_1, C_3) = (0.08382, 5.326)$; Table 1 puts it next to the earlier results. For $S(T)$, Bellotti and Wong [5, Cor. 1.5] prove $|S(T)| \leq 0.10076L + 0.24460 \log L + 7.20844$ for $T \geq e$, and [6, Cor. 1.3] gives, through $|S| \leq |N - \mathcal{M}| + \frac{7}{8} + |\varpi|/\pi$ (Lemma 4.8), $|S(T)| \leq 0.0946L + 6.691$ for $T \geq e$.

For the asymptotic coefficient

$$\mathfrak{a} = \limsup_{T \to \infty} \frac{|S(T)|}{\log T}, \tag{2.5}$$

the explicit bounds give $\mathfrak{a} \leq 0.0946$ [6] and $\mathfrak{a} \leq 0.08382$ (Theorem 2.1), and Backlund's method with the exponent pair of Bourgain gives $\mathfrak{a} \leq 0.0968160\ldots$ [8, Cor. 1.2].

Theorem 2.3 (Hybrid estimate)**.** $\mathfrak{a} \leq C_{\mathrm{H}}$*, where C_{H} is the number* (6.1) *defined by the data of Section 6, and*

$$C_{\mathrm{H}} = 0.0748596142710907\ldots < 0.0748596142711. \tag{2.6}$$

In particular $\mathfrak{a} < \Lambda(0)$, with $\Lambda(0)$ as in (1.7)*.*

Theorem 2.3 is asymptotic: it states that for every $\varepsilon > 0$ there is $T_0(\varepsilon)$ with $|S(T)| \leq (C_{\mathrm{H}} + \varepsilon) \log T$ for $T \geq T_0(\varepsilon)$, and $T_0(\varepsilon)$ is not computed, because the subconvexity bounds that enter the proof (Lemma 5.3) have no explicit constants. The explicit bounds of Theorem 2.1 are not improved by it at any finite height.

	C_1	C_2	C_3	T_0
von Mangoldt [32] (1905)	0.4320	1.9167	13.0788	28.5580
Grossmann [13] (1913)	0.2907	1.7862	7.0120	50
Backlund [3] (1918)	0.1370	0.4430	5.2250	200
Rosser [23] (1941)	0.1370	0.4430	2.4630	2
Trudgian [28] (2012)	0.1700	0	2.8730	e
Trudgian [29] (2014)	0.1120	0.2780	3.3850	e
Platt and Trudgian [20] (2015)	0.1100	0.2900	3.165	e
Hasanalizade, Shen and Wong [14] (2022)	0.1038	0.2573	9.4925	e
Bellotti and Wong [5] (2025)	0.10076	0.24460	8.08344	e
Amberger [1, Cor. 2] (2025, v5 of 2026)	0.097	0	4.807	1
Bellotti and Wong [6, Cor. 1.3] (2026)	0.0946	0	5.746	1
Theorem 2.1, line ℓ_{400}	0.08399	0	5.202	1
Theorem 2.1, line ℓ_{6000}	0.08382	0	5.326	1

TABLE 1. Explicit bounds $|N(T)-\mathcal{M}(T)| \le C_1 \log T + C_2 \log\log T + C_3$ for $T \ge T_0$. The rows up to 2025 are taken from [6, Table 3].

Remark 2.4 (Conventions). Amberger [1] and Bellotti and Wong [6] count a zero with imaginary part exactly T with weight $\frac{1}{2}$, that is, they use $\frac{1}{2}(N(T^+)+N(T^-))$. Since the bounds of Theorem 2.1 are proved for every T that is not an ordinate and $\mathcal{M}$ is continuous, they hold for $N(T^+)$ and for $N(T^-)$ at an ordinate, hence also for their mean. The comparisons of Corollary 2.2 are therefore between bounds for the same quantity.

Remark 2.5 (On the infimum of the leading constant). Bellotti and Wong [6, §3] ask whether the infimum of C_1 over all majorants of the type used in the method, with finitely many kernel functions, is 0. The program `kernel_only.py` of Appendix B (output in Appendix C) gives, for the family (3.13) with the constraint (K) imposed on a grid and refined by cutting planes, and d optimised over $\{0.7, 0.8, 0.9\}$, the leading constants 0.08831, 0.08071 and 0.07983 for $m = 4, 10, 20$ (grid optima, not certified; only runs whose cutting planes converged, with a residual violation below 10^{-5} on the fine grid, are quoted, and for $m = 40$ the iteration did not converge, the linear programme being badly conditioned), which suggest a positive limit near 0.079. The same computations with the constraint imposed only at $\beta = \frac{1}{2}$, as would suffice under the Riemann hypothesis, give 0.04375 and 0.02002 for $m = 4, 10$ (for $m = 20$ the iteration did not converge). The obstruction thus comes from hypothetical zeros near $\beta \in \{0, 1\}$, and it is exactly what the hybrid of Theorem 2.3 removes, at the price of using subconvexity bounds for ζ inside the strip.

3. The identity of Turing type

On the closed half-plane $\operatorname{Re} s \ge 1$ the function ξ has no zeros (ζ has none, Γ has none, and $\xi(1) = \frac{1}{2}$), and the half-plane is simply connected, so $\arg \xi$ has a continuous branch there; it is normalised by $\arg \xi(\sigma) = 0$ for real $\sigma \ge 1$, where $\xi(\sigma) > 0$. For $d > \frac{1}{2}$ put

$$\sigma_1 = \tfrac{1}{2} + d, \qquad \sigma_2 = \tfrac{1}{2} + 2d, \qquad s_1 = \sigma_1 + iT, \qquad s_2 = \sigma_2 + iT. \tag{3.1}$$

For $b \in \mathbb{R}$ with $|b| < d$ and $t \ne 0$ put

$$f(b,t) = 2\arctan\frac{d+b}{t} + 2\arctan\frac{d-b}{t} - \arctan\frac{2d+b}{t} - \arctan\frac{2d-b}{t}, \tag{3.2}$$

and for a zero $\rho = \beta + i\gamma$ and a height T write

$$b_\rho = \tfrac{1}{2} - \beta, \qquad t_\rho = T - \gamma. \tag{3.3}$$

Since $0 < \beta < 1$, one has $|b_\rho| < \frac{1}{2} < d$. The function f is even in b and odd in t.

Lemma 3.1 (Identity of Turing type [31, 1, 6])**.** *Let $d > \frac{1}{2}$ and let $T > 0$ not be an ordinate. Then*

$$N(T) = \frac{1}{\pi}\Big(2\arg\xi(s_1) - \arg\xi(s_2)\Big) + \frac{1}{2\pi}\sum_{\rho} f(b_\rho, t_\rho), \tag{3.4}$$

where the series converges absolutely.

Proof. The argument principle. Let $\mathcal{R}$ be the rectangle with vertices σ_2, σ_2+iT, $1-\sigma_2+iT$, $1-\sigma_2$. The function ξ has no zeros on $\partial\mathcal{R}$: none on the vertical sides, which lie outside the critical strip; none on the top side, because T is not an ordinate; and none on the real segment, because $\xi(\sigma) > 0$ for all real σ (for $\sigma \geq 1$ this was noted above; for $0 < \sigma < 1$ one has $\sigma(\sigma-1) < 0$ and $\zeta(\sigma) < 0$, the latter because $(1-2^{1-\sigma})\zeta(\sigma) = \sum_{n\geq 1}(-1)^{n-1}n^{-\sigma} > 0$ and $1-2^{1-\sigma} < 0$; the case $\sigma \leq 0$ follows from $\xi(\sigma) = \xi(1-\sigma)$). Since T is not an ordinate, the zeros of ξ inside $\mathcal{R}$, namely the non-trivial zeros with $0 < \gamma < T$, are exactly the $N(T)$ zeros counted in (1.2), so by the argument principle the variation of $\arg\xi$ along $\partial\mathcal{R}$, counterclockwise, is $2\pi N(T)$.

Let Δ_1 be the variation of $\arg\xi$ along the path π_1 from σ_2 to $\sigma_2 + iT$ and then to $\frac{1}{2} + iT$. The reflection $s \mapsto 1 - \bar{s}$ in the critical line maps π_1 onto the path π_1' from $1-\sigma_2$ to $1-\sigma_2+iT$ to $\frac{1}{2}+iT$, and $\xi(1-\bar{s}) = \xi(\bar{s}) = \overline{\xi(s)}$, so the variation of $\arg\xi$ along π_1' is $-\Delta_1$. The boundary $\partial\mathcal{R}$, counterclockwise, is π_1 followed by the reverse of π_1' followed by the real segment from $1-\sigma_2$ to σ_2, on which $\arg\xi$ is constant. Hence $2\pi N(T) = \Delta_1 + \Delta_1$, that is, $\Delta_1 = \pi N(T)$.

Along the vertical part of π_1 the variation is $\arg\xi(s_2)$, the branch defined above. Along the horizontal part, ξ has no zeros, and with

$$F(\sigma) = \operatorname{Im}\frac{\xi'}{\xi}(\sigma+iT) = \frac{\partial}{\partial\sigma}\arg\xi(\sigma+iT) \qquad (\tfrac{1}{2} \leq \sigma \leq \sigma_2), \tag{3.5}$$

a continuous function, the variation from $\sigma_2 + iT$ to $\frac{1}{2}+iT$ is $-\int_{1/2}^{\sigma_2} F(\sigma)\,d\sigma$. Therefore

$$\pi N(T) = \arg\xi(s_2) - \int_{1/2}^{\sigma_2} F(\sigma)\,d\sigma. \tag{3.6}$$

Since $\int_{\sigma_1}^{\sigma_2} F(\sigma)\,d\sigma = \arg\xi(s_2) - \arg\xi(s_1)$ (the segment lies in $\operatorname{Re} s \geq 1$, where the branch is the one fixed above) and $\int_{1/2}^{\sigma_1} F(\sigma+d)\,d\sigma = \int_{\sigma_1}^{\sigma_2} F$,

$$\pi N(T) = 2\arg\xi(s_1) - \arg\xi(s_2) - \int_{1/2}^{\sigma_1}\big(F(\sigma) - F(\sigma+d)\big)\,d\sigma. \tag{3.7}$$

The sum over the zeros. By the Hadamard product [9, Ch. 12],

$$\frac{\xi'}{\xi}(s) = B + \sum_{\rho}\Big(\frac{1}{s-\rho} + \frac{1}{\rho}\Big), \tag{3.8}$$

with a real constant B and absolute convergence for every s that is not a zero. Taking imaginary parts at $\sigma+iT$ and at $\sigma+d+iT$ and subtracting, the constants cancel and

$$F(\sigma) - F(\sigma+d) = \sum_{\rho}\operatorname{Im}\Big(\frac{1}{\sigma+iT-\rho} - \frac{1}{\sigma+d+iT-\rho}\Big), \tag{3.9}$$

an absolutely convergent series of differences (the two series (3.8), with their regularising terms $1/\rho$, are each absolutely convergent; nothing is asserted about $\sum_\rho (s-\rho)^{-1}$ alone). For $\frac{1}{2} \leq \sigma \leq \sigma_1$ the ρ-th term has modulus at most $d/|t_\rho|^2$, because $|\sigma+iT-\rho| \geq |t_\rho|$ and $|\sigma+d+iT-\rho| \geq |t_\rho|$; since T is not an ordinate, $t_\rho \neq 0$ for every ρ, only finitely many zeros have $|t_\rho| < 1$, and $\sum_{|t_\rho|\geq 1}|t_\rho|^{-2} < \infty$ [27, §9.4], the series (3.9) is dominated by a summable sequence independent of σ and may be integrated term by term over $[\frac{1}{2}, \sigma_1]$. With $\operatorname{Im}\frac{1}{x+it} = -\frac{t}{x^2+t^2}$ and $\int \frac{-t\,dx}{x^2+t^2} = -\arctan\frac{x}{t}$,

$$\int_{1/2}^{\sigma_1}\operatorname{Im}\Big(\frac{1}{\sigma+iT-\rho} - \frac{1}{\sigma+d+iT-\rho}\Big)d\sigma = g(b_\rho, t_\rho), \tag{3.10}$$

where

$$g(b,t) = -2\arctan\frac{b+d}{t} + \arctan\frac{b}{t} + \arctan\frac{b+2d}{t}, \tag{3.11}$$

since $\sigma-\beta$ runs from b_ρ to $b_\rho+d$ in the first term and $\sigma+d-\beta$ from $b_\rho+d$ to $b_\rho+2d$ in the second. For $|b|\le\frac12$ and $|t|\ge1$, the expansion $\arctan(y/t) = y/t - y^3/(3t^3) + O(|t|^{-5})$ gives $g(b,t) = O(|t|^{-3})$, because $-2(b+d)+b+(b+2d)=0$; hence $\sum_\rho g(b_\rho,t_\rho)$ converges absolutely.

The map $\rho\mapsto 1-\bar\rho$ is a bijection of the set of zeros that preserves multiplicities and maps (b_ρ,t_ρ) to $(-b_\rho,t_\rho)$. Hence

$$\sum_\rho g(b_\rho,t_\rho) = \frac12\sum_\rho\bigl(g(b_\rho,t_\rho)+g(-b_\rho,t_\rho)\bigr) = -\frac12\sum_\rho f(b_\rho,t_\rho),$$

because $g(b,t)+g(-b,t) = -f(b,t)$: the terms $\arctan\frac{b}{t}$ and $\arctan\frac{-b}{t}$ cancel, and the others give (3.2) with the opposite sign. Inserting this in (3.7) proves (3.4). □

For an index $\iota=(\mathrm{X},j)$ with $\mathrm{X}\in\{\mathrm{R},\mathrm{I}\}$ and $j\ge1$, and for $x>0$, $t\in\mathbb{R}$, put

$$F_{\mathrm{R},j}(x,t) = \operatorname{Re}(x+it)^{-j},\quad F_{\mathrm{I},j}(x,t) = -\operatorname{Im}(x+it)^{-j},\quad G_\iota(b,t) = F_\iota(d+b,t)+F_\iota(d-b,t)\ \ (|b|<d). \tag{3.12}$$

A *kernel* is a real linear combination

$$U = \sum_{\iota\in\mathcal{B}} c_\iota G_\iota,\qquad \mathcal{B}=\mathcal{B}_m = \{(\mathrm{R},j):1\le j\le m\}\cup\{(\mathrm{I},j):2\le j\le m\}. \tag{3.13}$$

Each G_ι is even in b; $G_{\mathrm{R},j}$ is even in t and $G_{\mathrm{I},j}$ is odd in t, because $(x-it)^{-j} = \overline{(x+it)^{-j}}$. Write $\Phi_{\mathrm{R}}(z)=\operatorname{Re}z$, $\Phi_{\mathrm{I}}(z)=-\operatorname{Im}z$, and

$$D_1(s) = \frac1s+\frac{1}{s-1}-\frac12\log\pi+\frac12\psi_\Gamma\Bigl(\frac{s}{2}\Bigr),\qquad D_j(s) = \frac{1}{s^j}+\frac{1}{(s-1)^j}+\frac{(-1)^{j-1}}{(j-1)!\,2^j}\psi_\Gamma^{(j-1)}\Bigl(\frac{s}{2}\Bigr)\quad(j\ge2), \tag{3.14}$$

where $\psi_\Gamma=\Gamma'/\Gamma$ is the digamma function (the letter ψ alone denotes the Chebyshev function $\psi(x)=\sum_{n\le x}\Lambda(n)$ in Section 4.2).

Lemma 3.2 (Sums of the kernel functions over the zeros)**.** *Let $d>\frac12$, $T>0$ and $s_1=\sigma_1+iT$. For every $\iota=(\mathrm{X},j)\in\mathcal{B}$ the series $\sum_\rho G_\iota(b_\rho,t_\rho)$ converges absolutely and*

$$\sum_\rho G_\iota(b_\rho,t_\rho) = 2\,\Phi_{\mathrm{X}}\bigl(D_j(s_1)\bigr) - \frac{2}{(j-1)!}\sum_{n\ge2}\Lambda(n)(\log n)^{j-1}\,\Phi_{\mathrm{X}}\bigl(n^{-s_1}\bigr), \tag{3.15}$$

where $\Phi_{\mathrm{R}}(n^{-s_1}) = n^{-\sigma_1}\cos(T\log n)$ and $\Phi_{\mathrm{I}}(n^{-s_1}) = n^{-\sigma_1}\sin(T\log n)$.

Proof. Since $s_1-\rho = (d+b_\rho)+it_\rho$ and $s_1-(1-\bar\rho) = (d-b_\rho)+it_\rho$, and $\rho\mapsto1-\bar\rho$ permutes the zeros, $\sum_\rho G_\iota(b_\rho,t_\rho) = 2\sum_\rho F_\iota(s_1-\rho)$, where $F_\iota(z)$ stands for $F_\iota(\operatorname{Re}z,\operatorname{Im}z)$, provided the series converge absolutely. For $j\ge2$: only finitely many zeros have $|s_1-\rho|<1$ (the zeros are isolated and s_1 is not a zero), for the others $|s_1-\rho|^{-j}\le|s_1-\rho|^{-2}$, and $\sum_\rho|s_1-\rho|^{-2}<\infty$ [27, §9.4]. For $j=1$, $\operatorname{Re}\frac{1}{s_1-\rho} = \frac{\sigma_1-\beta}{|s_1-\rho|^2}>0$, and by (3.8) together with $B=-\sum_\rho\operatorname{Re}\frac1\rho$ [9, Ch. 12], the series of positive terms $\sum_\rho\operatorname{Re}\frac{1}{s_1-\rho}$ converges to $\operatorname{Re}\frac{\xi'}{\xi}(s_1)$.

From (1.1), for $\operatorname{Re}s>1$,

$$\frac{\xi'}{\xi}(s) = \frac1s+\frac{1}{s-1}-\frac12\log\pi+\frac12\psi_\Gamma\Bigl(\frac{s}{2}\Bigr)+\frac{\zeta'}{\zeta}(s) = D_1(s)-\sum_{n\ge2}\Lambda(n)n^{-s}, \tag{3.16}$$

which gives (3.15) for $j=1$. For $j\ge2$, differentiating (3.8) $j-1$ times term by term (the differentiated series converge absolutely and locally uniformly away from the zeros) gives $\sum_\rho(s-$

$\rho)^{-j} = \frac{(-1)^{j-1}}{(j-1)!}\big(\frac{\xi'}{\xi}\big)^{(j-1)}(s)$, and differentiating (3.16) $j-1$ times gives

$$\frac{(-1)^{j-1}}{(j-1)!}\Big(\frac{\xi'}{\xi}\Big)^{(j-1)}(s) = D_j(s) - \frac{1}{(j-1)!}\sum_{n\geq 2}\Lambda(n)(\log n)^{j-1}n^{-s},$$

because $\frac{d^{j-1}}{ds^{j-1}}s^{-1} = (-1)^{j-1}(j-1)!\,s^{-j}$ and $\frac{d^{j-1}}{ds^{j-1}}\frac{1}{2}\psi_\Gamma(\frac{s}{2}) = 2^{-j}\psi_\Gamma^{(j-1)}(\frac{s}{2})$. Applying Φ_{X} and multiplying by 2 proves (3.15). □

4. The explicit estimate

4.1. **The decomposition.** Fix $d > \frac{1}{2}$, $m \geq 2$ and a kernel $U = \sum_{\iota\in\mathcal{B}_m} c_\iota G_\iota$ such that

$$f(b,t) \leq U(b,t) \qquad \text{for } 0 \leq b \leq \tfrac{1}{2},\ t \neq 0. \tag{K}$$

Since f and U are even in b, (K) holds for $|b| \leq \frac{1}{2}$. Put $\varepsilon_{\mathrm{R}} = 1$, $\varepsilon_{\mathrm{I}} = -1$ and, for $L > 0$,

$$a(L) = -\frac{1}{\pi}\sum_{(\mathrm{R},j)\in\mathcal{B}} c_{\mathrm{R},j}\frac{L^{j-1}}{(j-1)!}, \qquad b(L) = \frac{1}{\pi}\Big(\frac{e^{-dL}-2}{L} - \sum_{(\mathrm{I},j)\in\mathcal{B}} c_{\mathrm{I},j}\frac{L^{j-1}}{(j-1)!}\Big). \tag{4.1}$$

For $s = \sigma + iT$ with $\sigma > 1$ and $T > 0$ put

$$\mathcal{A}(s) = \arg s + \arg(s-1) - \frac{T}{2}\log\pi + \operatorname{Im}\operatorname{Log}\Gamma\Big(\frac{s}{2}\Big), \qquad \arg s = \arctan\frac{T}{\sigma},\quad \arg(s-1) = \arctan\frac{T}{\sigma-1}, \tag{4.2}$$

and

$$\begin{aligned} \mathcal{E}(T) &= \frac{1}{\pi}\big(2\mathcal{A}(s_1) - \mathcal{A}(s_2)\big) - \mathcal{M}(T), \\ \mathcal{E}_u(T) = \mathcal{E}(T) + \frac{1}{\pi}\sum_{\iota\in\mathcal{B}} c_\iota \Phi_{\mathrm{X}}\big(D_j(s_1)\big), &\qquad \mathcal{E}_\ell(T) = \mathcal{E}(T) - \frac{1}{\pi}\sum_{\iota\in\mathcal{B}} \varepsilon_{\mathrm{X}} c_\iota \Phi_{\mathrm{X}}\big(D_j(s_1)\big). \end{aligned} \tag{4.3}$$

Proposition 4.1 (Two-sided estimate)**.** *Under* (K)*, for every $T > 0$ that is not an ordinate,*

$$\begin{aligned} N(T) - \mathcal{M}(T) &\leq \mathcal{E}_u(T) + \sum_{n\geq 2}\Lambda(n)n^{-\sigma_1}\big(a(\log n)\cos(T\log n) + b(\log n)\sin(T\log n)\big), \\ N(T) - \mathcal{M}(T) &\geq \mathcal{E}_\ell(T) + \sum_{n\geq 2}\Lambda(n)n^{-\sigma_1}\big(-a(\log n)\cos(T\log n) + b(\log n)\sin(T\log n)\big), \end{aligned} \tag{4.4}$$

and both series converge absolutely.

Proof. For $\operatorname{Re} s > 1$, $\operatorname{Log}\zeta(s) = \sum_{n\geq 2}\frac{\Lambda(n)}{\log n}n^{-s}$ is the branch of $\log\zeta$ that is real on $(1,\infty)$ and continuous on the half-plane, and $\xi(s) = \frac{1}{2}\cdot s\cdot(s-1)\cdot\pi^{-s/2}\cdot\Gamma(\frac{s}{2})\cdot\zeta(s)$ is a product of non-vanishing factors each of which has a continuous argument on $\operatorname{Re} s \geq 1$, $T > 0$, vanishing on the real axis: $\arg s$ and $\arg(s-1)$ as in (4.2), $-\frac{T}{2}\log\pi$, $\operatorname{Im}\operatorname{Log}\Gamma(\frac{s}{2})$ and $\operatorname{Im}\operatorname{Log}\zeta(s)$. Their sum is a continuous argument of ξ vanishing on the real axis, hence it is the branch of Section 3:

$$\arg\xi(s) = \mathcal{A}(s) - \sum_{n\geq 2}\frac{\Lambda(n)}{\log n}n^{-\sigma}\sin(T\log n) \qquad (s = \sigma + iT,\ \sigma > 1,\ T > 0). \tag{4.5}$$

Since $n^{-\sigma_2} = n^{-\sigma_1}e^{-d\log n}$,

$$\frac{1}{\pi}\big(2\arg\xi(s_1) - \arg\xi(s_2)\big) = \frac{1}{\pi}\big(2\mathcal{A}(s_1) - \mathcal{A}(s_2)\big) + \frac{1}{\pi}\sum_{n\geq 2}\Lambda(n)n^{-\sigma_1}\frac{e^{-d\log n}-2}{\log n}\sin(T\log n). \tag{4.6}$$

Upper bound. By (K) at (b_ρ, t_ρ), $f(b_\rho, t_\rho) \le U(b_\rho, t_\rho)$ for every zero, and the series $\sum_\rho U(b_\rho, t_\rho)$ converges absolutely by Lemma 3.2. Hence, by Lemma 3.1 and (3.15),

$$\frac{1}{2\pi}\sum_\rho f(b_\rho,t_\rho) \le \frac{1}{2\pi}\sum_\iota c_\iota \sum_\rho G_\iota(b_\rho,t_\rho)$$
$$= \frac{1}{\pi}\sum_\iota c_\iota \Phi_{\mathrm{X}}\big(D_j(s_1)\big) - \frac{1}{\pi}\sum_{n\ge 2}\Lambda(n)n^{-\sigma_1}\sum_\iota c_\iota \frac{(\log n)^{j-1}}{(j-1)!}\Phi_{\mathrm{X}}\big(e^{-iT\log n}\big),$$

and $\Phi_{\mathrm{R}}(e^{-iT\log n}) = \cos(T\log n)$, $\Phi_{\mathrm{I}}(e^{-iT\log n}) = \sin(T\log n)$, so, with $Q(L) = \sum_{(\mathrm{I},j)} c_{\mathrm{I},j} L^{j-1}/(j-1)! = \frac{e^{-dL}-2}{L} - \pi b(L)$, the inner sum over ι is $-\pi a(\log n)\cos(T\log n) + Q(\log n)\sin(T\log n)$ and the right side equals

$$\frac{1}{\pi}\sum_\iota c_\iota \Phi_{\mathrm{X}}(D_j(s_1)) + \sum_n \Lambda(n)n^{-\sigma_1}\big(a(\log n)\cos(T\log n) + b(\log n)\sin(T\log n)\big)$$
$$- \frac{1}{\pi}\sum_n \Lambda(n)n^{-\sigma_1}\frac{e^{-d\log n}-2}{\log n}\sin(T\log n).$$

Adding (4.6) cancels the last sum and gives the first line of (4.4).

Lower bound. Since $f(b,-t) = -f(b,t)$, (K) at $(b_\rho, -t_\rho)$ gives $f(b_\rho,t_\rho) \ge -U(b_\rho,-t_\rho)$, and $U(b,-t) = \sum_\iota \varepsilon_{\mathrm{X}} c_\iota G_\iota(b,t)$ by the parity of the G_ι in t. Hence

$$\frac{1}{2\pi}\sum_\rho f(b_\rho,t_\rho) \ge -\frac{1}{\pi}\sum_\iota \varepsilon_{\mathrm{X}} c_\iota \Phi_{\mathrm{X}}\big(D_j(s_1)\big)$$
$$+ \frac{1}{\pi}\sum_{n\ge 2}\Lambda(n)n^{-\sigma_1}\sum_\iota \varepsilon_{\mathrm{X}} c_\iota \frac{(\log n)^{j-1}}{(j-1)!}\Phi_{\mathrm{X}}\big(e^{-iT\log n}\big).$$

Here the inner sum over ι is $-\pi a(\log n)\cos(T\log n) - Q(\log n)\sin(T\log n)$, by (4.1) and $\varepsilon_{\mathrm{I}} = -1$, so the right side equals

$$-\frac{1}{\pi}\sum_\iota \varepsilon_{\mathrm{X}} c_\iota \Phi_{\mathrm{X}}(D_j(s_1)) + \sum_n \Lambda(n)n^{-\sigma_1}\big(-a(\log n)\cos(T\log n) + b(\log n)\sin(T\log n)\big)$$
$$- \frac{1}{\pi}\sum_n \Lambda(n)n^{-\sigma_1}\frac{e^{-d\log n}-2}{\log n}\sin(T\log n).$$

Adding (4.6) cancels the last sum and gives the second line of (4.4). Absolute convergence follows from $\sigma_1 > 1$ and the polynomial growth of a and b. □

For a prime p write $\lambda_k = (\log p)\,p^{-k\sigma_1}$, $a_k = a(k\log p)$, $b_k = b(k\log p)$ and

$$f_p(x) = \sum_{k\ge 1}\lambda_k\big(a_k\cos kx + b_k\sin kx\big) \qquad (x\in\mathbb{R}), \qquad M_p = \max_{x\in\mathbb{R}} f_p(x). \tag{4.7}$$

The series converges absolutely and uniformly, so f_p is continuous and 2π-periodic, and M_p exists. Grouping the prime powers $n = p^k$ in (4.4) according to p, with the common phase $x = T\log p$, the series in the first line is $\sum_p f_p(T\log p) \le \sum_p M_p$, and the series in the second line is $\sum_p\big(-f_p(-T\log p)\big) \ge -\sum_p M_p$, because $\sum_k \lambda_k(-a_k\cos kx + b_k\sin kx) = -f_p(-x)$. This is the observation of Amberger [1, Lemma 9] that the lower bound is obtained from the upper one by changing the sign of the odd part. With

$$F_u(T) = \mathcal{E}_u(T) - A_1\log T, \qquad F_\ell(T) = -\mathcal{E}_\ell(T) - A_1\log T, \tag{4.8}$$

Proposition 4.1 gives, for every $T > 0$ not an ordinate and every A_1 and $C_\zeta \ge \sum_p M_p$,

$$\big|N(T) - \mathcal{M}(T)\big| \le A_1\log T + C_\zeta + \max\big(F_u(T), F_\ell(T)\big). \tag{4.9}$$

Sections 4.2–4.4 bound the three ingredients: C_ζ, $\sup_{T\ge1}\max(F_u,F_\ell)$, and the verification of (K).

4.2. **The Euler product.**

Lemma 4.2 (Bounds for M_p)**.** *Let $h(L)=\sqrt{a(L)^2+b(L)^2}$ and $H_{p,k}=\lambda_k h(k\log p)$. Then*

(a) $M_p\le\sum_{k\ge1}H_{p,k}$; more precisely, for every $K\ge1$, $M_p\le\max_x\sum_{k\le K}\lambda_k(a_k\cos kx+b_k\sin kx)+\sum_{k>K}H_{p,k}$;

(b) for every integer $\mathsf{M}\ge1$ and $x_i=2\pi i/\mathsf{M}$ $(0\le i<\mathsf{M})$, and every $K\ge1$,

$$\max_x\sum_{k\le K}\lambda_k(a_k\cos kx+b_k\sin kx)\le\max_{0\le i<\mathsf{M}}\sum_{k\le K}\lambda_k(a_k\cos kx_i+b_k\sin kx_i)+\frac{\pi}{\mathsf{M}}\sum_{k\le K}kH_{p,k}.$$

Proof. (a) $|a\cos\theta+b\sin\theta|\le\sqrt{a^2+b^2}$ for real a,b,θ, and the maximum of a sum is at most the sum of the maxima. (b) The finite sum $\phi(x)=\sum_{k\le K}\lambda_k(a_k\cos kx+b_k\sin kx)$ is differentiable with $|\phi'(x)|\le\sum_{k\le K}k\lambda_k\sqrt{a_k^2+b_k^2}$, every real x is within π/M of some x_i modulo 2π, and ϕ is 2π-periodic; the mean value theorem gives the claim. □

Lemma 4.3 (The tail)**.** *Let $N_0\ge2$, $g(x)=h(\log x)\,x^{-\sigma_1}$ for $x\ge N_0$, and let $G\colon[N_0,\infty)\to[0,\infty)$ be non-increasing, integrable, with $G\ge g$. Then*

$$\sum_{n>N_0}\Lambda(n)g(n)\le G(N_0)\bigl(1.03883\,N_0-\psi(N_0)\bigr)+1.03883\int_{N_0}^{\infty}G(x)\,dx,\tag{4.10}$$

where $\psi(x)=\sum_{n\le x}\Lambda(n)$.

Proof. For $0<y<G(N_0)$ put $X_y=\sup\{x\ge N_0:G(x)>y\}$, which is finite because G is integrable and non-increasing. If $n>N_0$ and $G(n)>y$ then $n\le X_y$, so $\sum_{n>N_0,\,G(n)>y}\Lambda(n)\le\psi(X_y)-\psi(N_0)<1.03883X_y-\psi(N_0)$ by $\psi(x)<1.03883x$ for all $x>0$ [24, Thm. 12]; for $y\ge G(N_0)$ the set is empty. By Tonelli's theorem,

$$\sum_{n>N_0}\Lambda(n)g(n)\le\sum_{n>N_0}\Lambda(n)G(n)=\int_0^{\infty}\sum_{\substack{n>N_0\\G(n)>y}}\Lambda(n)\,dy\le\int_0^{G(N_0)}\bigl(1.03883X_y-\psi(N_0)\bigr)dy.$$

Finally $\int_0^{G(N_0)}X_y\,dy=N_0G(N_0)+\int_0^{G(N_0)}(X_y-N_0)\,dy$, and for $x>N_0$ one has $X_y\ge x$ if $G(x)>y$, while $X_y\ge x$ implies $G(x)\ge y$ (there are $x'\ge x$ with $G(x')>y$, and $G(x)\ge G(x')$); hence $\{y:X_y\ge x\}$ and $\{y:y<G(x)\}$ differ by a set of measure zero, and $\int_0^{G(N_0)}(X_y-N_0)\,dy=\int_{N_0}^{\infty}\int_0^{G(N_0)}\mathbf{1}_{X_y\ge x}\,dy\,dx=\int_{N_0}^{\infty}G(x)\,dx$. This is (4.10). □

The majorant G is built as follows. Let $L_0=\log N_0$ and let $L_{\mathrm B}>L_0$ be such that g is non-increasing on $[e^{L_{\mathrm B}},\infty)$; then on $[e^{L_{\mathrm B}},\infty)$ take $G=g$. On $[L_0,L_{\mathrm B}]$ choose points $L_0=u_0<u_1<\cdots<u_r=L_{\mathrm B}$, let $\bar g_i$ be an upper bound of g on $[e^{u_i},e^{u_{i+1}}]$ computed in ball arithmetic, and put $G=\max(\bar g_i,\bar g_{i+1},\dots,\bar g_{r-1},g(e^{L_{\mathrm B}}))$ on $[e^{u_i},e^{u_{i+1}})$; this G is non-increasing and $G\ge g$, and $\int_{N_0}^{e^{L_{\mathrm B}}}G\,dx$ is a finite sum. The condition on $L_{\mathrm B}$ is verified by the next lemma. Write $a(L)=\frac{1}{\pi}P_a(L)$ and $b(L)=\frac{1}{\pi}\bigl(P_b(L)+\frac{e^{-dL}-2}{L}\bigr)$ with polynomials $P_a=\sum_i\mathsf{a}_iL^i$, $P_b=\sum_i\mathsf{b}_iL^i$ read off from (4.1), let D be the larger of their degrees, P the one of P_a, P_b of degree D (if both have degree D, take $P=P_a$), and $q_i=|\mathsf{a}_i|+|\mathsf{b}_i|$.

Lemma 4.4 (Monotonicity of g)**.** *Let $u=1/L$ and*

$$\mathrm{num}(u)=u\sum_{i\ge1}iq_iu^{D-i}+d\,u^{D+1}+2u^{D+2},\qquad\mathrm{low}(u)=|\mathsf{p}_D|-\sum_{i<D}|\mathsf{p}_i|u^{D-i}-\varepsilon_Pu^{D+1},$$

where p_i are the coefficients of P and $\varepsilon_P=2$ if $P=P_b$, $\varepsilon_P=0$ if $P=P_a$. If $\mathrm{low}(u)>0$ and $\mathrm{num}(u)/\mathrm{low}(u)<\sigma_1$ for all $u\in[0,1/L_{\mathrm B}]$, then g is strictly decreasing on $[e^{L_{\mathrm B}},\infty)$.

Proof. On $[e^{L_{\mathrm{B}}},\infty)$ one has $h>0$ (because $\mathrm{low}>0$ gives $P\neq 0$ there), so $\log g$ is differentiable in $L=\log x$ with $\frac{d}{dL}\log g=\frac{aa'+bb'}{a^2+b^2}-\sigma_1\le\frac{|a'|+|b'|}{h}-\sigma_1$, using $|a|,|b|\le h$. Further $h\ge\frac{1}{\pi}|P(L)|-\frac{\varepsilon_P}{\pi L}\ge\frac{L^D}{\pi}\mathrm{low}(1/L)$, since $|\frac{e^{-dL}-2}{L}|\le\frac{2}{L}$, and $|a'|+|b'|\le\frac{1}{\pi}(\sum_{i\ge1}iq_iL^{i-1}+\frac{d}{L}+\frac{2}{L^2})=\frac{L^D}{\pi}\mathrm{num}(1/L)$, because $|\frac{d}{dL}\frac{e^{-dL}-2}{L}|\le\frac{de^{-dL}}{L}+\frac{|e^{-dL}-2|}{L^2}\le\frac{d}{L}+\frac{2}{L^2}$. Hence $\frac{d}{dL}\log g\le\mathrm{num/low}-\sigma_1<0$. □

Proposition 4.5 (Certified value of C_ζ)**.** *Let $N_0=10^8$, $P_0=1000$, $\mathsf{M}=4096$. Then $\sum_p M_p\le C_\zeta$ with*

$$C_\zeta=\sum_{p\le P_0}\min\Big(\max_{i<\mathsf{M}}\sum_{k:\,p^k\le N_0}\lambda_k(a_k\cos kx_i+b_k\sin kx_i)+\frac{\pi}{\mathsf{M}}\sum_{k:\,p^k\le N_0}kH_{p,k},\ \sum_{k:\,p^k\le N_0}H_{p,k}\Big)+\sum_{\substack{P_0<p\le N_0\\ p^k\le N_0}}H_{p,k}+\mathrm{Tail},\tag{4.11}$$

where Tail *is the right side of* (4.10) *with L_{B} verified by Lemma 4.4. The program* `cz_gen_pp.py` *(Appendix B) evaluates every term of* (4.11) *in ball arithmetic and reports an upper bound of the resulting ball. For the six kernels of Theorem 2.1 the values are in Table 2.*

Proof. For $p\le P_0$, Lemma 4.2(a) with K the largest k with $p^k\le N_0$, then (b) for the finite sum, and the powers $p^k>N_0$ go to the tail; for $P_0<p\le N_0$, Lemma 4.2(a) with the same split; for $p>N_0$, Lemma 4.2(a) with $K=0$. All prime powers $n>N_0$ are covered by $\sum_{n>N_0}\Lambda(n)g(n)$, bounded by Lemma 4.3 with the majorant described before Lemma 4.4. The last integral $\int_{e^{L_{\mathrm{B}}}}^{\infty}g\,dx$ is bounded as follows. Let $L_1=800$ and let $\tilde g\ge g$ be the majorant of g on $[e^{L_1},\infty)$ given below; then

$$\int_{e^{L_{\mathrm{B}}}}^{\infty}g\,dx\le\int_{e^{L_{\mathrm{B}}}}^{e^{L_1}}g\,dx+\int_{e^{L_1}}^{\infty}\tilde g\,dx,$$

because the right side minus the left side equals $\int_{e^{L_1}}^{e^{L_{\mathrm{B}}}}(\tilde g-g)\,dx+\int_{e^{L_{\mathrm{B}}}}^{\infty}(\tilde g-g)\,dx\ge0$ when $L_{\mathrm{B}}\ge L_1$ (in the six runs $L_{\mathrm{B}}=2000$ or 5000, so the first integral on the right is negative), and equals $\int_{e^{L_1}}^{\infty}(\tilde g-g)\,dx\ge0$ when $L_{\mathrm{B}}<L_1$. The integral $\int_{e^{L_{\mathrm{B}}}}^{e^{L_1}}g\,dx$ is computed by the rigorous quadrature of Arb [17] in the variable L, which evaluates the integrand $\sqrt{a(L)^2+b(L)^2}\,e^{(1-\sigma_1)L}$ on complex balls and uses Gauss–Legendre rules only on sub-intervals where the implementation certifies that the integrand is holomorphic (the principal square root returns an indeterminate ball when its argument ball meets $(-\infty,0]$, and elsewhere it is holomorphic and coincides with h on the real segment, where $a^2+b^2\ge0$), falling back to the enclosure of the integrand on the sub-interval otherwise, and the majorant is $\tilde g(x)=\frac{1}{\pi}(\sum_i q_iL^i+3)e^{-\sigma_1L}$ with $L=\log x$, which satisfies $\tilde g\ge g$ on $[e^{L_1},\infty)$ because $h\le|a|+|b|\le\frac{1}{\pi}(\sum_i q_iL^i+\frac{2}{L})$ and $\frac{2}{L}\le3$; its integral $\int_{e^{L_1}}^{\infty}\tilde g\,dx=\frac{1}{\pi}\int_{L_1}^{\infty}(\sum_i q_iL^i+3)e^{(1-\sigma_1)L}\,dL$ is a finite combination of incomplete gamma functions evaluated in closed form. The partial sums over $n\le N_0$ are exact ball sums over the 5 761 455 primes below 10^8 and their powers. □

4.3. **The elementary part.**

Lemma 4.6 (Stirling's formula with remainder)**.** *For $|\operatorname{ph}z|<\pi$, $\operatorname{Log}\Gamma(z)=(z-\frac12)\operatorname{Log}z-z+\frac12\log(2\pi)+\mathcal{R}(z)$ with $|\mathcal{R}(z)|\le\sec^2(\frac12\operatorname{ph}z)/(12|z|)$* [10, §5.11(ii)], [19, (8.4.06)]*. Consequently, for $\operatorname{Re}z>0$ and every integer $k\ge0$,*

$$|\mathcal{R}^{(k+1)}(z)|\le(k+1)!\,\frac{2}{3|z|}\Big(\frac{2}{|z|}\Big)^{k+1}.\tag{4.12}$$

Proof. On the circle $|w-z|=\frac12|z|$ one has $|w|\ge\frac12|z|$ and $|\operatorname{ph}w-\operatorname{ph}z|\le\arcsin\frac12=\frac{\pi}{6}$, so $|\operatorname{ph}w|<\frac{\pi}{2}+\frac{\pi}{6}=\frac{2\pi}{3}$ and $\sec^2(\frac12\operatorname{ph}w)<\sec^2\frac{\pi}{3}=4$; hence $|\mathcal{R}(w)|\le\frac{4}{12\cdot|z|/2}=\frac{2}{3|z|}$ there, and Cauchy's estimate for the $(k+1)$-st derivative on this circle gives (4.12). □

Since $\psi_\Gamma = (\operatorname{Log}\Gamma)'$, Lemma 4.6 gives $\psi_\Gamma(z) = \operatorname{Log} z - \frac{1}{2z} + \mathcal{R}'(z)$ and, for $k \geq 1$,

$$\psi_\Gamma^{(k)}(z) = \frac{(-1)^{k-1}(k-1)!}{z^k} + \frac{(-1)^k k!}{2z^{k+1}} + \mathcal{R}^{(k+1)}(z). \tag{4.13}$$

Lemma 4.7 (The elementary part for large T)**.** *Let $T \geq T_0 > 0$, $z_1 = s_1/2$, $C_1 = c_{\mathrm{R},1}/(2\pi)$ and suppose $A_1 \geq C_1$. Then*

$$F_u(T) \leq \frac{7}{8} - C_1 \log(2\pi) + \frac{\mathsf{K}}{T}, \qquad F_\ell(T) \leq -\frac{7}{8} - C_1 \log(2\pi) + \frac{\mathsf{K}}{T}, \tag{4.14}$$

where

$$\begin{aligned}\mathsf{K} = {} & \frac{1}{\pi}\Big(4d + \frac{\sigma_1^2}{2} + \frac{\sigma_2(\sigma_2-1)}{2} + 1 + \frac{\sigma_2^2}{2} + \sigma_1(\sigma_1-1)\Big) + \frac{|c_{\mathrm{R},1}|}{\pi T_0}\Big((2\sigma_1-1) + \frac{\sigma_1^2}{4} + \frac{8}{3}\Big) \\ & + \frac{T_0}{\pi} \sum_{\substack{\iota=(\mathrm{X},j)\in\mathcal{B} \\ j\geq 2}} |c_\iota| \Big(\frac{2}{T_0^j} + \frac{1}{(j-1)!\,2^j}\Big[(j-2)!\Big(\frac{2}{T_0}\Big)^{j-1} + \frac{(j-1)!}{2}\Big(\frac{2}{T_0}\Big)^j + j!\,\frac{4}{3T_0}\Big(\frac{4}{T_0}\Big)^j\Big]\Big).\end{aligned} \tag{4.15}$$

Proof. The term $\mathcal{E}(T)$. For $s = \sigma + iT$ with $\sigma > 1$, $\arg s = \frac{\pi}{2} - \arctan\frac{\sigma}{T}$ and $\arg(s-1) = \frac{\pi}{2} - \arctan\frac{\sigma-1}{T}$, so $\arg s + \arg(s-1) = \pi - \theta_\sigma$ with $0 \leq \theta_\sigma \leq \frac{2\sigma-1}{T}$. With $z = s/2$, Lemma 4.6 gives

$$\operatorname{Im}\operatorname{Log}\Gamma(z) = \Big(\frac{\sigma}{2} - \frac{1}{2}\Big)\arg z + \frac{T}{2}\log|z| - \frac{T}{2} + \operatorname{Im}\mathcal{R}(z),$$

with $\arg z = \frac{\pi}{2} - \arctan\frac{\sigma}{T}$ and $\log|z| = \log\frac{T}{2} + \frac{1}{2}\log(1 + \frac{\sigma^2}{T^2})$, and $|\mathcal{R}(z)| \leq \frac{1}{3T}$ since $|\operatorname{ph} z| < \frac{\pi}{2}$ and $|z| \geq \frac{T}{2}$. Hence

$$\mathcal{A}(s) = \pi + \frac{T}{2}\log\frac{T}{2\pi} - \frac{T}{2} + \frac{(\sigma-1)\pi}{4} + \delta_\sigma, \quad \delta_\sigma = -\theta_\sigma - \frac{\sigma-1}{2}\arctan\frac{\sigma}{T} + \frac{T}{4}\log\Big(1 + \frac{\sigma^2}{T^2}\Big) + \operatorname{Im}\mathcal{R}(z),$$

with $0 \leq \frac{\sigma-1}{2}\arctan\frac{\sigma}{T} \leq \frac{\sigma(\sigma-1)}{2T}$ and $0 \leq \frac{T}{4}\log(1 + \frac{\sigma^2}{T^2}) \leq \frac{\sigma^2}{4T}$. Since $2(\sigma_1-1) - (\sigma_2-1) = -\frac{1}{2}$ and $2(2\sigma_1 - 1) = 2\sigma_2 - 1 = 4d$,

$$\frac{1}{\pi}\big(2\mathcal{A}(s_1) - \mathcal{A}(s_2)\big) = 1 + \mathcal{M}(T) - \frac{1}{8} + \frac{2\delta_{\sigma_1} - \delta_{\sigma_2}}{\pi},$$

and, collecting the bounds (the arguments contribute at most $\frac{4d}{T}$ in absolute value, the three remainders at most $\frac{1}{T}$),

$$|\mathcal{E}(T) - \tfrac{7}{8}| \leq \frac{1}{\pi T}\Big(4d + \sigma_1(\sigma_1-1) + \frac{\sigma_1^2}{2} + \frac{\sigma_2(\sigma_2-1)}{2} + \frac{\sigma_2^2}{4} + 1\Big) \leq \frac{\mathsf{K}_\Gamma}{T},$$

where K_Γ is the first bracket of (4.15) (which is larger, having $\frac{\sigma_2^2}{2}$ in place of $\frac{\sigma_2^2}{4}$).

The term $j = 1$. By $\psi_\Gamma(z) = \operatorname{Log} z - \frac{1}{2z} + \mathcal{R}'(z)$ (the formula preceding (4.13)), $\operatorname{Re} D_1(s_1) = \operatorname{Re}\frac{1}{s_1} + \operatorname{Re}\frac{1}{s_1-1} - \frac{1}{2}\log\pi + \frac{1}{2}\log|z_1| - \frac{1}{2}\operatorname{Re}\frac{1}{s_1} + \frac{1}{2}\operatorname{Re}\mathcal{R}'(z_1)$, and $\log|z_1| = \log\frac{T}{2} + \frac{1}{2}\log(1 + \frac{\sigma_1^2}{T^2})$, so

$$\operatorname{Re} D_1(s_1) = \frac{1}{2}\log\frac{T}{2\pi} + \delta_1, \qquad \delta_1 = \frac{1}{2}\operatorname{Re}\frac{1}{s_1} + \operatorname{Re}\frac{1}{s_1 - 1} + \frac{1}{4}\log\Big(1 + \frac{\sigma_1^2}{T^2}\Big) + \frac{1}{2}\operatorname{Re}\mathcal{R}'(z_1),$$

and, using $0 \leq \operatorname{Re}\frac{1}{s} \leq \frac{\sigma}{T^2}$, $0 \leq \operatorname{Re}\frac{1}{s-1} \leq \frac{\sigma-1}{T^2}$, $0 \leq \log(1+y) \leq y$ and $|\mathcal{R}'(z_1)| \leq \frac{4}{3|z_1|^2} \leq \frac{16}{3T^2}$ from (4.12),

$$|\delta_1| \leq \frac{1}{T^2}\Big(\frac{3\sigma_1}{2} - 1 + \frac{\sigma_1^2}{4} + \frac{8}{3}\Big) \leq \frac{1}{T^2}\Big((2\sigma_1-1) + \frac{\sigma_1^2}{4} + \frac{8}{3}\Big) \leq \frac{1}{T_0 T}\Big((2\sigma_1-1) + \frac{\sigma_1^2}{4} + \frac{8}{3}\Big)$$

for $T \geq T_0$, so $|c_{\mathrm{R},1}|\,|\delta_1|/\pi$ is at most the second term of (4.15) divided by T.

The terms $j \ge 2$. $|D_j(s_1)| \le \frac{2}{T^j} + \frac{|\psi_\Gamma^{(j-1)}(z_1)|}{(j-1)!\,2^j}$, and by (4.13) with $k = j-1 \ge 1$, (4.12) and $|z_1| \ge \frac{T}{2}$,

$$|\psi_\Gamma^{(j-1)}(z_1)| \le (j-2)!\Big(\frac{2}{T}\Big)^{j-1} + \frac{(j-1)!}{2}\Big(\frac{2}{T}\Big)^{j} + j!\,\frac{4}{3T}\Big(\frac{4}{T}\Big)^{j}.$$

The resulting bound for $|D_j(s_1)|$ is of the form $\sum_{\nu\ge1}\alpha_\nu T^{-\nu}$ with $\alpha_\nu \ge 0$ (for $j=2$ the first term is $\frac{1}{2T}$), so T times it is non-increasing in T and $|D_j(s_1)| \le T_0\,\mathrm{bound}(T_0)/T$ for $T \ge T_0$; this gives the third term of (4.15).

Conclusion. $\mathcal{E}_u(T) = \mathcal{E}(T) + \frac{c_{\mathrm{R},1}}{\pi}\operatorname{Re} D_1(s_1) + \frac{1}{\pi}\sum_{j\ge2} c_\iota \Phi_{\mathrm{X}}(D_j(s_1)) = \frac{7}{8} + C_1\log\frac{T}{2\pi} + \Delta(T)$ with $|\Delta(T)| \le \mathsf{K}/T$, so $F_u(T) = \frac{7}{8} - C_1\log(2\pi) + (C_1 - A_1)\log T + \Delta(T) \le \frac{7}{8} - C_1\log(2\pi) + \mathsf{K}/T$, because $A_1 \ge C_1$ and $\log T \ge 0$. Likewise $\mathcal{E}_\ell(T) = \frac{7}{8} - C_1\log\frac{T}{2\pi} + \Delta_\ell(T)$ with $|\Delta_\ell| \le \mathsf{K}/T$, and $F_\ell(T) = -\frac{7}{8} - C_1\log(2\pi) + (C_1 - A_1)\log T - \Delta_\ell(T) \le -\frac{7}{8} - C_1\log(2\pi) + \mathsf{K}/T$. □

Lemma 4.8 (The theta function)**.** *For $T > 0$, $\frac{1}{\pi}\vartheta(T) + 1 = \mathcal{M}(T) + \frac{7}{8} + \frac{\varpi(T)}{\pi}$ with $|\varpi(T)| \le (\frac{1}{16} + \frac{1}{8} + \frac{1}{3})\frac{1}{T}$.*

Proof. With $z = \frac{1}{4} + \frac{iT}{2}$, Lemma 4.6 gives $\vartheta(T) = -\frac{1}{4}\arg z + \frac{T}{2}\log|z| - \frac{T}{2} - \frac{T}{2}\log\pi + \operatorname{Im}\mathcal{R}(z)$, and $\arg z = \frac{\pi}{2} - \arctan\frac{1}{2T}$, $\log|z| = \log\frac{T}{2} + \frac{1}{2}\log(1 + \frac{1}{4T^2})$. Hence $\vartheta(T) = \frac{T}{2}\log\frac{T}{2\pi} - \frac{T}{2} - \frac{\pi}{8} + \varpi(T)$ with $\varpi = \frac{1}{4}\arctan\frac{1}{2T} + \frac{T}{4}\log(1+\frac{1}{4T^2}) + \operatorname{Im}\mathcal{R}(z)$, and the three terms are at most $\frac{1}{8T}$, $\frac{1}{16T}$ and $\frac{1}{3T}$ ($|z| \ge \frac{T}{2}$, $|\operatorname{ph} z| < \frac{\pi}{2}$). Dividing by π and using $\pi\mathcal{M}(T) = \frac{T}{2}\log\frac{T}{2\pi} - \frac{T}{2}$ gives the claim. □

For the argument, write $\mathcal{E}_u^S = \mathcal{E}_u - \frac{7}{8} - \frac{\varpi}{\pi}$, $\mathcal{E}_\ell^S = \mathcal{E}_\ell - \frac{7}{8} - \frac{\varpi}{\pi}$ and $F_u^S = \mathcal{E}_u^S - A_1\log T$, $F_\ell^S = -\mathcal{E}_\ell^S - A_1\log T$. Since $S(T) = N(T) - \mathcal{M}(T) - \frac{7}{8} - \frac{\varpi(T)}{\pi}$ by (1.5) and Lemma 4.8, Proposition 4.1 gives

$$|S(T)| \le A_1\log T + C_\zeta + \max\big(F_u^S(T), F_\ell^S(T)\big), \tag{4.16}$$

and Lemma 4.7 gives, for $T \ge T_0$, $\max(F_u^S, F_\ell^S) \le -C_1\log(2\pi) + (\mathsf{K} + \frac{1}{\pi}(\frac{1}{16} + \frac{1}{8} + \frac{1}{3}))/T$.

Proposition 4.9 (Certified suprema of the elementary part)**.** *Let $T_0 = 10^4$. The program* `elem_gen.py` *(Appendix B) covers $[1, T_0]$ by 9215 intervals $[\alpha, \beta]$ with $\beta/\alpha \le 1.001$ and, on each, encloses F_u and F_ℓ (or F_u^S, F_ℓ^S) by the centred form $F(\frac{\alpha+\beta}{2}) + F'([\alpha,\beta])\cdot[-\frac{\beta-\alpha}{2}, \frac{\beta-\alpha}{2}]$, where $F'([\alpha,\beta])$ is a ball enclosing the derivative on the interval; for $T \ge T_0$ it uses Lemma 4.7. The resulting upper bounds for $\sup_{T\ge1}\max(F_u, F_\ell)$ and $\sup_{T\ge1}\max(F_u^S, F_\ell^S)$ for the six kernels are in Table 2.*

Proof. The functions F_u, F_ℓ are real-analytic in $T > 0$; the centred form is the mean value theorem $F(T) = F(T_{\mathrm{m}}) + F'(T_*)(T - T_{\mathrm{m}})$ for some T_* between T and T_{m}. The derivative is computed from $\frac{d}{dT}\arg s = \operatorname{Re}\frac{1}{s}$, $\frac{d}{dT}\operatorname{Im}\operatorname{Log}\Gamma(\frac{s}{2}) = \frac{1}{2}\operatorname{Re}\psi_\Gamma(\frac{s}{2})$, $\frac{d}{dT}\mathcal{M}(T) = \frac{1}{2\pi}\log\frac{T}{2\pi}$, $\frac{d}{dT}D_j(s) = iD_j'(s)$ with $D_1'(s) = -s^{-2} - (s-1)^{-2} + \frac{1}{4}\psi_\Gamma'(\frac{s}{2})$ and $D_j'(s) = -js^{-j-1} - j(s-1)^{-j-1} + \frac{(-1)^{j-1}}{(j-1)!\,2^{j+1}}\psi_\Gamma^{(j)}(\frac{s}{2})$ for $j \ge 2$, and, in the S-mode, $\frac{d}{dT}\vartheta(T) = \frac{1}{2}\operatorname{Re}\psi_\Gamma(\frac{1}{4} + \frac{iT}{2}) - \frac{1}{2}\log\pi$. The polygamma values on an interval are enclosed by $\psi_\Gamma^{(k)}(z_{\mathrm{m}}) + \psi_\Gamma^{(k+1)}(Z)\cdot i[-\frac{r}{2}, \frac{r}{2}]$, where Z is the ball of the segment and r its length in T, which is valid because $\psi_\Gamma^{(k)}(z+\eta) - \psi_\Gamma^{(k)}(z) = \int_0^1 \psi_\Gamma^{(k+1)}(z+u\eta)\eta\,du$ lies in η times the convex hull of the enclosure. All other operations are ball operations, so each interval bound is rigorous, and the supremum over $[1, T_0]$ is the maximum of the 9215 upper bounds. For $T \ge T_0$ the bound (4.14) is monotone in T, so it is evaluated at T_0. □

4.4. **The kernel inequality.** Condition (K) is an inequality between explicit elementary functions of two real variables on a half-strip. It is certified by bisection, as follows. For $t \ne 0$, by $\arctan y + \arctan\frac{1}{y} = \operatorname{sgn}(y)\frac{\pi}{2}$ and since $d \pm b > 0$ and $2d \pm b > 0$ for $0 \le b \le \frac{1}{2}$,

$$f(b,t) = \operatorname{sgn}(t)\,\pi - 2\arctan\frac{t}{d+b} - 2\arctan\frac{t}{d-b} + \arctan\frac{t}{2d+b} + \arctan\frac{t}{2d-b}, \tag{4.17}$$

so on each closed half-box $[0,\frac12]\times[0,T_1]$ (with $\operatorname{sgn}=+1$) and $[0,\frac12]\times[-T_1,0]$ (with $\operatorname{sgn}=-1$) the right side of (4.17) is real-analytic, including at $t=0$, and so is U, because $d\pm b\ge d-\frac12>0$.

Lemma 4.10 (Tail of the kernel inequality)**.** *Let $T_1\ge 2d+\frac12$, $0\le b\le\frac12$, $|t|\ge T_1$ and $u=1/t$. Then*

$$t^2 f(b,t)\le 4d^3u+\mathsf{K}_5|u|^3,\qquad \mathsf{K}_5=\frac15\Big(2\big(d+\tfrac12\big)^5+2d^5+\big(2d+\tfrac12\big)^5+(2d)^5\Big),\tag{4.18}$$

and

$$t^2U(b,t)=c_{\mathrm{R},1}\Big(\frac{d+b}{1+(d+b)^2u^2}+\frac{d-b}{1+(d-b)^2u^2}\Big)+\sum_{\substack{\iota\in\mathcal{B}\\ j\ge2}}c_\iota\sum_{x=d\pm b}\Phi_{\mathrm{X}}\Big(u^{j-2}(xu+i)^{-j}\Big),\tag{4.19}$$

a real-analytic function of (b,u) on $[0,\frac12]\times[-1/T_1,1/T_1]$ with value $2dc_{\mathrm{R},1}-2c_{\mathrm{R},2}$ at $u=0$.

Proof. Write $f=\sum_{i=1}^4\alpha_i\arctan(y_iu)$ with $(\alpha_i)=(2,2,-1,-1)$ and $(y_i)=(d+b,d-b,2d+b,2d-b)$; then $|y_iu|\le(2d+\frac12)/T_1\le1$. For $|y|\le1$, $\arctan y=y-\frac{y^3}{3}+r(y)$ with $|r(y)|\le\frac{|y|^5}{5}$ (alternating series). Since $\sum\alpha_iy_i=0$ and $\sum\alpha_iy_i^3=2(d+b)^3+2(d-b)^3-(2d+b)^3-(2d-b)^3=-12d^3$,

$$f=4d^3u^3+\sum_i\alpha_ir(y_iu),\qquad \Big|\sum_i\alpha_ir(y_iu)\Big|\le\frac{|u|^5}{5}\sum_i|\alpha_i||y_i|^5\le\mathsf{K}_5|u|^5,$$

using $|y_1|\le d+\frac12$, $|y_2|\le d$, $|y_3|\le 2d+\frac12$, $|y_4|\le 2d$. Multiplying by $t^2=u^{-2}$ gives (4.18). For (4.19): $t^2\operatorname{Re}(x+it)^{-1}=t^2\frac{x}{x^2+t^2}=\frac{x}{1+x^2u^2}$, and for $j\ge2$, $x+it=(xu+i)/u$, so $t^2(x+it)^{-j}=u^{j-2}(xu+i)^{-j}$. At $u=0$ the terms with $j=1$ give $2dc_{\mathrm{R},1}$, the terms with $j=2$ give $c_{\mathrm{R},2}\cdot2\operatorname{Re}(i^{-2})+c_{\mathrm{I},2}\cdot2(-\operatorname{Im}i^{-2})=-2c_{\mathrm{R},2}$, and the terms with $j\ge3$ vanish. □

Proposition 4.11 (Certification of (K))**.** *The program* `certify_hybrid.py` *(Appendix B), run with an empty list of discs, certifies* (K) *for a kernel as follows. (a) On $[0,\frac12]\times[0,T_1]$ and on $[0,\frac12]\times[-T_1,0]$ it evaluates $H=U-f$, with f in the form* (4.17)*, on boxes by the centred form $H(\text{centre})+\nabla H(\text{box})\cdot(\text{half-widths})$, and bisects every box on which the ball of H does not lie in $(0,\infty)$; it stops when all boxes are certified. (b) For $|t|\ge T_1$ it certifies $t^2U-4d^3u-\mathsf{K}_5|u|^3>0$ on $[0,\frac12]\times(0,1/T_1]$ and on $[0,\frac12]\times[-1/T_1,0)$ in the same way, with the expressions of Lemma 4.10; by* (4.18) *this implies $U>f$ there. The driver tries $T_1=100$ first and $T_1=2000$ if the first run fails; both satisfy $T_1\ge2d+\frac12$. For each of the six kernels of Theorem 2.1 the run terminated with every box certified (Appendix C).*

Proof. The centred form is the two-variable mean value theorem for the real-analytic function H on a convex box, with the gradient enclosed by balls; a positive lower bound of the enclosure on a box proves $U>f$ on the box. The union of the certified boxes is the whole half-strip $0\le b\le\frac12$, $0<|t|\le T_1$ (the boxes are closed and include $t=0$, where (4.17) is the continuous extension), together with $|t|\ge T_1$ by (b). □

4.5. **Proof of Theorem 2.1 and of Corollary 2.2.**

Proof of Theorem 2.1. Each line A_1L+A_3 of (2.2) corresponds to a kernel listed in Table 2 and Appendix A, with $A_1=\lceil C_1\rceil_5\ge C_1=c_{\mathrm{R},1}/(2\pi)$. For it, (K) holds by Proposition 4.11; $\sum_pM_p\le C_\zeta$ by Proposition 4.5; and $\sup_{T\ge1}\max(F_u,F_\ell)$ is bounded by Proposition 4.9, with $A_1\ge C_1$ as required in Lemma 4.7. By (4.9), for every $T\ge1$ that is not an ordinate, $|N(T)-\mathcal{M}(T)|\le A_1\log T+C_\zeta+\sup\max(F_u,F_\ell)\le A_1\log T+A_3^N$. At an ordinate T, $N(T)=\lim_{t\to T^+}N(t)$ and $\mathcal{M}$ is continuous, so the inequality persists. The same argument with (4.16) gives (2.3), with $S(T)=S(T^+)$ at the ordinates. □

line	d	m	A_1	$C_\zeta <$	$\sup\max(F_u, F_\ell) \le$	A_3^N	$\sup\max(F_u^S, F_\ell^S) \le$	A_3^S
ℓ_5	1	5	0.20176	0.70360002	0.66536440	1.369	0.13949090	0.844
ℓ_{20}	1.2	5	0.10823	1.46807534	0.70207071	2.171	−0.18278649	1.286
ℓ_{60}	1	5	0.09473	2.10426949	0.72894156	2.834	−0.15847575	1.946
ℓ_{150}	0.85	5	0.09349	2.25578346	0.73031685	2.987	−0.15757634	2.099
ℓ_{400}	0.85	10	0.08399	4.46374163	0.73756551	5.202	−0.14582673	4.318
ℓ_{6000}	0.85	10	0.08382	4.58951035	0.73573753	5.326	−0.14641036	4.444

TABLE 2. The six kernels of Theorem 2.1: parameters, certified upper bounds for C_ζ and for the suprema of the elementary part, and the resulting constants $A_3^N = \lceil C_\zeta + \sup\max(F_u, F_\ell)\rceil_3$, $A_3^S = \lceil C_\zeta + \sup\max(F_u^S, F_\ell^S)\rceil_3$ (rounded up to three decimals); $A_1 = \lceil c_{\mathrm{R},1}/(2\pi)\rceil_5$. The coefficients c_ι are in Appendix A.

Proof of Corollary 2.2. The difference between the minimum of finitely many lines and the minimum of finitely many other lines is a piecewise linear function of $L \ge 0$, whose minimum over $[0,\infty)$ is attained at $L = 0$, at a breakpoint of one of the two envelopes, or, if the slopes allow, approached as $L \to \infty$. Here the smallest slope of the bounds of [6] and of [1] is 0.0946 and 0.097 respectively, larger than 0.08382, so the differences tend to $+\infty$. Evaluating at $L = 0$ and at the finitely many breakpoints, in exact rational arithmetic, gives the minimal differences 2.231 (at $L = 49.11$) and 0.291 (at $L = 8.57$), both positive. The statements about single lines are the comparisons of two lines each. When both the slope and the constant of the new line are smaller, the inequality holds for all $L \ge 0$: this is the case for $0.08399L + 5.202$ and for $0.08382L + 5.326$ against $0.0946L + 5.746$. When the new line has the smaller slope but the larger constant, the inequality holds exactly for L larger than the quotient of the difference of the constants by the difference of the slopes: $(5.202 - 4.954)/(0.097 - 0.08399)$ and $(5.326 - 4.954)/(0.097 - 0.08382)$ against $0.097L + 4.954$. □

5. Jensen discs and admissible profiles

Definition 5.1 (Admissible profile)**.** An *admissible profile* is a continuous function $\mathfrak{e}\colon [\frac12, 1] \to [0,\infty)$ with $\mathfrak{e}(1) = 0$ such that, for every $\varepsilon > 0$,

$$|\zeta(\sigma + it)| \le K_\varepsilon |t|^{\mathfrak{e}(\sigma)+\varepsilon} \qquad \text{for } \tfrac12 \le \sigma \le 1,\ |t| \ge 2, \tag{5.1}$$

with a constant K_ε independent of σ and t. It is extended by $\mathfrak{e}(\sigma) = 0$ for $\sigma > 1$.

For $\frac12 \le \sigma \le 1$ let $\mu(\sigma)$ be the infimum of the $\xi \ge 0$ such that $\zeta(\sigma + it) \ll |t|^\xi$ as $|t| \to \infty$ [27, §5.1]; μ is convex, hence continuous, on $[\frac12, 1]$, and $\mu(1) = 0$. By definition, for every $\varepsilon > 0$ and each fixed σ, $\zeta(\sigma + it) \ll_{\sigma,\varepsilon} |t|^{\mu(\sigma)+\varepsilon}$ for $|t| \ge 2$.

Lemma 5.2 (Interpolation of pointwise bounds)**.** *Let $\frac12 = \varsigma_0 < \varsigma_1 < \cdots < \varsigma_n = 1$ and $e_0 \ge e_1 \ge \cdots \ge e_n = 0$ be such that, for each i and each $\varepsilon > 0$, $|\zeta(\varsigma_i + it)| \le K_{i,\varepsilon}|t|^{e_i+\varepsilon}$ for $|t| \ge 2$. Then the piecewise linear function $\mathfrak{e}$ with $\mathfrak{e}(\varsigma_i) = e_i$ is an admissible profile.*

Proof. Fix $\varepsilon > 0$ and $i < n$, and put $\mathcal{Z}(s) = (s-1)\zeta(s)$, $Q = 10$. On the strip $\varsigma_i \le \operatorname{Re} s \le \varsigma_{i+1}$, $\mathcal{Z}$ is holomorphic and of polynomial growth: by [27, (2.1.4)], $\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \{x\} x^{-s-1}\,dx$ for $\operatorname{Re} s > 0$, so $|\mathcal{Z}(s)| \le |s| + |s||s-1|/\operatorname{Re} s \le 3|s|(1+|s|)$ there. On the line $\operatorname{Re} s = \varsigma_i$: for $|t| \ge 2$, $|s-1| \le |Q+s|$ (as $Q + \varsigma_i \ge |\varsigma_i - 1|$) and $|t| \le |Q+s|$, so $|\mathcal{Z}(s)| \le K_{i,\varepsilon}|Q+s|^{1+e_i+\varepsilon}$; for $|t| \le 2$, $\mathcal{Z}$ is bounded on the compact segment and $|Q+s| \ge 10$, so the same bound holds with a larger constant A_i. Likewise $|\mathcal{Z}(s)| \le B_i|Q+s|^{1+e_{i+1}+\varepsilon}$ on $\operatorname{Re} s = \varsigma_{i+1}$. Since $1 + e_i + \varepsilon \ge 1 + e_{i+1} + \varepsilon \ge 0$, the theorem of Phragmén–Lindelöf type of Rademacher [21] (also [11, Cor. 3] with $\alpha_2 = 0$) gives,

on the strip,

$$|\mathcal{Z}(s)| \le \big(A_i|Q+s|^{1+e_i+\varepsilon}\big)^{\frac{\varsigma_{i+1}-\sigma}{\varsigma_{i+1}-\varsigma_i}} \big(B_i|Q+s|^{1+e_{i+1}+\varepsilon}\big)^{\frac{\sigma-\varsigma_i}{\varsigma_{i+1}-\varsigma_i}} \le \max(A_i,B_i)\,|Q+s|^{1+\mathfrak{e}(\sigma)+\varepsilon},$$

because the exponent interpolates linearly. For $|t|\ge 2$, $|Q+s|^2=(Q+\sigma)^2+t^2\le 121+t^2\le 36t^2$ and $|s-1|\ge|t|$, so $|\zeta(s)|\le \max(A_i,B_i)6^{1+e_0+\varepsilon}|t|^{\mathfrak{e}(\sigma)+\varepsilon}$ on the strip. Taking the maximum over the n strips gives (5.1); $\mathfrak{e}$ is continuous, non-negative, and $\mathfrak{e}(1)=0$. □

An *exponent pair* (k,ℓ) in the sense of [27, §5.20] satisfies $0\le k\le\frac12\le\ell\le 1$, and

$$\mu(\ell-k)\le k \tag{5.2}$$

[25, Lemma 23], [27, (5.20.1)].

Lemma 5.3 (The profile $\mathfrak{e}_{\mathrm{P}}$)**.** *Let $\mathcal{N}$ be the set of the following points (ς,e) with $\frac12\le\varsigma\le 1$, for each of which $\mu(\varsigma)\le e$ is a theorem:*

- *(i)* $(\frac12,\frac{13}{84})$ [7, Thm. 5]*;*
- *(ii) the breakpoints of the piecewise linear bound for μ of Trudgian and Yang* [30, Thm. 2.4]*, that is, $(\varsigma,\mu^{\mathrm{TY}}(\varsigma))$ for $\varsigma\in\{\frac{88225}{153852},\frac{521}{796},\frac{53141}{76066},\frac{3620}{5119},\frac{52209}{69128},\frac{1389}{1736},\frac{134765}{163248},\frac{18193}{21906},\frac{249}{280},\frac{277}{300}\}$, where μ^{TY} is the bound of that theorem (its ten pieces agree at the common endpoints, and the last piece is stated there up to $\frac{277}{300}$);*
- *(iii) $(\ell-k,k)$ for the four exponent pairs (k,ℓ) of Tao, Trudgian and Yang* [25, Thm. 20]*: $(\frac{89}{1282},\frac{997}{1282})$, $(\frac{652397}{9713986},\frac{7599781}{9713986})$, $(\frac{10769}{351096},\frac{609317}{702192})$, $(\frac{89}{3478},\frac{15327}{17390})$, by* (5.2)*;*
- *(iv) (ς_n,μ_n) with $\varsigma_n=1-\frac{3n^2-3n+2}{n(n-1)^2(n+2)}$ and $\mu_n=\frac{2}{(n-1)^2(n+2)}$ for $6\le n\le 399$: by Heath-Brown* [15, Thm. 2]*, $\big(\frac{2}{(n-1)^2(n+2)},1-\frac{3n-2}{n(n-1)(n+2)}+\varepsilon'\big)$ is an exponent pair for every $n\ge 3$ and $\varepsilon'>0$, so* (5.2) *gives $\mu(\varsigma_n+\varepsilon')\le\mu_n$ for every $\varepsilon'>0$, and $\mu(\varsigma_n)\le\mu_n$ by the continuity of μ;*
- *(v) $(1,0)$, since $\zeta(1+it)\ll\log|t|$* [27, Thm. 3.5].

Let $\mathfrak{e}_{\mathrm{P}}$ be the piecewise linear function through the vertices of the lower convex hull of $\mathcal{N}$ (Table 5 in Appendix A lists them, except the vertices of (iv) with $10\le n\le 398$, which are given by the formula in (iv) and generated by `profile_published.py`*; the four points of (ii) with $\varsigma\in\{\frac{3620}{5119},\frac{134765}{163248},\frac{18193}{21906},\frac{249}{280}\}$ lie on or above chords of other points and are not vertices). Then $\mathfrak{e}_{\mathrm{P}}$ is an admissible profile.*

Proof. At each vertex (ς_i,e_i) of the hull, $\mu(\varsigma_i)\le e_i$, so $|\zeta(\varsigma_i+it)|\le K_{i,\varepsilon}|t|^{e_i+\varepsilon}$ for $|t|\ge 2$ and every $\varepsilon>0$, by the definition of μ. The vertices have $\varsigma_0=\frac12$, $\varsigma_{\mathrm{last}}=1$, and non-increasing ordinates (the values in Table 5 decrease, and the last is 0). Lemma 5.2 applies. □

The points of $\mathcal{N}$ that are not vertices are not needed: the chord between two vertices is a valid bound by Lemma 5.2, and it is smaller. No use is made of the convexity of μ beyond its continuity in (iv).

A *disc datum* is a triple (c,R,v) of real numbers with $c>1$, $R>0$ and $c-R\ge\frac12$. Its weight is

$$W(b,t)=\log^+\frac{R}{\sqrt{(c-\frac12-|b|)^2+(t+v)^2}}\qquad(|b|<\tfrac12,\ t\in\mathbb{R}), \tag{5.3}$$

and, for an admissible profile $\mathfrak{e}$,

$$J(\mathfrak{e};c,R)=\frac{1}{2\pi}\int_0^{2\pi}\mathfrak{e}(c+R\cos\theta)\,d\theta=\frac{1}{\pi}\int_0^{\pi}\mathfrak{e}(c+R\cos\theta)\,d\theta. \tag{5.4}$$

Since $\mathfrak{e}=0$ on $[1,\infty)$ and $c-R\ge\frac12$, only the values of $\mathfrak{e}$ on $[c-R,1]\subset[\frac12,1]$ enter.

Lemma 5.4 (Jensen bound)**.** *Let $\mathfrak{e}$ be an admissible profile, (c,R,v) a disc datum and $\varepsilon>0$. There is T_ε such that for all $T\ge T_\varepsilon$,*

$$\sum_\rho W(b_\rho,t_\rho)\le 2\big(J(\mathfrak{e};c,R)+\varepsilon\big)\log T, \tag{5.5}$$

and the same bound holds for $\sum_\rho W(b_\rho,-t_\rho)$.

Proof. Let $s_0=c+i(T+v)$ and, for a zero ρ, let ρ^* be the one of ρ, $1-\bar\rho$ with real part $\ge\frac12$, so $\rho^*=\frac12+|b_\rho|+i\gamma$. Then $|\rho^*-s_0|^2=(c-\frac12-|b_\rho|)^2+(t_\rho+v)^2$ and $W(b_\rho,t_\rho)=\log^+(R/|\rho^*-s_0|)$. The point ρ^* is a zero of ζ, and each zero w with $\operatorname{Re}w>\frac12$ is ρ^* for exactly two zeros (w and $1-\bar w$), which have the same multiplicity, while a zero w with $\operatorname{Re}w=\frac12$ is ρ^* only for $\rho=w$. Since the open disc $\mathbb{D}=\{|s-s_0|<R\}$ lies in $\operatorname{Re}s>c-R\ge\frac12$,

$$\sum_\rho W(b_\rho,t_\rho)\le 2\sum_{w\in\mathbb{D}}\log\frac{R}{|w-s_0|}, \tag{5.6}$$

the sum over the zeros of ζ in $\mathbb{D}$ with multiplicity. For $T>R+|v|+1$ the closed disc $\bar{\mathbb{D}}$ does not contain $s=1$, so ζ is holomorphic on it, and $\zeta(s_0)\ne0$ because $\operatorname{Re}s_0=c>1$. Jensen's formula [26, §3.61], [22, Thm. 15.18] (which allows zeros on the circle) gives

$$\sum_{w\in\mathbb{D}}\log\frac{R}{|w-s_0|}=\frac{1}{2\pi}\int_0^{2\pi}\log|\zeta(s_0+Re^{i\theta})|\,d\theta-\log|\zeta(s_0)|. \tag{5.7}$$

By the Euler product, $|\zeta(s_0)|\ge\prod_p(1+p^{-c})^{-1}=\zeta(2c)/\zeta(c)$. On the circle, $s=\sigma+it$ with $\sigma=c+R\cos\theta\ge\frac12$ and $t=T+v+R\sin\theta\in[T-R-|v|,T+R+|v|]$, so $\log t=\log T+O(1/T)$ and $t\ge2$ for large T. Where $\sigma\le1$, (5.1) gives $\log|\zeta(s)|\le(\mathfrak{e}(\sigma)+\varepsilon/2)\log t+\log K_{\varepsilon/2}$. Where $1\le\sigma\le2$, $|\zeta(s)|\le K\log t$ [27, Thm. 3.5], and where $\sigma\ge2$, $|\zeta(s)|\le\zeta(2)$; in both cases $\log|\zeta(s)|\le\log\log t+O(1)=o(\log T)$. Inserting in (5.7) and (5.6),

$$\sum_\rho W(b_\rho,t_\rho)\le 2\big(J(\mathfrak{e};c,R)+\tfrac{\varepsilon}{2}\big)\log T+O_\varepsilon(\log\log T)+2\log\frac{\zeta(c)}{\zeta(2c)},$$

and the last two terms are at most $\varepsilon\log T$ for T large. Replacing t_ρ by $-t_\rho$ in W amounts to replacing v by $-v$, which changes neither J nor the argument. □

6. The asymptotic estimate

Lemma 6.1 (Main term, asymptotic form)**.** *For fixed $d>\frac12$, as $T\to\infty$, $\frac1\pi\big(2\arg\xi(s_1)-\arg\xi(s_2)\big)=\frac1\pi\vartheta(T)+1+O(1)$.*

Proof. By (4.5), $\arg\xi(s)=\mathcal{A}(s)+O(1)$ for $\sigma\in\{\sigma_1,\sigma_2\}$, since $|\sum_n\frac{\Lambda(n)}{\log n}n^{-\sigma}\sin(T\log n)|\le\log\zeta(\sigma)$. The proof of Lemma 4.7 gives $\frac1\pi(2\mathcal{A}(s_1)-\mathcal{A}(s_2))=\mathcal{M}(T)+\frac78+O(1/T)$, and Lemma 4.8 gives $\mathcal{M}(T)+\frac78=\frac1\pi\vartheta(T)+1+O(1/T)$. □

Lemma 6.2 (Sums of the kernel functions, asymptotic form)**.** *For fixed $d>\frac12$ and a kernel U, as $T\to\infty$,*

$$\sum_\rho U(b_\rho,t_\rho)=c_{\mathrm{R},1}\log T+O(1),\qquad\sum_\rho U(b_\rho,-t_\rho)=c_{\mathrm{R},1}\log T+O(1).$$

Proof. By Lemma 3.2, $\sum_\rho G_\iota(b_\rho,t_\rho)=2\Phi_{\mathrm{X}}(D_j(s_1))+O(1)$, the arithmetic sum being bounded by $\frac{2}{(j-1)!}\sum_n\Lambda(n)(\log n)^{j-1}n^{-\sigma_1}<\infty$. By the proof of Lemma 4.7, $\operatorname{Re}D_1(s_1)=\frac12\log\frac{T}{2\pi}+O(T^{-2})$, $\operatorname{Im}D_1(s_1)=O(1)$ and $D_j(s_1)=O(T^{-(j-1)})=o(1)$ for $j\ge2$, because $|\psi_\Gamma^{(j-1)}(z_1)|=O(T^{-(j-1)})$ by (4.13) and (4.12); for $j=2$ this is only $O(T^{-1})$, which suffices. Hence $\sum_\rho U(b_\rho,t_\rho)=c_{\mathrm{R},1}\log T+O(1)$. The second statement follows from $U(b,-t)=\sum_\iota\varepsilon_{\mathrm{X}}c_\iota G_\iota(b,t)$ and the same bounds. □

Fix $d = \frac{9}{10}$, $m = 10$, the 19 coefficients c_ι ($\iota \in \mathcal{B}_{10}$) of Table 3, and the $K = 90$ disc data (c_k, R_k, v_k) with weights $\kappa_k \geq 0$ of Table 4 (Appendix A). The c_ι and κ_k are the exact binary numbers of the file `hyb_m10_P.json` (Appendix D); the disc data are the rational numbers of the table, obtained from the file by rounding to denominators $\leq 10^4$ and replacing R_k by $\min(R_k, c_k - \frac{1}{2})$, so that $c_k - R_k \geq \frac{1}{2}$ exactly. Let W_k be the weight (5.3) of (c_k, R_k, v_k) and put

$$U = \sum_{\iota \in \mathcal{B}_{10}} c_\iota G_\iota, \qquad C_{\mathrm{H}} = \frac{c_{\mathrm{R},1}}{2\pi} + \frac{1}{\pi} \sum_{k=1}^{K} \kappa_k J(\mathfrak{e}_{\mathrm{P}}; c_k, R_k). \tag{6.1}$$

Proposition 6.3 (Certified hybrid inequality)**.** *For $0 \leq b \leq \frac{1}{2}$ and $t \neq 0$,*

$$f(b,t) \leq U(b,t) + \sum_{k=1}^{K} \kappa_k W_k(b,t). \tag{6.2}$$

Moreover $C_{\mathrm{H}} \in [0.07485961427109072 \pm 1.25 \cdot 10^{-18}]$, so $C_{\mathrm{H}} < 0.0748596142711$; its two parts are $c_{\mathrm{R},1}/(2\pi) = 0.0655821827845\ldots$ and $\frac{1}{\pi}\sum_k \kappa_k J_k = 0.009277431487\ldots$.

Proof. The program `certify_hybrid.py` with the file `hyb_m10_P.json` and $T_1 = 40$ (Appendix C) proceeds as in Proposition 4.11, with these changes. (a) On the two half-boxes $[0, \frac{1}{2}] \times [0, T_1]$ and $[0, \frac{1}{2}] \times [-T_1, 0]$ it bounds $H = U - f + \sum_k \kappa_k W_k^{\mathrm{low}}$ from below, where $U - f$ is enclosed by the centred form and W_k^{low} is the lower endpoint of the ball of $\log \frac{R_k}{\sqrt{(c_k - \frac{1}{2} - b)^2 + (t + v_k)^2}}$ over the box when that endpoint is positive, and 0 otherwise; since $\kappa_k \geq 0$ and $W_k \geq W_k^{\mathrm{low}}$ on the box, a positive lower bound for H proves (6.2) on the box. It terminated with all boxes certified (2 028 149 boxes for $t > 0$, 2 294 869 for $t < 0$). (b) For $|t| \geq T_1 = 40$ every W_k vanishes, because $W_k(b,t) > 0$ requires $|t + v_k| < R_k$, hence $|t| < R_k + |v_k| \leq 2.2 < T_1$ (Table 4), and $T_1 \geq 2d + \frac{1}{2} = 2.3$; the tail is certified as in Proposition 4.11(b) (277 098 boxes). (c) The profile $\mathfrak{e}_{\mathrm{P}}$ is piecewise linear with rational vertices and $c_k - R_k \geq \frac{1}{2}$, so by (5.4) each $J(\mathfrak{e}_{\mathrm{P}}; c_k, R_k)$ is a finite sum, over the pieces $[\varsigma_i, \varsigma_{i+1}] \cap [c_k - R_k, 1] = [p, q]$ with $\mathfrak{e}_{\mathrm{P}} = \alpha + \beta\sigma$ there, of

$$\frac{1}{\pi}\Big[(\alpha + \beta c_k)(\theta_p - \theta_q) + \beta R_k(\sin\theta_p - \sin\theta_q)\Big], \qquad \theta_\sigma = \arccos\frac{\sigma - c_k}{R_k} \quad (\theta_p \geq \theta_q),$$

since $\sigma = c_k + R_k\cos\theta$ and $d\sigma = -R_k \sin\theta\, d\theta$; the arguments of arccos lie in $[-1, 1]$ because $p, q \in [c_k - R_k, c_k + R_k]$. The program evaluates these sums in ball arithmetic and reports the ball of C_{H} stated. □

Proof of Theorem 2.3. Let $\varepsilon > 0$ and let $T \to \infty$ through values that are not ordinates. All functions in (6.2) are even in b, and $|b_\rho| < \frac{1}{2}$, so (6.2) holds at (b_ρ, t_ρ) and at $(b_\rho, -t_\rho)$ for every zero; with $f(b, -t) = -f(b, t)$,

$$-U(b_\rho, -t_\rho) - \sum_k \kappa_k W_k(b_\rho, -t_\rho) \leq f(b_\rho, t_\rho) \leq U(b_\rho, t_\rho) + \sum_k \kappa_k W_k(b_\rho, t_\rho).$$

Summing over ρ (all series converge absolutely: those of U by Lemma 3.2, those of W_k because only finitely many zeros lie in each disc) and using Lemmas 6.2 and 5.4 with the admissible profile $\mathfrak{e}_{\mathrm{P}}$ of Lemma 5.3,

$$\Big|\frac{1}{2\pi}\sum_\rho f(b_\rho, t_\rho)\Big| \leq \Big(\frac{c_{\mathrm{R},1}}{2\pi} + \frac{1}{\pi}\sum_k \kappa_k\big(J(\mathfrak{e}_{\mathrm{P}}; c_k, R_k) + \varepsilon\big)\Big)\log T + O(1)$$

for $T \geq \max_k T_\varepsilon(k)$. By Lemma 3.1 and Lemma 6.1, $S(T) = \frac{1}{2\pi}\sum_\rho f(b_\rho, t_\rho) + O(1)$, so $|S(T)| \leq (C_{\mathrm{H}} + \varepsilon\sum_k \kappa_k/\pi)\log T + O(1)$. Dividing by $\log T$, letting $T \to \infty$ (at the ordinates $S(T) = S(T^+)$ is a limit of values at non-ordinates, so the limit superior over non-ordinates is $\mathfrak{a}$) and then $\varepsilon \to 0$ gives $\mathfrak{a} \leq C_{\mathrm{H}}$, and Proposition 6.3 gives (2.6). Finally $0.0748596142711 < 0.0786 < \Lambda(0)$. □

7. Methods and remarks

7.1. **Where the kernel inequality binds.** For fixed coefficients, $U - f$ is harmonic in (b,t) on each of the half-strips $|b| < d$, $t > 0$ and $|b| < d$, $t < 0$: each $\arctan\frac{y}{t}$ is an argument of $t + iy$, hence harmonic in (y,t) and in (b,t), and each $F_\iota(d \pm b, t)$ is the real or imaginary part of a holomorphic function of $(d \pm b) + it$. By the minimum principle on the rectangles $[-\frac12,\frac12] \times [\varepsilon, R]$ and $[-\frac12,\frac12] \times [-R, -\varepsilon]$, followed by $\varepsilon \to 0$ and $R \to \infty$, $f \le U$ on $|b| \le \frac12$, $t \ne 0$ holds as soon as $U - f \ge 0$ on the lines $b = \pm\frac12$ and $\liminf(U - f) \ge 0$ as $t \to 0^{\pm}$ and as $|t| \to \infty$, uniformly in b. The last condition is automatic, because $U - f \to 0$ uniformly in b as $|t| \to \infty$; more precisely $t^2(U - f) \to 2dc_{\mathrm{R},1} - 2c_{\mathrm{R},2}$ (Lemma 4.10), so the condition on the lines $b = \pm\frac12$ forces this constant to be non-negative, and for every kernel certified here it is positive, since the boxes of part (b) of the certification contain $u = 0$. The validity of the kernel inequality is thus controlled by its boundary $b = \pm\frac12$, that is, by hypothetical zeros with β arbitrarily close to 0 or 1; in the optimised kernels the active and nearly active constraints indeed lie near $b = \pm\frac12$. The values of ζ on $\operatorname{Re} s > 1$, which are all the method uses, cannot exclude clusters of order $\log T$ of such zeros near height T, and the numerical optima of Remark 2.5 reflect this. The weights W_k relax exactly these constraints: in Table 4 every disc has $c_k - R_k \ge \frac12$, and only the values of $\mathfrak{e}_{\mathrm{P}}$ on $[\frac12, 1]$ enter C_{H}.

7.2. **Optimisation.** For fixed d the set of admissible (c_ι) in (K), and of admissible (c_ι, κ_k) in (6.2), is convex, being defined by linear inequalities indexed by (b,t). For Theorem 2.1, the quantity minimised at a target height L^* is $A_1 L^* + C_\zeta + \sup F$, where C_ζ is approximated by $\sum_n \Lambda(n) n^{-\sigma_1} h(\log n)$ over the prime powers $n \le 2 \cdot 10^4$ plus a quadrature of the tail (for ℓ_{6000}, with the maximum over a grid of 96 phases in place of h for the primes $p \le 300$), a convex function of the coefficients (a sum of Euclidean norms and of pointwise maxima of linear functions), and $\sup F$ by its maximum on a grid of 260 heights in $[1, 10^6]$ and its limit at $T = \infty$. The problem is a second-order cone programme, solved with `cvxpy` and Clarabel; the constraint (K) is imposed on a grid of 26×2200 points (b,t) with a margin $\eta/(1+t^2)$, $\eta = 2 \cdot 10^{-4}$, and the points where the current solution violates it on a finer grid (201×24000) are added as cutting planes, until no violation remains. The optimisation was run for $d \in \{0.65, 0.75, 0.8, 0.85, 0.9, 0.95, 1, 1.2\}$, $m \in \{2,3,4,5,7,10\}$ and $L^* \in \{5, 20, 60, 150, 400, 1000, 2500, 6000, 20000\}$; the six kernels of Theorem 2.1 are the best found for $L^* = 5, 20, 60, 150, 400, 6000$. Longer kernels did not lower the leading constant below 0.0838: for $L^* = 20000$ the optimum is again $A_1 = 0.08381$.

For Theorem 2.3 the objective is (6.1), linear in (c_ι, κ_k), so the problem is a linear programme (HiGHS through `scipy`; program `explore_discs.py` of Appendix B), with the profile $\mathfrak{e}_{\mathrm{P}}$ in the coefficients J_k, a margin $\eta = 10^{-3}$, the constraint (6.2) imposed on a grid of 26×1600 points and refined by cutting planes. The functions W_k have a logarithmic singularity at the centre of the disc and a kink on its boundary, and $U + \sum_k \kappa_k W_k - f$ can have narrow dips that a grid misses: the violations are therefore searched on a finer grid (101 values of b; in t, 12 000 geometrically spaced points in $10^{-7} \le |t| \le 3000$ together with 60 000 equally spaced points in $|t| \le 3$) and the largest values found are refined by a local maximisation in (b,t) (Nelder–Mead), the maxima being added as cutting planes. The candidate discs were all (c, R, v) with

$$c \in \{1.03, 1.06, 1.1, \log 3, 1.15, 1.2, 1.25, 1.3, 1.4, 1.5, \log 5, 2, e, 3\},$$
$$R = c - 1 + x, \quad x \in \{0.05, 0.1, \dots, 0.5\}, \quad R \le c - \tfrac12,$$

and $v \in \{0, \pm0.2, \pm0.35, \pm0.5, \pm0.65, \pm0.8, \pm1, \pm1.3, \pm1.6, \pm2\}$; the optimum uses 90 of the 2 660 candidates, its value on the grid is 0.07486, and the certified value of (6.1) for the same data is $C_{\mathrm{H}} = 0.0748596142710907\dots$ (the difference comes from the rational rounding of the disc data and from the exact J_k). The certification of this solution succeeded at the first attempt. (An earlier run of the same programme, with the coarser violation search of a grid alone and with the profile of the table of [2] in place of $\mathfrak{e}_{\mathrm{P}}$, had produced a solution with a true violation of (6.2) of size $-8.9 \cdot 10^{-5}$

at $(b,t)\approx(0.467,-0.450)$, between two cutting planes, which had to be repaired by adding $2\cdot10^{-4}$ to $c_{\mathrm{R},1}$; the refinement of the violation search described above was introduced for this reason, and the result is $2.9\cdot10^{-5}$ smaller.) With the Bourgain profile (the piecewise linear function through the points $(\lambda_j-\kappa_j,\kappa_j)$, $(\kappa_j,\lambda_j)=A^j(\frac{13}{84},\frac{55}{84})$, $0\le j\le 7$, and $(1,0)$, where A is the A-process) in place of $\mathfrak{e}_{\mathrm{P}}$ the same programme gives, on the grid and without margin, 0.07531; with $\mathfrak{e}_{\mathrm{P}}$ and without margin it gives 0.07478; neither is certified.

7.3. **Certification.** All certificates use `python-flint` 0.9.0 (Arb [18], FLINT [12]) at 256 bits of precision; a ball computed by Arb contains the exact value of the expression evaluated, and the programs only ever use the endpoints of balls in the direction that makes the final statement weaker. The endpoints are binary numbers; when a bound is printed or written to a file it is converted to decimal with directed rounding (towards $+\infty$ for an upper bound), so that every decimal number in the outputs of Appendix C is itself a valid bound; the constants of Theorem 2.1 are sums of such decimals, rounded upwards. The kernel inequality for the hybrid ($T_1=40$) needed 2 028 149, 2 294 869 and 277 098 boxes for the three parts of Proposition 4.11 and 822 seconds; for the kernel of ℓ_{400} the three parts needed 11 835 193, 9 801 985 and 4 194 302 boxes and 2272 seconds, and for that of ℓ_{6000}, 14 760 921, 15 866 017 and 4 194 302 boxes and 3551 seconds. The constant C_ζ for each kernel takes between 52 and 164 seconds (the sum over the 5 761 455 primes below 10^8 and their powers, and the tail), and the elementary part between 124 and 162 seconds per mode. The coefficients are passed to the certifiers as the exact binary numbers written by the optimiser (the JSON files of Appendix D, whose decimal strings round-trip), so there is no rounding between search and certificate other than the rational rounding of the disc data described before (6.1), which is itself part of the certified data.

7.4. **Comparison of the two methods.** Backlund's method bounds the variation of $\arg\zeta$ on a horizontal segment ending at $\frac12+iT$ by the number of zeros of $\operatorname{Re}\zeta$ on it, which Jensen's formula controls through the growth of ζ on a disc centred at $c+iT$, $c>1$, that reaches $\frac12-\delta$; the growth to the left of the critical line is supplied by the functional equation, and the factor $\frac{1}{\log 2}$ in (1.7) comes from the geometry of a disc that reaches $\frac12$ from a centre at distance about 1. The method of Turing type bounds a sum over all the zeros, each weighted by $f(b_\rho,t_\rho)$, by its exact evaluation at points of $\operatorname{Re}s>1$, and needs no information inside the strip; its limit, in the computations of Remark 2.5, is near 0.079. The hybrid keeps the exact evaluation for most of the sum and uses Jensen's formula only where the kernel fails, on small discs that lie entirely in $\operatorname{Re}s\ge\frac12$, so that only the growth of ζ to the right of the critical line, where it is smallest, is paid for. Any improvement of the bounds for $\mu(\sigma)$ on $(\frac12,1)$ lowers C_{H} through $J(\mathfrak{e}_{\mathrm{P}};c_k,R_k)$ without any other change in the proof. Making Theorem 2.3 explicit requires versions of (5.1) with explicit constants, for which the available bounds are those of Hiary, Patel and Yang [16] on the critical line, of Yang [33] on the lines $\sigma=1-m/(2^m-2)$, and of Bellotti [4] in the strip; with them the method gives bounds of the form $A_1\log T+B\log\log T+C$ with A_1 near 0.076, valid from a computable height, and this is left for a later article.

Data availability. The programs, the coefficient files and the outputs of all certificates are reproduced in Appendices B–D and are available at `https://github.com/vcasana1/PaperMath/tree/explicit-S-turing/numerics/explicit_S_turing`.

AI declaration. The ideas of this article, the constructions and the strategy of the proofs are the author's. Generative AI tools were used during the work and the preparation of the manuscript: Claude (Anthropic), models Claude Fable 5.1, Claude Opus 5.5 and Claude Sonnet 5.5; ChatGPT (OpenAI), model Astra 6; and Microsoft 365 Copilot, based on the GPT-5 reasoning model. Under the author's direction, Claude and ChatGPT were used to draft and revise the LaTeX text, to check proofs, constants and citations against the sources, and to search the literature. The Python

programs of Appendix B were written with Claude. Microsoft 365 Copilot was used only to review drafts of the manuscript. The author has checked the whole article and takes full responsibility for its content. No AI tool is an author.

Mathematical software. The computations of Appendices B–C use Python 3.13 with `numpy`, `scipy`, `cvxpy` (solver Clarabel), `mpmath` and `python-flint` 0.9.0 (Arb ball arithmetic [18], FLINT 3.6.0 [12]).

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Appendix A. Data tables

ι	c_ι	ι	c_ι
$(\mathrm{R},1)$	0.412065007285	$(\mathrm{I},2)$	0.733460748935
$(\mathrm{R},2)$	0.370154287951	$(\mathrm{I},3)$	−0.136130059477
$(\mathrm{R},3)$	0.28440158115	$(\mathrm{I},4)$	0.151360502416
$(\mathrm{R},4)$	0.158618575787	$(\mathrm{I},5)$	−0.116929743682
$(\mathrm{R},5)$	0.0695422300653	$(\mathrm{I},6)$	−0.0365689491866
$(\mathrm{R},6)$	−0.0615776499388	$(\mathrm{I},7)$	0.106014270436
$(\mathrm{R},7)$	−0.040570135781	$(\mathrm{I},8)$	−0.0622716309967
$(\mathrm{R},8)$	0.075267182926	$(\mathrm{I},9)$	0.0166909270918
$(\mathrm{R},9)$	−0.0323507305873	$(\mathrm{I},10)$	−0.00179414655284
$(\mathrm{R},10)$	0.00563260523722		

Table 3. Coefficients c_ι of the hybrid kernel ($d = \frac{9}{10}$, $m = 10$), to twelve significant digits; the exact binary values are in Appendix D.

Table 4: The 90 disc data (c_k, R_k, v_k) (rational, as used in the certificate), the weights κ_k (ten significant digits; exact binary values in Appendix D), $J_k = J(\mathfrak{e}_{\mathrm{P}}; c_k, R_k)$ and the contributions $\kappa_k J_k/\pi$ to C_{H}.

k	c_k	R_k	v_k	κ_k	J_k	$\kappa_k J_k/\pi$
1	103/100	9/50	−1/2	0.04645216861	0.0069138770	0.000102230
2	103/100	9/50	1/2	0.04300701195	0.0069138770	$9.46479 \cdot 10^{-5}$
3	103/100	23/100	−13/20	0.04706304324	0.010786963	0.000161596
4	103/100	23/100	13/20	0.1466908127	0.010786963	0.000503677
5	103/100	7/25	−13/20	0.02597995484	0.015048973	0.000124450
6	103/100	7/25	13/20	0.1037916655	0.015048973	0.000497187

k	c_k	R_k	v_k	κ_k	J_k	$\kappa_k J_k/\pi$
7	103/100	33/100	−13/20	0.02381800473	0.019571517	0.000148382
8	103/100	33/100	13/20	0.1041736978	0.019571517	0.000648982
9	103/100	19/50	−13/20	0.0145395318	0.024737703	0.000114488
10	103/100	19/50	13/20	0.06487216198	0.024737703	0.000510820
11	103/100	43/100	1	0.01585298753	0.030311094	0.000152955
12	103/100	12/25	−1	0.003361796325	0.036143828	$3.86773 \cdot 10^{-5}$
13	103/100	12/25	1	0.006488928258	0.036143828	$7.46547 \cdot 10^{-5}$
14	103/100	53/100	−8/5	$5.414487565 \cdot 10^{-5}$	0.042238079	$7.27967 \cdot 10^{-7}$
15	103/100	53/100	−13/10	0.000301871607	0.042238079	$4.05860 \cdot 10^{-6}$
16	103/100	53/100	−1	0.0009151843659	0.042238079	$1.23045 \cdot 10^{-5}$
17	103/100	53/100	1	0.004457480243	0.042238079	$5.99299 \cdot 10^{-5}$
18	103/100	53/100	13/10	0.001466000015	0.042238079	$1.97101 \cdot 10^{-5}$
19	53/50	4/25	−1/2	0.04752584139	0.0032593311	$4.93070 \cdot 10^{-5}$
20	53/50	4/25	1/2	0.159072944	0.0032593311	0.000165035
21	53/50	21/100	−1/2	0.02538968149	0.0063616476	$5.14135 \cdot 10^{-5}$
22	53/50	21/100	1/2	0.2223325253	0.0063616476	0.000450218
23	53/50	13/50	−1/2	0.08631620266	0.010091730	0.000277273
24	53/50	13/50	1/2	0.1538221037	0.010091730	0.000494122
25	53/50	13/50	13/20	0.01149832481	0.010091730	$3.69360 \cdot 10^{-5}$
26	53/50	41/100	−4/5	0.01072845742	0.023727069	$8.10273 \cdot 10^{-5}$
27	53/50	41/100	−13/20	0.02409446047	0.023727069	0.000181975
28	53/50	41/100	13/20	0.08996010049	0.023727069	0.000679429
29	53/50	41/100	4/5	0.01877223007	0.023727069	0.000141778
30	53/50	23/50	4/5	0.01359832436	0.029208025	0.000126426
31	53/50	14/25	−8/5	$9.669233393 \cdot 10^{-5}$	0.040975795	$1.26116 \cdot 10^{-6}$
32	53/50	14/25	−13/10	$4.802780697 \cdot 10^{-5}$	0.040975795	$6.26427 \cdot 10^{-7}$
33	53/50	14/25	13/10	0.0001824982988	0.040975795	$2.38033 \cdot 10^{-6}$
34	11/10	3/20	−7/20	0.05671657488	0.00078557774	$1.41824 \cdot 10^{-5}$
35	11/10	1/4	−1/2	0.05147231926	0.0057973918	$9.49853 \cdot 10^{-5}$
36	11/10	1/4	1/2	0.1883380387	0.0057973918	0.000347553
37	11/10	2/5	4/5	0.07545736056	0.017623649	0.000423299
38	11/10	9/20	−4/5	0.001627486755	0.022555747	$1.16849 \cdot 10^{-5}$
39	11/10	9/20	4/5	0.009838802011	0.022555747	$7.06398 \cdot 10^{-5}$
40	11/10	1/2	−4/5	0.001256180045	0.027910450	$1.11601 \cdot 10^{-5}$
41	11/10	1/2	4/5	0.005913769697	0.027910450	$5.25389 \cdot 10^{-5}$
42	11/10	3/5	−1	0.0006183338066	0.039458156	$7.76622 \cdot 10^{-6}$
43	11/10	3/5	1	0.002725978105	0.039458156	$3.42381 \cdot 10^{-5}$
44	11/10	3/5	13/10	0.0002112464591	0.039458156	$2.65324 \cdot 10^{-6}$
45	5225/4756	229/1153	−1/2	0.05954976707	0.0029087233	$5.51357 \cdot 10^{-5}$
46	5225/4756	229/1153	1/2	0.2546846193	0.0029087233	0.000235806
47	5225/4756	2329/9368	−1/2	0.01670054759	0.0058145056	$3.09096 \cdot 10^{-5}$
48	5225/4756	2329/9368	1/2	0.07440766183	0.0058145056	0.000137715
49	5225/4756	1743/5837	−1/2	0.01266957078	0.0093687838	$3.77829 \cdot 10^{-5}$
50	5225/4756	1743/5837	1/2	0.1891814931	0.0093687838	0.000564173
51	5225/4756	2667/5945	−4/5	0.001477775558	0.022593493	$1.06278 \cdot 10^{-5}$
52	5225/4756	2667/5945	4/5	0.006728141614	0.022593493	$4.83870 \cdot 10^{-5}$
53	5225/4756	4132/8287	−4/5	0.001479398897	0.027952580	$1.31631 \cdot 10^{-5}$
54	5225/4756	4132/8287	4/5	0.005460180548	0.027952580	$4.85824 \cdot 10^{-5}$
55	5225/4756	2847/4756	−13/10	$5.562368483 \cdot 10^{-5}$	0.039507983	$6.99511 \cdot 10^{-7}$
56	23/20	1/5	−1/2	0.0542433989	0.00067784859	$1.17039 \cdot 10^{-5}$
57	23/20	1/5	−7/20	0.03475948982	0.00067784859	$7.49991 \cdot 10^{-6}$
58	23/20	1/5	7/20	0.2143837181	0.00067784859	$4.62567 \cdot 10^{-5}$

k	c_k	R_k	v_k	κ_k	J_k	$\kappa_k J_k/\pi$
59	23/20	9/20	$-4/5$	0.006316370699	0.016543682	$3.32621 \cdot 10^{-5}$
60	23/20	1/2	$-4/5$	0.003332642063	0.021311247	$2.26072 \cdot 10^{-5}$
61	23/20	11/20	$-4/5$	0.006954420904	0.026509943	$5.86840 \cdot 10^{-5}$
62	23/20	11/20	$4/5$	0.02859280204	0.026509943	0.000241277
63	23/20	13/20	$-13/10$	$5.258824036 \cdot 10^{-5}$	0.037780806	$6.32426 \cdot 10^{-7}$
64	23/20	13/20	-1	0.0004219252908	0.037780806	$5.07408 \cdot 10^{-6}$
65	23/20	13/20	1	0.001832673786	0.037780806	$2.20397 \cdot 10^{-5}$
66	23/20	13/20	13/10	0.0002062650784	0.037780806	$2.48054 \cdot 10^{-6}$
67	6/5	1/4	$-1/2$	0.03706538771	0.00060498120	$7.13774 \cdot 10^{-6}$
68	6/5	1/4	$-7/20$	0.0305999968	0.00060498120	$5.89269 \cdot 10^{-6}$
69	6/5	1/4	7/20	0.06841653548	0.00060498120	$1.31751 \cdot 10^{-5}$
70	6/5	1/4	1/2	0.3323254238	0.00060498120	$6.39964 \cdot 10^{-5}$
71	6/5	7/10	$-13/10$	$4.579439123 \cdot 10^{-5}$	0.036302130	$5.29169 \cdot 10^{-7}$
72	6/5	7/10	-1	0.0003653681966	0.036302130	$4.22195 \cdot 10^{-6}$
73	6/5	7/10	1	0.00157216072	0.036302130	$1.81668 \cdot 10^{-5}$
74	6/5	7/10	13/10	0.0001785005556	0.036302130	$2.06263 \cdot 10^{-6}$
75	5/4	3/10	$-7/20$	0.05227534138	0.00055148537	$9.17658 \cdot 10^{-6}$
76	5/4	3/10	7/20	0.2040492823	0.00055148537	$3.58195 \cdot 10^{-5}$
77	5/4	3/4	$-13/10$	$6.871033612 \cdot 10^{-5}$	0.034985511	$7.65174 \cdot 10^{-7}$
78	5/4	3/4	-1	0.000318403076	0.034985511	$3.54581 \cdot 10^{-6}$
79	5/4	3/4	1	0.001362286751	0.034985511	$1.51707 \cdot 10^{-5}$
80	5/4	3/4	13/10	0.0002746990133	0.034985511	$3.05911 \cdot 10^{-6}$
81	13/10	4/5	-1	0.0008212433178	0.033803191	$8.83649 \cdot 10^{-6}$
82	13/10	4/5	1	0.003961667359	0.033803191	$4.26271 \cdot 10^{-5}$
83	7/5	9/20	$-7/20$	0.3861919019	0.00044923233	$5.52235 \cdot 10^{-5}$
84	7/5	9/20	7/20	0.8551654873	0.00044923233	0.000122284
85	7/5	9/10	$-13/10$	0.0001493724531	0.031760037	$1.51009 \cdot 10^{-6}$
86	7/5	9/10	13/10	0.00143209911	0.031760037	$1.44779 \cdot 10^{-5}$
87	3/2	11/20	$-7/20$	0.0979914707	0.00040600316	$1.26639 \cdot 10^{-5}$
88	3/2	11/20	7/20	0.3388231655	0.00040600316	$4.37878 \cdot 10^{-5}$
89	9993/6209	2980/4519	$-7/20$	0.1671407202	0.00037055405	$1.97144 \cdot 10^{-5}$
90	9993/6209	2980/4519	7/20	0.5061513201	0.00037055405	$5.97011 \cdot 10^{-5}$
					$\frac{1}{\pi}\sum_k \kappa_k J_k =$	0.009277431487

Table 5: The vertices (ς_i, e_i) of the profile $\mathfrak{e}_{\mathrm{P}}$ of Lemma 5.3, and the source of the bound $\mu(\varsigma_i) \le e_i$. The Heath-Brown vertices $n = 10, \dots, 398$ are omitted from the table; they are generated by `profile_published.py` (Appendix B).

i	ς_i	e_i	source
0	1/2	13/84	Bourgain [7, Thm. 5]
1	88225/153852	4742/38463	Trudgian–Yang [30, Thm. 2.4]
2	521/796	18/199	Trudgian–Yang [30, Thm. 2.4]
3	53141/76066	2779/38033	Trudgian–Yang [30, Thm. 2.4]
4	454/641	89/1282	Tao–Trudgian–Yang [25, Thm. 20, Lemma 23]
5	3473692/4856993	652397/9713986	Tao–Trudgian–Yang [25, Thm. 20, Lemma 23]
6	52209/69128	2371/43205	Trudgian–Yang [30, Thm. 2.4]
7	1389/1736	9/217	Trudgian–Yang [30, Thm. 2.4]
8	587779/702192	10769/351096	Tao–Trudgian–Yang [25, Thm. 20, Lemma 23]

i	ς_i	e_i	source
9	7441/8695	89/3478	Tao–Trudgian–Yang [25, Thm. 20, Lemma 23]
10	277/300	1/100	Heath-Brown [15, Thm. 2], $n = 6$
11	535/567	1/162	Heath-Brown [15, Thm. 2], $n = 7$
12	375/392	1/245	Heath-Brown [15, Thm. 2], $n = 8$
13	3059/3168	1/352	Heath-Brown [15, Thm. 2], $n = 9$
	⋮	(Heath-Brown nodes $n = 10, \dots, 398$)	⋮
403	6336001297/6336120399	1/31760002	Heath-Brown [15, Thm. 2], $n = 399$
404	1	0	$\zeta(1 + it) \ll \log t$ [27, Thm. 3.5]

ι	c_ι	ι	c_ι
(R, 1)	1.26767379327	(I, 2)	−0.37840733351
(R, 2)	0.700929451456	(I, 3)	0.134730628931
(R, 3)	−0.494633048662	(I, 4)	−0.017083827788
(R, 4)	0.097515550525	(I, 5)	0.000100343368477
(R, 5)	−0.000589396293163		

TABLE 6. Coefficients of the kernel of the line ℓ_5 ($d = 1$, $m = 5$), to twelve significant digits.

ι	c_ι	ι	c_ι
(R, 1)	0.679978474949	(I, 2)	1.18435020971
(R, 2)	0.804110639165	(I, 3)	−0.576537489831
(R, 3)	0.777427090742	(I, 4)	0.00610293149501
(R, 4)	−0.0084486134969	(I, 5)	$-2.10769863064 \cdot 10^{-5}$
(R, 5)	$2.95597215845 \cdot 10^{-5}$		

TABLE 7. Coefficients of the kernel of the line ℓ_{20} ($d = 1.2$, $m = 5$), to twelve significant digits.

ι	c_ι	ι	c_ι
(R, 1)	0.595153203515	(I, 2)	0.80794661471
(R, 2)	0.591131249545	(I, 3)	−0.34004800791
(R, 3)	0.388921642826	(I, 4)	0.00364916717989
(R, 4)	−0.00432516763692	(I, 5)	$-1.27504674131 \cdot 10^{-5}$
(R, 5)	$1.53789024752 \cdot 10^{-5}$		

TABLE 8. Coefficients of the kernel of the line ℓ_{60} ($d = 1$, $m = 5$), to twelve significant digits.

APPENDIX B. PROGRAMS

The programs are written in Python 3 and use `python-flint` (Arb ball arithmetic [18], FLINT [12]) at 256 bits, `numpy`, `scipy` and `cvxpy` with the solver Clarabel. The certification programs are those of Sections B–B; every bound they print or write to a file is converted to decimal with

ι	c_ι	ι	c_ι
$(\mathrm{R},1)$	0.587356111095	$(\mathrm{I},2)$	0.529298607056
$(\mathrm{R},2)$	0.495527358923	$(\mathrm{I},3)$	−0.269742540483
$(\mathrm{R},3)$	0.171910239516	$(\mathrm{I},4)$	0.0395067096637
$(\mathrm{R},4)$	−0.0451542915632	$(\mathrm{I},5)$	−0.000234590737695
$(\mathrm{R},5)$	0.000271746755402		

TABLE 9. Coefficients of the kernel of the line ℓ_{150} ($d = 0.85$, $m = 5$), to twelve significant digits.

ι	c_ι	ι	c_ι
$(\mathrm{R},1)$	0.527699716861	$(\mathrm{I},2)$	0.579501653809
$(\mathrm{R},2)$	0.448444622334	$(\mathrm{I},3)$	−0.243041549546
$(\mathrm{R},3)$	0.2073491544	$(\mathrm{I},4)$	0.0959846058781
$(\mathrm{R},4)$	−0.0517088136271	$(\mathrm{I},5)$	−0.0223162904392
$(\mathrm{R},5)$	0.0427743920458	$(\mathrm{I},6)$	0.00101667452746
$(\mathrm{R},6)$	−0.00216867871949	$(\mathrm{I},7)$	$-2.32679097042 \cdot 10^{-5}$
$(\mathrm{R},7)$	$5.22536634852 \cdot 10^{-5}$	$(\mathrm{I},8)$	$3.09645235738 \cdot 10^{-7}$
$(\mathrm{R},8)$	$-7.19471178778 \cdot 10^{-7}$	$(\mathrm{I},9)$	$-2.32748949201 \cdot 10^{-9}$
$(\mathrm{R},9)$	$5.54823288096 \cdot 10^{-9}$	$(\mathrm{I},10)$	$7.769753796 \cdot 10^{-12}$
$(\mathrm{R},10)$	$-1.89056202487 \cdot 10^{-11}$		

TABLE 10. Coefficients of the kernel of the line ℓ_{400} ($d = 0.85$, $m = 10$), to twelve significant digits.

ι	c_ι	ι	c_ι
$(\mathrm{R},1)$	0.526619965261	$(\mathrm{I},2)$	0.583424601581
$(\mathrm{R},2)$	0.44752559746	$(\mathrm{I},3)$	−0.23667212703
$(\mathrm{R},3)$	0.217633359331	$(\mathrm{I},4)$	0.0956784075046
$(\mathrm{R},4)$	−0.0663261021081	$(\mathrm{I},5)$	−0.0185896622954
$(\mathrm{R},5)$	0.0492247519465	$(\mathrm{I},6)$	0.000831384847612
$(\mathrm{R},6)$	−0.0024545979738	$(\mathrm{I},7)$	$-1.89384002059 \cdot 10^{-5}$
$(\mathrm{R},7)$	$5.84151825308 \cdot 10^{-5}$	$(\mathrm{I},8)$	$2.51756719451 \cdot 10^{-7}$
$(\mathrm{R},8)$	$-7.9615067149 \cdot 10^{-7}$	$(\mathrm{I},9)$	$-1.8927048877 \cdot 10^{-9}$
$(\mathrm{R},9)$	$6.08645994128 \cdot 10^{-9}$	$(\mathrm{I},10)$	$6.32254880932 \cdot 10^{-12}$
$(\mathrm{R},10)$	$-2.0583832928 \cdot 10^{-11}$		

TABLE 11. Coefficients of the kernel of the line ℓ_{6000} ($d = 0.85$, $m = 10$), to twelve significant digits.

directed rounding (`up_str` in `certutil.py`), so that the printed decimal is itself a valid bound. The search programs that produced the coefficients are reproduced in Section B for completeness, but no statement of this article depends on them. All files are in the repository cited in Section 7, in the folders `hybrid_asymptotic/` and `family/`; the deposited archive is flat, and the programs locate their helper modules in either layout.

`certutil.py`: shared helpers, directed decimal rounding of bounds

```
# Shared helpers of the certification programs (python-flint / Arb ball arithmetic at 256 bits).
#   A(q)            exact ball of a Fraction or of a Python number
#   box(lo, hi)     ball containing the interval [lo, hi]
#   exact(x)        the rational value of an exact (zero-radius) arb
```

```
#   up_str(x, n)    decimal string with n significant digits that is >= every point of the ball x (directed
     rounding)
#   down_str(x, n)  decimal string with n significant digits that is <= every point of the ball x
# The .str() method of arb rounds to the nearest decimal, so it must not be used to print an upper or lower
     bound;
# every bound printed or written to a JSON file by the certification programs goes through up_str / down_str.
from fractions import Fraction as Fr
from decimal import Decimal, ROUND_CEILING, ROUND_FLOOR, localcontext
from flint import arb, ctx
ctx.prec = 256

def A(q):
    return arb(q.numerator)/arb(q.denominator) if isinstance(q, Fr) else arb(q)

def box(lo, hi):
    return A(lo).union(A(hi))

def exact(x):
    assert x.is_exact() and x.is_finite(), x
    m, e = x.man_exp()
    return Fr(int(m))*Fr(2)**int(e)

def _dir_str(q, n, rounding):
    with localcontext() as c:
        c.prec = n; c.rounding = rounding
        dec = Decimal(q.numerator)/Decimal(q.denominator)      # correctly rounded to n significant digits
    return format(dec, 'f')

def up_str(x, n):
    u = exact(x.upper()); s = _dir_str(u, n, ROUND_CEILING)
    assert Fr(s) >= u, (s, u)
    return s

def down_str(x, n):
    l = exact(x.lower()); s = _dir_str(l, n, ROUND_FLOOR)
    assert Fr(s) <= l, (s, l)
    return s

if __name__ == '__main__':
    x = arb(1)/3
    print(x.str(12, radius=False), up_str(x, 12), down_str(x, 12))
    y = -arb(1)/7
    print(y.str(12, radius=False), up_str(y, 12), down_str(y, 12))
```

`certify_hybrid.py`: certification of the kernel inequality (with or without discs) and of C_{H}

```
# Certification (Arb ball arithmetic, python-flint, 256 bits) of the kernel inequality
#   f(b,t) <= U(b,t) + sum_k kappa_k W_k(b,t)     for 0 <= b <= 1/2, t != 0,
# with f(b,t) = 2 atan((d+b)/t) + 2 atan((d-b)/t) - atan((2d+b)/t) - atan((2d-b)/t),
#      U(b,t) = sum_iota c_iota [F_iota(d+b,t) + F_iota(d-b,t)],  F_Rj = Re (x+it)^(-j),  F_Ij = -Im (x+it)
     ^(-j),
#      W_k(b,t) = log^+( R_k / sqrt((c_k-1/2-b)^2 + (t+v_k)^2) ),
# and of the constant C_H = c_R1/(2 pi) + (1/pi) sum_k kappa_k J_k,  J_k = (1/pi) int_0^pi e_P(c_k + R_k cos
     th) dth,
# where e_P is the piecewise linear profile of profile_published.py.  With an empty list of discs the program
      certifies
# the kernel inequality f <= U alone and prints C_1 = c_R1/(2 pi).
#
# Usage:  python3 certify_hybrid.py <solution.json> [T1]
#   solution.json: d, B (basis), c (coefficients, exact binary values), discs [(c_k,R_k,v_k)], kappa; T1
     defaults to 40.
# Method (Proposition "Certification of the kernel inequality" and Section "The hybrid" of the article):
```

```
#   (a) on [0,1/2] x (0,T1] and [0,1/2] x [-T1,0): centred form H(centre) + grad H(box) . half-widths,
     bisection;
#   (b) for |t| >= T1, u = 1/t: t^2 U - 4 d^3 u - K5 |u|^3 > 0 on [0,1/2] x [0,1/T1] and [0,1/2] x [-1/T1,0],
      bisection;
#       this needs T1 >= 2d + 1/2 (convergence of the arctan series) and W_k = 0 for |t| >= T1.
import sys, json, time
from fractions import Fraction as Fr
from flint import arb, acb
from certutil import A, box, up_str

sol = json.load(open(sys.argv[1])); T1 = Fr(sys.argv[2]) if len(sys.argv) > 2 else Fr(40)
d = Fr(sol['d']).limit_denominator(10**6); B = [tuple(x) for x in sol['B']]
c = [Fr(x) for x in sol['c']]                                          # exact binary values written by the
     optimiser
discs = [(Fr(cc).limit_denominator(10**4), Fr(R).limit_denominator(10**4), Fr(v).limit_denominator(10**4))
     for cc, R, v in sol['discs']]
discs = [(cc, min(R, cc-Fr(1, 2)), v) for cc, R, v in discs]            # rational rounding must keep c_k -
     R_k >= 1/2 (a smaller R_k only lowers W_k)
kap = [Fr(x) for x in sol['kappa']]
# integrity of the data
assert len(c) == len(B) and len(discs) == len(kap)
assert B[0] == ('R', 1) and ('R', 2) in B and all(k in ('R', 'I') and j >= 1 for k, j in B) and ('I', 1) not
     in B
assert d > Fr(1, 2)
assert all(k >= 0 for k in kap) and all(R > 0 and cc-R >= Fr(1, 2) and cc > 1 for cc, R, v in discs)
assert T1 >= 2*d+Fr(1, 2)
assert all(abs(v)+R <= T1 for cc, R, v in discs)                         # then W_k = 0 for |t| >= T1
JMAX = max(j for _, j in B)
dA = A(d); half = A(Fr(1, 2)); PI = arb.pi()
cA = [A(x) for x in c]; kA = [A(x) for x in kap]; dsA = [(A(cc), A(R).log(), A(v)) for cc, R, v in discs]

def powers(x, t, n):
    # [(x+it)^0, (x+it)^-1, ..., (x+it)^-n]
    w = acb(1)/acb(x, t); pw = [acb(1)]
    for _ in range(n): pw.append(pw[-1]*w)
    return pw
def kernel(b, t):
    U = arb(0)
    for x in (dA+b, dA-b):
        pw = powers(x, t, JMAX)
        for i, (k, j) in enumerate(B):
            U += cA[i]*(pw[j].real if k == 'R' else -pw[j].imag)
    return U
def fbox(b, t, sgn):
    # f on the closure of t > 0 (sgn = +1) or of t < 0 (sgn = -1), in the form regular at t = 0
    return sgn*PI-2*(t/(dA+b)).atan()-2*(t/(dA-b)).atan()+(t/(2*dA+b)).atan()+(t/(2*dA-b)).atan()
def Wlow(b, t):
    # lower bound of sum_k kappa_k W_k on the box (W_k >= 0, so dropping a term is safe)
    s = arb(0)
    for (cc, lR, v), k in zip(dsA, kA):
        val = lR-((cc-half-b)**2+(t+v)**2).log()/2
        lo = val.lower()
        if lo > 0: s += k*lo
    return s
def grad(b, t):
    # partial derivatives of U - f with respect to b and t, on a box
    gb = arb(0); gt = arb(0)
    for x, sg in ((dA+b, 1), (dA-b, -1)):
        pw = powers(x, t, JMAX+1)
        for i, (k, j) in enumerate(B):
            dx = -j*pw[j+1]; dt = acb(0, -j)*pw[j+1]                     # d/dx (x+it)^-j,  d/dt (x+it)^-j
            if k == 'R': gb += cA[i]*sg*dx.real; gt += cA[i]*dt.real
            else: gb -= cA[i]*sg*dx.imag; gt -= cA[i]*dt.imag
    y1 = dA+b; y2 = dA-b; y3 = 2*dA+b; y4 = 2*dA-b; t2 = t*t
    ft = -2*y1/(y1*y1+t2)-2*y2/(y2*y2+t2)+y3/(y3*y3+t2)+y4/(y4*y4+t2)
```

```
    fb = 2*t/(y1*y1+t2)-2*t/(y2*y2+t2)-t/(y3*y3+t2)+t/(y4*y4+t2)
    return gb-fb, gt-ft
def Hcentered(b0, b1, t0, t1, sgn):
    bm = (b0+b1)/2; tm = (t0+t1)/2; b = box(b0, b1); t = box(t0, t1)
    H0 = kernel(A(bm), A(tm))-fbox(A(bm), A(tm), sgn)
    gb, gt = grad(b, t)
    H = H0+gb*box(-(b1-b0)/2, (b1-b0)/2)+gt*box(-(t1-t0)/2, (t1-t0)/2)
    return H+Wlow(b, t)
def cert_main(blo, bhi, tlo, thi, sgn, maxdepth=60):
    stack = [(blo, bhi, tlo, thi, 0)]; n = 0
    while stack:
        b0, b1, t0, t1, dep = stack.pop(); n += 1
        H = Hcentered(b0, b1, t0, t1, sgn)
        if H > 0: continue
        if dep >= maxdepth: raise RuntimeError('fail at b=[%s,%s] t=[%s,%s] H=%s' % (float(b0), float(b1),
     float(t0), float(t1), H))
        wb = (b1-b0)/Fr(1, 20); tm = max(abs(t0), abs(t1)); wt = (t1-t0)/max(Fr(1, 20), tm*Fr(3, 10))
        if wb >= wt: m_ = (b0+b1)/2; stack += [(b0, m_, t0, t1, dep+1), (m_, b1, t0, t1, dep+1)]
        else: m_ = (t0+t1)/2; stack += [(b0, b1, t0, m_, dep+1), (b0, b1, m_, t1, dep+1)]
    return n

# tail |t| >= T1, u = 1/t:  t^2 (U - f) >= t^2 U - 4 d^3 u - K5 |u|^3,  with t^2 U = sum_iota c_iota [
     Phi_iota(d+b,u) + Phi_iota(d-b,u)],
# Phi_R1 = x/(1+x^2 u^2),  Phi_iota = Re / -Im of u^(j-2) (xu+i)^(-j) for j >= 2;  K5 = (1/5) sum_i |alpha_i|
      y_i^5 (Lemma "The tail of the kernel").
def tailval(b, u):
    V = arb(0)
    for x in (dA+b, dA-b):
        for i, (k, j) in enumerate(B):
            if j == 1: V += cA[i]*x/(1+x*x*u*u)
            else:
                z = acb(u)**(j-2)*(acb(1)/acb(x*u, 1))**j
                V += cA[i]*(z.real if k == 'R' else -z.imag)
    return V
K5 = (2*(dA+half)**5+2*dA**5+(2*dA+half)**5+(2*dA)**5)/5
def cert_tail(u0, u1, maxdepth=40):
    assert max(abs(u0), abs(u1))*(2*d+Fr(1, 2)) <= 1                    # |y u| <= 1 for every y in {d+b, d-b,
      2d+b, 2d-b}
    stack = [(Fr(0), Fr(1, 2), u0, u1, 0)]; n = 0
    while stack:
        b0, b1, a0, a1, dep = stack.pop(); n += 1
        b = box(b0, b1); u = box(a0, a1)
        H = tailval(b, u)-4*dA**3*u-K5*abs(u)**3
        if H > 0: continue
        if dep >= maxdepth: raise RuntimeError('tail fail b=[%s,%s] u=[%s,%s] H=%s' % (float(b0), float(b1),
     float(a0), float(a1), H))
        if (b1-b0)*8 >= (a1-a0)*T1: m_ = (b0+b1)/2; stack += [(b0, m_, a0, a1, dep+1), (m_, b1, a0, a1, dep
     +1)]
        else: m_ = (a0+a1)/2; stack += [(b0, b1, a0, m_, dep+1), (b0, b1, m_, a1, dep+1)]
    return n

def Jexact(nodes, cc, R):
    # (1/pi) int_0^pi e(cc + R cos th) dth for e piecewise linear through nodes, e = 0 for sigma >= 1; needs
     cc - R >= 1/2
    assert nodes[0][0] == Fr(1, 2) and nodes[-1] == (Fr(1), Fr(0)) and all(p[0] < q[0] for p, q in zip(nodes,
      nodes[1:]))
    tot = arb(0); ccA = A(cc); RA = A(R)
    lo, hi = cc-R, min(cc+R, Fr(1))
    for (p, ep), (q, eq) in zip(nodes, nodes[1:]):
        a, bq = max(p, lo), min(q, hi)
        if a >= bq: continue
        Bs = (eq-ep)/(q-p); As = ep-Bs*p                                # e = As + Bs sigma on [p, q]
        tha = A((a-cc)/R).acos(); thb = A((bq-cc)/R).acos()              # tha >= thb
        tot += (A(As)+A(Bs)*ccA)*(tha-thb)+A(Bs)*RA*(tha.sin()-thb.sin())
    return tot/PI
```

```
if __name__ == '__main__':
    t_ = time.time()
    if discs:
        assert sol['prof'] == 'P', 'only the published-source profile e_P is supported'
        from profile_published import nodes_published
        nodes = nodes_published()
        J = [Jexact(nodes, cc, R) for cc, R, v in discs]
    else: J = []
    C = cA[0]/(2*PI)+sum((k*j for k, j in zip(kA, J)), arb(0))/PI
    print('d =', d, ' m =', JMAX, ' discs used:', len(discs), (' profile e_P (profile_published.py)' if discs
      else ''))
    print('C = c_R1/(2pi) + sum kappa_k J_k/pi =', C.str(20, radius=False), ' (ball', C.str(5), ')')
    print('kernel part c_R1/(2pi) =', (cA[0]/(2*PI)).str(16, radius=False))
    lim = 2*dA*cA[0]-2*cA[B.index(('R', 2))]; print('limit of t^2 U at |t| -> oo = 2 d c_R1 - 2 c_R2 =', lim.
     str(10), ' positive:', lim > 0)
    n1 = cert_main(Fr(0), Fr(1, 2), Fr(0), T1, +1); print('t in (0,T1]: certified, boxes', n1, flush=True)
    n2 = cert_main(Fr(0), Fr(1, 2), -T1, Fr(0), -1); print('t in [-T1,0): certified, boxes', n2, flush=True)
    n3 = cert_tail(Fr(0), 1/T1); n4 = cert_tail(-1/T1, Fr(0)); print('|t|>=T1: certified, boxes', n3+n4)
    print('all certified;  C <', up_str(C, 12), '  time %.0fs' % (time.time()-t_))
```

`profile_published.py`: the nodes of the profile $\mathfrak{e}_{\mathrm{P}}$

```
# Upper bound for mu(sigma) on [1/2,1] from published sources only, as a piecewise linear function through
     nodes
# (sigma_i, e_i) with mu(sigma_i) <= e_i a theorem in each case:
#  - 1/2: 13/84 (Bourgain 2017, Thm 5);
#  - breakpoints of Trudgian-Yang, Toward optimal exponent pairs, arXiv:2306.05599v3, Thm 2.4 (pieces are
     continuous);
#  - l-k for the exponent pairs (k,l) of Tao-Trudgian-Yang, arXiv:2501.16779v1, Thm 20, via Lemma 23 (mu(l-k)
      <= k);
#  - Heath-Brown 2017 (Proc. Steklov Inst. Math. 296), Thm 2: pairs (2/((n-1)^2(n+2)), 1-(3n-2)/(n(n-1)(n+2))
     +eps), n >= 3,
#    giving mu(sigma_n) <= mu_n, sigma_n = 1-(3n^2-3n+2)/(n(n-1)^2(n+2)), mu_n = 2/((n-1)^2(n+2)) (continuity
      of mu), 6 <= n <= 399;
#  - (1, 0) (zeta(1+it) << log t).
# The profile is the lower convex hull of these points (nodes above a chord of two others are dropped: the
     chord is a
# valid bound by the Phragmen-Lindelof interpolation lemma, and smaller).
from fractions import Fraction as Fr
TY = [(Fr(1,2),Fr(88225,153852),Fr(31,84),Fr(3,7)),(Fr(88225,153852),Fr(521,796),Fr(220633,620612),Fr
     (62831,155153)),
 (Fr(521,796),Fr(53141,76066),Fr(1333,3825),Fr(1508,3825)),(Fr(53141,76066),Fr(3620,5119),Fr(405,1202),Fr
     (227,601)),
 (Fr(3620,5119),Fr(52209,69128),Fr(49318855,170145110),Fr(52938216,170145110)),(Fr(52209,69128),Fr(1389,1736)
     ,Fr(471957,1682490),Fr(502648,1682490)),
 (Fr(1389,1736),Fr(134765,163248),Fr(2841,10316),Fr(3016,10316)),(Fr(134765,163248),Fr(18193,21906),Fr
     (859,3214),Fr(908,3214)),
 (Fr(18193,21906),Fr(249,280),Fr(5*8707,180277),Fr(5*9067,180277)),(Fr(249,280),Fr(277,300),Fr(29,130),Fr
     (30,130))]
PAIRS = [(Fr(89,1282),Fr(997,1282)),(Fr(652397,9713986),Fr(7599781,9713986)),(Fr(10769,351096),Fr
     (609317,702192)),(Fr(89,3478),Fr(15327,17390))]
def hb(n): return 1-Fr(3*n*n-3*n+2, n*(n-1)**2*(n+2)), Fr(2, (n-1)**2*(n+2))
def raw_nodes(nmax=400):
    for (l1,h1,a1,b1),(l2,h2,a2,b2) in zip(TY,TY[1:]): assert a1-b1*h1 == a2-b2*h1
    nodes = {}
    for lo,hi,a,b in TY: nodes[lo] = a-b*lo; nodes[hi] = a-b*hi
    for k,l in PAIRS: nodes[l-k] = min(nodes.get(l-k, Fr(1)), k)
    for n in range(6, nmax): s,m = hb(n); nodes[s] = min(nodes.get(s, Fr(1)), m)
    nodes[Fr(1)] = Fr(0)
    return sorted(nodes.items())
def nodes_published(nmax=400):
    P = raw_nodes(nmax); hull = []
    def cross(o,a,b): return (a[0]-o[0])*(b[1]-o[1])-(a[1]-o[1])*(b[0]-o[0])
```

```
    for p in P:
        while len(hull) >= 2 and cross(hull[-2], hull[-1], p) <= 0: hull.pop()
        hull.append(p)
    return hull
if __name__ == '__main__':
    H = nodes_published()
    print(len(H), 'nodes; first ones:')
    for s,e in H[:14]: print('  sigma = %-20s  e = %-18s  (%.6f, %.6f)' % (s, e, s, e))
```

`cz_gen_pp.py`: certification of C_ζ (per-prime phase maximum)

```
# Rigorous upper bound (Arb ball arithmetic, 256 bits) for the Euler-product constant of a kernel,
#   C_zeta = sum_{p <= P0} M_p + sum_{n = p^k, p > P0} Lambda(n) n^(-sigma1) h(log n) + tail,
# with h(L) = sqrt(a(L)^2 + b(L)^2),  a(L) = -(1/pi) sum_{(R,j)} c_{R,j} L^(j-1)/(j-1)!,
# b(L) = (1/pi) [ (e^(-dL) - 2)/L - sum_{(I,j)} c_{I,j} L^(j-1)/(j-1)! ],  and M_p the maximum over the phase
      x of
#   f_p(x) = sum_{k: p^k <= N0} (log p) p^(-k sigma1) [ a(k log p) cos(kx) + b(k log p) sin(kx) ]
# (per-prime phase maximum, Lemma "The contribution of one prime"), bounded by the maximum on a grid of M
     points plus a
# Lipschitz correction.  The sum over the prime powers n <= N0 is exact; the tail n > N0 uses Lemma "The tail
     " with the
# non-increasing step majorant G of g(x) = h(log x) x^(-sigma1) on [N0, e^LB], the certified monotonicity of
     g beyond e^LB
# (Lemma "Monotonicity of g") and psi(x) < 1.03883 x (Rosser-Schoenfeld 1962, Thm 12).
#
# Usage:  python3 cz_gen_pp.py <solution.json> <N0> [P0]      (the article uses N0 = 1e8, P0 = 1000)
# Output: <solution>_czpp.json with the upper bound 'Czeta_upper' (decimal rounded upwards).
import sys, os, json, time, numpy as np
from fractions import Fraction as Fr
from math import factorial
from flint import arb, acb
sys.path.append(os.path.join(os.path.dirname(os.path.abspath(__file__)), '..', 'hybrid_asymptotic'))   #
     certutil.py lives there (or next to this file in the flat archive)
from certutil import A, up_str

sol = json.load(open(sys.argv[1])); N0 = int(float(sys.argv[2])); P0 = int(sys.argv[3]) if len(sys.argv) > 3
     else 1000
d = Fr(sol['d']).limit_denominator(10**6); B = [tuple(b) for b in sol['B']]; c = [A(Fr(x)) for x in sol['c']]
assert len(c) == len(B) and B[0] == ('R', 1) and ('I', 1) not in B and d > Fr(1, 2) and P0 <= N0
PI = arb.pi(); dA = A(d); s1 = A(Fr(1, 2)+d); alpha = s1-1                 # alpha = sigma1 - 1 > 0
# polynomial coefficients (in L) of pi*a and of the polynomial part of pi*b
DEG = max(j for _, j in B)-1
pa = [arb(0)]*(DEG+1); pb = [arb(0)]*(DEG+1)
for (k, j), cc in zip(B, c):
    if k == 'R': pa[j-1] -= cc/factorial(j-1)
    else: pb[j-1] -= cc/factorial(j-1)
def poly(p, L): return sum((p[i]*L**i for i in range(len(p))), arb(0))
def h(L):
    a0 = poly(pa, L)/PI; b0 = (((-dA*L).exp()-2)/L+poly(pb, L))/PI
    return (a0*a0+b0*b0).sqrt()

t0 = time.time()
# primes up to N0 (sieve of Eratosthenes)
s = np.ones(N0+1, np.bool_); s[:2] = False
for i in range(2, int(N0**0.5)+1):
    if s[i]: s[i*i::i] = False
P = np.nonzero(s)[0]; del s
# exact sum over the prime powers n <= N0, with the per-prime phase maximum for p <= P0
M = 4096
xs = [2*PI*i/M for i in range(M)]; hx = PI/M                               # grid points and half-spacing
tot = arb(0); psi = arb(0); gain = arb(0)
for p in P.tolist():
    lp = arb(p).log(); n = p; k = 1; coef = []; hyp = arb(0)
    while n <= N0:
```

```
        L = k*lp; w = lp*(-s1*L).exp(); a0 = poly(pa, L)/PI; b0 = (((-dA*L).exp()-2)/L+poly(pb, L))/PI
        coef.append((k, w*a0, w*b0)); hyp += w*(a0*a0+b0*b0).sqrt(); psi += lp; n *= p; k += 1
    if p <= P0:
        # max_x f_p(x) <= max_i f_p(x_i) + hx * sum_k k |(a_k, b_k)|   (Lipschitz constant of f_p)
        Lip = sum((kk*(a*a+b*b).sqrt() for kk, a, b in coef), arb(0))
        best = None
        for x in xs:
            v = sum((a*(kk*x).cos()+b*(kk*x).sin() for kk, a, b in coef), arb(0))
            best = v.upper() if best is None or v.upper() > best else best
        mb = best+hx*Lip
        mb = mb if mb < hyp else hyp                                    # the trivial bound sum_k |(a_k,
     b_k)| is always valid
        gain += hyp-mb; tot += mb
    else:
        tot += hyp
print('per-prime phase maximum for p <=', P0, ': gain', gain.str(8), flush=True)
print('primes', len(P), ' partial sum (n<=%d) =' % N0, tot.str(15), ' psi(N0) =', psi.str(15), ' %.0fs' % (
     time.time()-t0), flush=True)

# monotonicity of g on [e^LB, oo) (Lemma "Monotonicity of g"): in u = 1/L in [0, 1/LB], num(u)/low(u) <
     sigma1 with
#   num(u) = u sum_{i>=1} i q_i u^(D-i) + 2 u^(D+2) + d u^(D+1),   low(u) = |p_D| - sum_{i<D} |p_i| u^(D-i) -
      eps_P u^(D+1),
# where P is the one of pi*a, pi*b of larger degree D (pi*a if equal), q_i = |a_i| + |b_i|, eps_P = 2 if P =
     pi*b else 0.
L0 = arb(N0).log()
DA = max([i for i in range(DEG+1) if pa[i] != 0], default=-1); DB = max([i for i in range(DEG+1) if pb[i] !=
     0], default=-1)
q = [abs(pa[i])+abs(pb[i]) for i in range(DEG+1)]
if DA >= DB: D, pp, extra = DA, pa, 0
else: D, pp, extra = DB, pb, 2
assert D == DEG, 'the top coefficient of the kernel vanishes'
LB = None
for LBc in [2000.0, 5000.0, 2e4, 1e5, 1e6, 1e7]:
    u = arb(0).union(1/arb(LBc))
    num = u*sum((i*q[i]*u**(D-i) for i in range(1, DEG+1)), arb(0))+2*u**(D+2)+dA*u**(D+1)
    low = abs(pp[D])-sum((abs(pp[i])*u**(D-i) for i in range(D)), arb(0))-extra*u**(D+1)
    if low > 0 and num/low < s1: LB = LBc; ratio = num/low; break
assert LB is not None, 'monotonicity of g could not be certified'
print('g monotone on [%g, oo): crude bound %s < sigma1' % (LB, ratio.str(6)))

# step majorant G >= g on [N0, e^LB]: on [e^u_i, e^u_{i+1}] the running maximum, from the right, of upper
     bounds of g on
# the small boxes, started at g(e^LB) (g is non-increasing beyond e^LB), so that G is non-increasing on [N0,
     oo)
def cpoly(p, Lb):   # centred form of a polynomial on a ball: p(m) + p'(Lb)(Lb - m)
    m = arb(Lb.mid()); dp = [p[i]*i for i in range(1, len(p))]
    return poly(p, m)+poly(dp, Lb)*(Lb-m)
def h_up(Lb):
    a0 = cpoly(pa, Lb)/PI; b0 = (((-dA*Lb).exp()-2)/Lb+cpoly(pb, Lb))/PI
    au = abs(a0).upper(); bu = abs(b0).upper()
    return (au*au+bu*bu).sqrt().upper()
edges = [float(L0.lower())-1e-9]
while edges[-1] < LB: edges.append(min(edges[-1]+max(0.003, 0.0002*edges[-1]), LB))
gup = []
for i in range(len(edges)-1):
    Lb = arb(edges[i]).union(arb(edges[i+1])); gup.append(h_up(Lb)*(-s1*arb(edges[i])).exp())
run = h(arb(LB))*(-s1*arb(LB)).exp(); G = [None]*len(gup)
for i in range(len(gup)-1, -1, -1):
    run = run if run > gup[i] else (gup[i] if gup[i] > run else run.union(gup[i]).upper()); G[i] = run.upper
     ()
Istep = sum((G[i]*(arb(edges[i+1]).exp()-arb(edges[i]).exp()) for i in range(len(G))), arb(0))   # int_{N0}^{
     e^LB} G dx

# int_{e^LB}^oo g dx  <=  int_{e^LB}^{e^L1} g dx + int_{e^L1}^oo gtilde dx   with L1 = 800 and the majorant
```

```
#   gtilde = (1/pi) (sum_i q_i L^i + 3) e^(-sigma1 L) >= g on [e^L1, oo)   (from h <= |a| + |b| and 2/L <= 3)
    ;
# when LB > L1 the first integral is negative, and the inequality holds because gtilde >= g on [e^L1, e^LB]
    as well.
L1 = arb(800)
assert LB >= 800
def fz(z, analytic):
    a0 = sum((acb(pa[i])*z**i for i in range(len(pa))), acb(0))/PI
    b0 = (((-acb(dA)*z).exp()-2)/z+sum((acb(pb[i])*z**i for i in range(len(pb))), acb(0)))/PI
    return (a0*a0+b0*b0).sqrt(analytic=analytic)*((1-acb(s1))*z).exp()
Icancel = acb.integral(fz, acb(L1), acb(LB))                              # int_{L1}^{LB} h(L) e^{(1-sigma1)
     L} dL = int_{e^L1}^{e^LB} g dx
assert Icancel.is_finite() and Icancel.imag.contains(0)
def inc(k, X): return (-alpha*X).exp()*sum((arb(factorial(k))/factorial(i)*X**i/alpha**(k-i+1) for i in range
    (k+1)), arb(0))   # int_X^oo L^k e^{-alpha L} dL
Ifar = (sum((q[i]*inc(i, L1) for i in range(DEG+1)), arb(0))+3*inc(0, L1))/PI   # int_{e^L1}^oo gtilde dx
Itail = Istep-Icancel.real+Ifar                                           # >= int_{N0}^oo G dx
# Lemma "The tail": sum_{n > N0} Lambda(n) g(n) <= G(N0) (1.03883 N0 - psi(N0)) + 1.03883 int_{N0}^oo G dx
tail = G[0]*(arb('1.03883')*N0-psi)+arb('1.03883')*Itail
Cz = tot+tail
print('tail <=', tail.str(10), ' (int_{N0}^{e^LB} G =', Istep.str(10), ', int_{e^L1}^{e^LB} g =', Icancel.
    real.str(10), ', far tail <=', Ifar.str(5), ')')
print('C_zeta <', up_str(Cz, 12), '   time %.0fs' % (time.time()-t0))
json.dump({'Czeta_upper': up_str(Cz, 15), 'N0': N0, 'P0': P0}, open(sys.argv[1].replace('.json', '_czpp.json
    '), 'w'))
```

`elem_gen.py`: certification of $\sup F_u$, $\sup F_\ell$ (modes N and S)

```
# Rigorous upper bound (Arb ball arithmetic, 256 bits) for sup_{T >= 1} F_u(T) and sup_{T >= 1} F_l(T), where
#   F_u(T) = E_u(T) - A1 log T,   F_l(T) = -E_l(T) - A1 log T,
#   E_u(T) = (1/pi)(2 Ael(s1) - Ael(s2)) - main(T) + (1/pi) sum_iota c_iota Phi_iota(D_j(s1)),
#   E_l(T) = (1/pi)(2 Ael(s1) - Ael(s2)) - main(T) - (1/pi) [ sum_{(R,j)} c Re D_j(s1) - sum_{(I,j)} c (-Im
    D_j(s1)) ],
#   Ael(s) = arg s + arg(s-1) - (T/2) log pi + Im log Gamma(s/2),   main(T) = (T/2pi) log(T/(2 pi e)),
#   D_1(s) = 1/s + 1/(s-1) - (1/2) log pi + (1/2) psi(s/2),   D_j(s) = s^-j + (s-1)^-j + ((-1)^(j-1)/(j-1)!)
    2^-j psi^(j-1)(s/2) (j >= 2),
#   s1 = 1/2 + d + iT,  s2 = 1/2 + 2d + iT,  Phi_R = Re,  Phi_I = -Im.
# Mode S replaces main(T) by theta(T)/pi + 1 (then the bounds are for |S(T)|).
# [1, T0]: centred form F(mid) + F'(box) * [-half-width, half-width] on geometric intervals of ratio
    1001/1000;
# T >= T0: Lemma "The elementary part for large T": F_u <= 7/8 - C1 log(2 pi) + K/T, F_l <= -7/8 - C1 log(2
    pi) + K/T
# (mode S: -C1 log(2 pi) + (K + (1/16 + 1/8 + 1/3)/pi)/T), with the constant K of that lemma; needs A1 >= C1
    = c_R1/(2 pi).
#
# Usage:  python3 elem_gen.py <solution.json> <A1> [T0] [N|S]      (the article uses T0 = 10000)
# Output: <solution>_elem.json or <solution>_elemS.json with 'Fsup_upper' (decimal rounded upwards).
import sys, os, json, time
from fractions import Fraction as Fr
from math import factorial
from flint import arb, acb
sys.path.append(os.path.join(os.path.dirname(os.path.abspath(__file__)), '..', 'hybrid_asymptotic'))   #
    certutil.py lives there (or next to this file in the flat archive)
from certutil import A, up_str

sol = json.load(open(sys.argv[1])); A1 = A(Fr(sys.argv[2])); T0 = int(sys.argv[3]) if len(sys.argv) > 3 else
    10**4
MODE = sys.argv[4] if len(sys.argv) > 4 else 'N'
assert MODE in ('N', 'S') and T0 >= 1
d = Fr(sol['d']).limit_denominator(10**6); B = [tuple(b) for b in sol['B']]; c = [A(Fr(x)) for x in sol['c']]
assert len(c) == len(B) and B[0] == ('R', 1) and ('I', 1) not in B and d > Fr(1, 2)
J = max(j for _, j in B)
PI = arb.pi(); s1r = A(Fr(1, 2)+d); s2r = A(Fr(1, 2)+2*d); LOGPI = PI.log()
cR1 = c[0]; C1 = cR1/(2*PI)
```

```
assert A1 >= C1, 'A1 must be >= c_R1/(2 pi)'

def psi(k, z): return z.polygamma(k)
def Dj(j, s, ps):   # ps[k] = psi^(k)(s/2)
    if j == 1: return 1/s+1/(s-1)-LOGPI/2+ps[0]/2
    return s**(-j)+(s-1)**(-j)+arb((-1)**(j-1))/factorial(j-1)/2**j*ps[j-1]
def dDj(j, s, ps):  # d/ds D_j(s)
    if j == 1: return -s**-2-(s-1)**-2+ps[1]/4
    return -j*s**(-j-1)-j*(s-1)**(-j-1)+arb((-1)**(j-1))/factorial(j-1)/2**(j+1)*ps[j]
def combine(vals):  # sum_{(R,j)} c Re(.) and sum_{(I,j)} c (-Im(.))
    R = sum((cc*vals[i].real for i, ((k, j), cc) in enumerate(zip(B, c)) if k == 'R'), arb(0))
    I = sum((cc*(-vals[i].imag) for i, ((k, j), cc) in enumerate(zip(B, c)) if k == 'I'), arb(0))
    return R, I
def F_at(T):
    # (F_u(T), F_l(T)) at an exact T
    s1 = acb(s1r, T); s2 = acb(s2r, T)
    Ael = lambda s: (s.log().imag+(s-1).log().imag)-T/2*LOGPI+(s/2).lgamma().imag
    base = (2*Ael(s1)-Ael(s2))/PI-T/(2*PI)*(T/(2*PI*arb(1).exp())).log()
    if MODE == 'S':   # replace main(T) by theta(T)/pi + 1
        theta = acb(arb(1)/4, T/2).lgamma().imag-T/2*LOGPI
        base = base+T/(2*PI)*(T/(2*PI*arb(1).exp())).log()-theta/PI-1
    ps = [psi(k, s1/2) for k in range(J)]
    R, I = combine([Dj(j, s1, ps) for (_, j) in B])
    return base+(R+I)/PI-A1*T.log(), -(base-(R-I)/PI)-A1*T.log()
def dF_ball(Tb):
    # (F_u'(T), F_l'(T)) on a ball Tb of heights; the polygammas on the ball by the mean value form psi^(k)(
     z_m) + psi^(k+1)(z_ball) (z - z_m)
    Tm = arb(Tb.mid()); rad = Tb-Tm
    def P(k, sr):
        zm = acb(sr, Tm)/2; zb = acb(sr, Tb)/2
        return psi(k, zm)+psi(k+1, zb)*acb(0, rad/2)
    s1 = acb(s1r, Tb); s2 = acb(s2r, Tb)
    dAel = lambda s, sr: ((1/s).real+(1/(s-1)).real)-LOGPI/2+P(0, sr).real/2      # d/dT Ael(sr + iT)
    dbase = (2*dAel(s1, s1r)-dAel(s2, s2r))/PI-(Tb/(2*PI)).log()/(2*PI)
    if MODE == 'S':   # d/dT [main - theta/pi] = (1/2pi) log(T/2pi) - (1/2pi) Re psi(1/4+iT/2) + (1/2pi) log
     pi
        zm = acb(arb(1)/4, Tm/2); zb = acb(arb(1)/4, Tb/2)
        P0 = psi(0, zm)+psi(1, zb)*acb(0, rad/2)
        dbase = dbase+(Tb/(2*PI)).log()/(2*PI)-(P0.real-LOGPI)/(2*PI)
    ps = [P(k, s1r) for k in range(J+1)]
    R, I = combine([acb(0, 1)*dDj(j, s1, ps) for (_, j) in B])              # d/dT = i d/ds
    return dbase+(R+I)/PI-A1/Tb, -(dbase-(R-I)/PI)-A1/Tb

# [1, T0]
t0 = time.time(); ratio = Fr(1001, 1000); a = Fr(1); n = 0; supu = None; supl = None
while a < T0:
    b = min(a*ratio, Fr(T0)); m = (a+b)/2; dl = (b-a)/2
    fu, fl = F_at(A(m)); gu, gl = dF_ball(A(a).union(A(b)))
    eu = (fu+gu*A(dl)*arb(0, 1)).upper(); el = (fl+gl*A(dl)*arb(0, 1)).upper()
    supu = eu if supu is None or eu > supu else supu; supl = el if supl is None or el > supl else supl
    a = b; n += 1
print('intervals', n, ' sup F_u on [1,T0] <=', up_str(supu, 10), '  sup F_l <=', up_str(supl, 10))
# T >= T0 (Lemma "The elementary part for large T")
T = A(Fr(T0)); s1 = s1r; s2 = s2r
K = (2*s2-1)/PI+(s1*s1/2+(s2-1)*s2/2+1)/PI+(2*s2*s2/4+(s1-1)*s1)/PI        # arguments and Gamma factors
K += abs(cR1)/PI*((2*s1-1)+s1*s1/4+arb(8)/3)/T                             # j = 1: |delta_1| <= B/T^2 <= B/(
     T0 T)
for (k, j), cc in zip(B, c):
    if j == 1: continue
    # |D_j(s1)| <= 2/T^j + |psi^(j-1)(z1)| / ((j-1)! 2^j),  |psi^(k)(z1)| <= (k-1)! (2/T)^k + k! (2/T)^(k+1)
     /2 + (k+1)! (4/(3T)) (4/T)^(k+1);
    # T times the bound is non-increasing in T, so the bound at T0 times T0/T is valid for T >= T0
    kk = j-1
    ps = arb(factorial(kk-1))*(2/T)**kk+arb(factorial(kk))*(2/T)**(kk+1)/2+arb(factorial(kk+1))*(4/(3*T))*(4/
     T)**(kk+1)
```

```
    Dbound = 2/T**j+ps/(factorial(j-1)*2**j)
    K += abs(cc)/PI*Dbound*T
Finf_u = arb(7)/8-C1*(2*PI).log(); Finf_l = -arb(7)/8-C1*(2*PI).log()
if MODE == 'S':   # theta/pi + 1 - main = 7/8 + varpi/pi, |varpi(T)| <= (1/16 + 1/8 + 1/3)/T (Lemma "The
     theta function")
    Finf_u = -C1*(2*PI).log(); Finf_l = -C1*(2*PI).log(); K = K+(arb(1)/16+arb(1)/8+arb(1)/3)/PI
tu = (Finf_u+K/T).upper(); tl = (Finf_l+K/T).upper()
S = max(supu, supl, tu, tl)
print('T>=T0: F_u <=', up_str(tu, 10), ' F_l <=', up_str(tl, 10), ' (K =', K.str(6), ')')
print('SUP over T>=1 of max(F_u,F_l) <=', up_str(S, 12), '   time %.0fs' % (time.time()-t0))
json.dump({'A1': sys.argv[2], 'T0': T0, 'mode': MODE, 'Fsup_upper': up_str(S, 15)}, open(sys.argv[1].replace
     ('.json', '_elem%s.json' % ('' if MODE == 'N' else 'S')), 'w'))
```

certify_line_pp.py: driver for one explicit line

```
# Driver: certify one explicit line |N(T) - main(T)| <= A1 log T + A3_N and |S(T)| <= A1 log T + A3_S for all
      T >= 1.
#   1. certify_hybrid.py (no discs): the kernel inequality f <= U, with T1 = 100 (T1 = 2000 if that run fails
     );
#   2. cz_gen_pp.py: C_zeta < Czeta_upper  (N0 = 1e8, per-prime phase maximum for p <= 1000);
#   3. elem_gen.py, modes N and S: sup_{T>=1} max(F_u, F_l) < Fsup_upper  (T0 = 1e4);
#   4. A1 = C1 rounded upwards to 5 decimals, A3 = Czeta_upper + Fsup_upper rounded upwards to 3 decimals (
     exact decimal arithmetic).
# Usage:  python3 certify_line_pp.py <tag> <solution.json>                  (writes line_<tag>/sol.json and
     the certificates there)
#         python3 certify_line_pp.py <tag> <L*> [sweep*.jsonl pattern]      (picks the best sweep candidate
     for the target height L*)
import json, sys, os, subprocess, glob
from decimal import Decimal, ROUND_CEILING, localcontext
HERE = os.path.dirname(os.path.abspath(__file__))
def locate(name):   # repository layout (../hybrid_asymptotic/) or flat archive (same folder)
    for p in (os.path.join(HERE, name), os.path.join(HERE, '..', 'hybrid_asymptotic', name)):
        if os.path.exists(p): return p
    raise FileNotFoundError(name)

tag = sys.argv[1]; src = sys.argv[2]; pat = sys.argv[3] if len(sys.argv) > 3 else 'sweep*.jsonl'
if src.endswith('.json'): r = json.load(open(src))
else:
    L = float(src); rs = [json.loads(l) for f in glob.glob(pat) for l in open(f)]
    r = min((x for x in rs if x['Lstar'] == L), key=lambda x: x['val'])
D = 'line_'+tag; os.makedirs(D, exist_ok=True); sol = D+'/sol.json'; json.dump(r, open(sol, 'w'))
A1s = str(Decimal(repr(r['C1'])).quantize(Decimal('0.00001'), rounding=ROUND_CEILING))   # elem_gen.py checks
      A1 >= C1 exactly
print('candidate', tag, 'd =', r['d'], 'C1 =', r['C1'], 'A3 (estimate) =', r['A3_est'], '-> A1 =', A1s, flush
     =True)

def run(cmd, log):
    with open(D+'/'+log, 'w') as fh: p = subprocess.run(cmd, stdout=fh, stderr=subprocess.STDOUT)
    out = open(D+'/'+log).read(); lines = out.strip().splitlines(); print(lines[-1] if lines else '(no output
     )', flush=True)
    return p.returncode, out
rc1, o1 = run(['python3', locate('certify_hybrid.py'), sol, '100'], 'cert_kernel.txt')
if not (rc1 == 0 and 'all certified' in o1): rc1, o1 = run(['python3', locate('certify_hybrid.py'), sol,
     '2000'], 'cert_kernel.txt')
rc2, o2 = run(['python3', os.path.join(HERE, 'cz_gen_pp.py'), sol, '1e8', '1000'], 'cert_czeta_pp.txt')
rc3, o3 = run(['python3', os.path.join(HERE, 'elem_gen.py'), sol, A1s, '10000', 'N'], 'cert_elem.txt')
rc4, o4 = run(['python3', os.path.join(HERE, 'elem_gen.py'), sol, A1s, '10000', 'S'], 'cert_elem_S.txt')

def finish():
    # the certified upper bounds are decimal strings rounded upwards; add them in exact decimal arithmetic
     and round upwards
    with localcontext() as dc:
        dc.prec = 60
        cz = Decimal(json.load(open(sol.replace('.json', '_czpp.json')))['Czeta_upper'])
```

```
        fN = Decimal(json.load(open(sol.replace('.json', '_elem.json')))['Fsup_upper'])
        fS = Decimal(json.load(open(sol.replace('.json', '_elemS.json')))['Fsup_upper'])
        AN = (cz+fN).quantize(Decimal('0.001'), rounding=ROUND_CEILING); AS = (cz+fS).quantize(Decimal
    ('0.001'), rounding=ROUND_CEILING)
    json.dump({'A1': A1s, 'A3_N': str(AN), 'A3_S': str(AS), 'Czeta_pp_upper': str(cz), 'Fsup_N': str(fN), '
    Fsup_S': str(fS),
               'd': r['d'], 'B': r['B'], 'Lstar': r['Lstar']}, open(D+'/summary_pp.json', 'w'), indent=1)
    print('LINE %s CERTIFIED: |N(T)-main(T)| <= %s log T + %s ;  |S(T)| <= %s log T + %s   (T >= 1)' % (tag,
    A1s, AN, A1s, AS), flush=True)
if rc1 == 0 and 'all certified' in o1 and rc2 == 0 and rc3 == 0 and rc4 == 0: finish()
else: print('LINE %s FAILED' % tag, flush=True); sys.exit(1)
```

Search programs

lpcommon.py: helpers of the search programs

```
# Floating-point helpers shared by the search programs (explore_discs.py, kernel_only.py, opt_line.py).
# Nothing here is used by the certification programs.
import numpy as np
from fractions import Fraction as Fr

def f(b, t, d):
    # f(b,t) = 2 atan((d+b)/t) + 2 atan((d-b)/t) - atan((2d+b)/t) - atan((2d-b)/t)
    return 2*np.arctan((b+d)/t)+2*np.arctan((d-b)/t)-np.arctan((b+2*d)/t)-np.arctan((2*d-b)/t)

def F(x, t, k, j):
    # F_Rj = Re (x+it)^-j,  F_Ij = -Im (x+it)^-j
    z = (x+1j*t)**(-j); return z.real if k == 'R' else -z.imag

def basis(m): return [('R', j) for j in range(1, m+1)]+[('I', j) for j in range(2, m+1)]

def tgrid(tmax, ngeom=700, nlin=400):
    # symmetric grid of heights t: geometric on [1e-4, tmax] plus a linear part on (0, 4], and a point near 0
    p = np.unique(np.concatenate([[1e-9], np.geomspace(1e-4, tmax, ngeom), np.linspace(0, 4, nlin)[1:]]));
     return np.concatenate([-p[::-1], p])

def bourgain_nodes():
    # piecewise linear profile through (lambda_j - kappa_j, kappa_j), (kappa_j, lambda_j) = A^j (13/84,
    55/84), j = 0..7, and (1, 0)
    k, l = Fr(13, 84), Fr(55, 84); nodes = []
    for _ in range(8):
        nodes.append((l-k, k)); k, l = k/(2*k+2), Fr(1, 2)+l/(2*k+2)
    nodes.append((Fr(1), Fr(0))); return nodes

def J_float(nodes, c, R, n=40001):
    # (1/pi) int_0^pi e(c + R cos th) dth, e piecewise linear through nodes on [1/2, 1], e = 0 for sigma >= 1
      (trapezoidal rule)
    xs = np.array([float(a) for a, _ in nodes]); es = np.array([float(e) for _, e in nodes])
    th = np.linspace(0, np.pi, n); s = c+R*np.cos(th); v = np.where(s >= 1, 0, np.interp(s, xs, es))
    return np.trapezoid(v, th)/np.pi
```

explore_discs.py: the hybrid linear programme

```
# Search program (not a certificate): the hybrid of the kernel majorant with Jensen discs.
# Linear programme in (c_iota, kappa_k):
#   minimise  c_R1/(2 pi) + (1/pi) sum_k kappa_k J_k
#   subject to  f(b,t) + eta/(1+t^2) <= U(b,t) + sum_k kappa_k W_k(b,t)  on a grid of (b,t), refined by
    cutting planes,
#               2 d c_R1 - 2 c_R2 >= eta  (limit of t^2 U at |t| -> oo),  kappa_k >= 0,
# with J_k = (1/pi) int_0^pi e(c_k + R_k cos th) dth for the profile e (PROF=P: e_P of profile_published.py;
    PROF=B: the
# Bourgain profile of lpcommon.bourgain_nodes).  Candidate discs: the family 'vwide' described in the article
    .
```

```
# Usage:  PROF=P python3 explore_discs.py <m> <d> [eta]      -> expl_m<m>_d<d>_<PROF>_eta<eta>.json (input of
      certify_hybrid.py)
import numpy as np, sys, os, json, time, math
from scipy.optimize import linprog, minimize
from scipy import sparse
from lpcommon import f, F, basis, tgrid, bourgain_nodes, J_float

PROF = os.environ.get('PROF', 'P')
if PROF == 'P':
    from profile_published import nodes_published; nodes = nodes_published()
elif PROF == 'B': nodes = bourgain_nodes()
else: raise SystemExit('PROF must be P or B')

def Wmat(discs, bb, tt):
    # sparse matrix of W_k(b,t) = log^+(R_k / sqrt((c_k - 1/2 - b)^2 + (t + v_k)^2))
    rows = []; cols = []; vals = []
    for k, (c, R, v) in enumerate(discs):
        dist = np.sqrt((c-0.5-bb)**2+(tt+v)**2)
        w = np.log(R/np.maximum(dist, 1e-300)); i = np.nonzero(w > 0)[0]
        rows.append(i); cols.append(np.full(len(i), k)); vals.append(w[i])
    r = np.concatenate(rows) if rows else np.zeros(0, int)
    return sparse.csr_matrix((np.concatenate(vals) if vals else [], (r, np.concatenate(cols) if cols else [])
     ), shape=(len(bb), len(discs)))
def Gmat(d, B, bb, tt): return np.array([F(d+bb, tt, k, j)+F(d-bb, tt, k, j) for (k, j) in B]).T

def solve(d, m, discs, eta=0.0, maxit=30, nb=26):
    B = basis(m); nB = len(B)
    sc = np.array([(d-0.5)**j for (_, j) in B])                 # scaling of the coefficients, for
     conditioning
    J = np.array([J_float(nodes, c, R) for c, R, v in discs])
    obj = np.concatenate([[sc[0]/(2*np.pi)], np.zeros(nB-1), J/np.pi])
    bb, tt = [a.ravel() for a in np.meshgrid(np.linspace(0, 0.5, nb), tgrid(300., 500, 300), indexing='ij')]
    def block(bb, tt): return sparse.hstack([sparse.csr_matrix(Gmat(d, B, bb, tt)*sc), Wmat(discs, bb, tt)]).
     tocsr(), f(bb, tt, d)+eta/(1+tt**2)
    A, Fv = block(bb, tt)
    arow = np.zeros(nB+len(discs)); arow[0] = 2*d*sc[0]; arow[1] = -2*sc[1]
    bf = np.linspace(0, 0.5, 101); tp = np.unique(np.concatenate([np.geomspace(1e-7, 3000., 6000), np.
     linspace(1e-4, 3., 30000)])); tf = np.concatenate([-tp[::-1], tp])
    def violation(x, used, du, bb, tt):   # f + margin - U - sum kappa W at the points (bb, tt); positive =
     violated
        v = f(bb, tt, d)+eta/(1+tt**2)-Gmat(d, B, bb, tt)@(x[:nB]*sc)
        if len(used): v = v-Wmat(du, bb, tt)@x[nB+used]
        return v
    for it in range(maxit):
        for method in ('highs', 'highs-ds', 'highs-ipm'):   # the dual simplex and the interior point method
     are tried if HiGHS' default fails
            r = linprog(obj, A_ub=-sparse.vstack([A, sparse.csr_matrix(arow)]), b_ub=-np.concatenate([Fv, [
     eta]]),
                        bounds=[(None, None)]*nB+[(0, None)]*len(discs), method=method)
            if r.status == 0: break
        assert r.status == 0, 'LP solver: '+r.message
        x = r.x; used = np.nonzero(x[nB:] > 1e-12)[0]; du = [discs[k] for k in used]
        nbl = []; ntl = []; worst = -1; cand = []
        for b in bf:   # violations on a finer grid -> new cutting planes (local maxima of the violation
     along t)
            v = violation(x, used, du, np.full_like(tf, b), tf)
            worst = max(worst, v.max()); idx = np.nonzero((v > 0) & (v >= np.roll(v, 1)) & (v >= np.roll(v,
     -1)))[0]
            nbl += [b]*len(idx); ntl += list(tf[idx])
            top = np.argsort(v)[-3:]; cand += [(v[i], b, tf[i]) for i in top]
        # the grid can miss narrow dips of U + sum kappa W - f (near the edges of the discs): refine the
     largest grid values
        # of the violation, including negative ones, by a local maximisation in (b, t), and add the maxima
     found as cutting planes
        cand.sort(reverse=True)
```

```
        for v0, b0, t0 in cand[:40]:
            if t0 == 0: continue
            g = lambda p: -violation(x, used, du, np.array([min(max(p[0], 0.0), 0.5)]), np.array([p[1] if p
    [1]*t0 > 0 else t0]))[0]
            res = minimize(g, [b0, t0], method='Nelder-Mead', options={'xatol': 1e-7, 'fatol': 1e-12, '
    maxiter': 400})
            bb_, tt_ = min(max(res.x[0], 0.0), 0.5), (res.x[1] if res.x[1]*t0 > 0 else t0)
            vv = -res.fun; worst = max(worst, vv)
            if vv > 0: nbl.append(bb_); ntl.append(tt_)
        print('  it %d  objective %.6f  worst violation %.2e  new cutting planes %d  rows %d' % (it, r.fun,
    worst, len(nbl), A.shape[0]), flush=True)
        if not nbl: break
        # new cutting planes, without near-duplicates (which make the linear programme ill-conditioned)
        pts = []
        for b_, t_ in zip(nbl, ntl):
            if all(abs(b_-b2) > 1e-6 or abs(t_-t2) > 1e-6*max(1.0, abs(t_)) for b2, t2 in pts): pts.append((
    b_, t_))
        A2, F2 = block(np.array([p[0] for p in pts]), np.array([p[1] for p in pts])); A = sparse.vstack([A,
    A2]).tocsr(); Fv = np.concatenate([Fv, F2])
    xs = x.copy(); xs[:nB] *= sc
    return r.fun, xs, B, J, worst, it

def family_vwide():
    # centres c, radii R = c - 1 + x with R <= c - 1/2, vertical offsets v
    C = [1.03, 1.06, 1.1, math.log(3), 1.15, 1.2, 1.25, 1.3, 1.4, 1.5, math.log(5), 2.0, math.e, 3.0]
    out = []
    for c in C:
        for R in [c-1+x for x in [0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5]]:
            if R <= 0 or R > c-0.5+1e-12: continue
            for v in [-2, -1.6, -1.3, -1.0, -0.8, -0.65, -0.5, -0.35, -0.2, 0, 0.2, 0.35, 0.5, 0.65, 0.8,
    1.0, 1.3, 1.6, 2]:
                if abs(v) <= R+2.5: out.append((c, R, float(v)))
    return out

if __name__ == '__main__':
    m = int(sys.argv[1]); d = float(sys.argv[2]); eta = float(sys.argv[3]) if len(sys.argv) > 3 else 0.0
    discs = family_vwide(); t0 = time.time()
    val, x, B, J, w, it = solve(d, m, discs, eta)
    nB = len(B); keep = [k for k in range(len(discs)) if x[nB+k] > 1e-10]
    used_c = sorted(set(round(discs[k][0], 4) for k in keep))
    print('m=%d d=%.3f prof=%s eta=%g ndiscs=%d C(grid)=%.6f kernel=%.6f worst violation=%.2e it=%d used=%d
    centres=%s  %.0fs'
          % (m, d, PROF, eta, len(discs), val, x[0]/(2*np.pi), w, it, len(keep), used_c, time.time()-t0),
    flush=True)
    json.dump({'m': m, 'd': d, 'prof': PROF, 'eta': eta, 'B': B, 'c': list(x[:nB]), 'discs': [discs[k] for k
    in keep], 'kappa': [x[nB+k] for k in keep],
               'J_grid': [J[k] for k in keep], 'C_grid_objective': val}, open('expl_m%d_d%.2f_%s_eta%g.json'
    % (m, d, PROF, eta), 'w'))
```

kernel_only.py: kernel-only optima (Remark on the infimum)

```
# Search program (not a certificate): the smallest leading constant c_R1/(2 pi) of a kernel majorant without
    discs,
#   minimise c_R1/(2 pi)  subject to  f(b,t) <= U(b,t) on a grid of (b,t), refined by cutting planes, and 2 d
     c_R1 - 2 c_R2 >= 0,
# for the basis of order m and the given d; with the option 'rh' the constraint is imposed only at b = 0 (
    beta = 1/2), which is
# what would be needed if all zeros were on the critical line.  Used for Remark "On the infimum of the
    leading constant".
# Usage:  python3 kernel_only.py <m> <d1,d2,...> [rh]
import numpy as np, sys, time
from scipy.optimize import linprog
from lpcommon import f, F, basis, tgrid
```

```
def Gmat(d, B, bb, tt): return np.array([F(d+bb, tt, k, j)+F(d-bb, tt, k, j) for (k, j) in B]).T

def solve(d, m, rh=False, maxit=60, nb=26, tol=1e-9):
    B = basis(m); nB = len(B)
    sc = np.array([(d-0.5)**j for (_, j) in B])                # scaling of the coefficients, for
     conditioning
    obj = np.zeros(nB); obj[0] = sc[0]/(2*np.pi)
    bgrid = np.array([0.0]) if rh else np.linspace(0, 0.5, nb)
    bb, tt = [a.ravel() for a in np.meshgrid(bgrid, tgrid(300.), indexing='ij')]
    A = Gmat(d, B, bb, tt)*sc; Fv = f(bb, tt, d)
    arow = np.zeros(nB); arow[0] = 2*d*sc[0]; arow[1] = -2*sc[1]
    bf = np.array([0.0]) if rh else np.linspace(0, 0.5, 201); tp = np.geomspace(1e-8, 3000., 12000); tf = np.
     concatenate([-tp[::-1], tp])
    for it in range(maxit):
        r = linprog(obj, A_ub=-np.vstack([A, arow]), b_ub=-np.concatenate([Fv, [0.0]]), bounds=[(None, None)
     ]*nB, method='highs')
        assert r.status == 0, 'LP solver: '+r.message
        x = r.x; nbl = []; ntl = []; worst = -1
        for b in bf:   # violations on a finer grid -> new cutting planes (local maxima of the violation)
            v = f(b, tf, d)-Gmat(d, B, np.full_like(tf, b), tf)@(x*sc)
            worst = max(worst, v.max()); idx = np.nonzero((v > tol) & (v >= np.roll(v, 1)) & (v >= np.roll(v,
      -1)))[0]
            nbl += [b]*len(idx); ntl += list(tf[idx])
        if not nbl: break
        A = np.vstack([A, Gmat(d, B, np.array(nbl), np.array(ntl))*sc]); Fv = np.concatenate([Fv, f(np.array(
     nbl), np.array(ntl), d)])
    return r.fun, worst, it

if __name__ == '__main__':
    m = int(sys.argv[1]); ds = [float(x) for x in sys.argv[2].split(',')]; rh = len(sys.argv) > 3 and sys.
     argv[3] == 'rh'
    for d in ds:
        t0 = time.time(); val, w, it = solve(d, m, rh)
        print('m=%d d=%.2f %s C1(grid)=%.5f worst violation=%.1e it=%d  %.0fs' % (m, d, 'constraint at beta
     =1/2 only' if rh else 'all beta', val, w, it, time.time()-t0), flush=True)
```

opt_line.py: the second-order cone programme of one explicit line

```
# Search program (not a certificate): optimise one explicit line |N(T)-main(T)| <= A1 log T + A3 for a target
      height L* = log T*.
# Kernel U = sum c_iota G_iota (basis R1..R_mR, I2..I_mI), d fixed. Convex programme (second-order cone,
     Clarabel through cvxpy):
#   minimise  (c_R1/2pi) L* + C_zeta(c) + z
#   s.t.      f(b,t) + eta/(1+t^2) <= U(b,t)      on a (b,t) grid, refined by cutting planes
#             2d c_R1 - 2 c_R2 >= eta              (limit of t^2 U at |t| -> oo)
#             z >= F_u(T;c), z >= F_l(T;c)         on a T grid in [1,1e6], and z >= the limit at T = oo
# C_zeta(c) = sum_n Lambda(n) n^{-sigma1} || (a(log n), b(log n)) ||_2  (n <= n0) + quadrature of the tail;
     with PPMAX > 0 the
# primes p <= PPMAX enter through the maximum of f_p over a grid of 96 phases (per-prime phase maximum).
# Usage:  [PPMAX=300] python3 opt_line.py <d> <mR> <mI> <L*> <out.json>
# Output: JSON with the coefficients as exact binary numbers, the input of certify_line_pp.py.
import sys, json, math, time, os, numpy as np, cvxpy as cp, mpmath as mp
PPMAX = int(os.environ.get('PPMAX', '0'))   # per-prime phase maximum in the objective for the primes p <=
     PPMAX (0: none)
sys.path.append(os.path.join(os.path.dirname(os.path.abspath(__file__)), '..', 'hybrid_asymptotic'))   #
     lpcommon.py lives there (or next to this file in the flat archive)
from lpcommon import f, F, tgrid

def basis(mR, mI): return [('R', j) for j in range(1, mR+1)] + [('I', j) for j in range(2, mI+1)]

def kernel_rows(d, B, bb, tt):
    G = np.array([F(d+bb, tt, k, j) + F(d-bb, tt, k, j) for (k, j) in B]).T
    return G, f(bb, tt, d)
```

```
def primes(N):
    s = np.ones(N+1, bool); s[:2] = False
    for i in range(2, int(N**0.5)+1):
        if s[i]: s[i*i::i] = False
    return np.nonzero(s)[0]

def czeta_data(d, B, n0):
    # rows (one per prime power n <= n0, then one per quadrature node of the tail) such that
    # C_zeta ~ sum_rows w * || (rows_a @ c, rows_b @ c + beta0) ||_2;  prime_of[row] = p for the row of n = p
     ^k, 0 for the tail
    s1 = 0.5+d; rows_a = []; rows_b = []; beta0 = []; w = []; prime_of = []
    def add(L, wt, p):
        rows_a.append([-(L**(j-1)/math.factorial(j-1))/math.pi if kk == 'R' else 0 for (kk, j) in B])
        rows_b.append([-(L**(j-1)/math.factorial(j-1))/math.pi if kk == 'I' else 0 for (kk, j) in B])
        beta0.append((math.exp(-d*L)-2)/(L*math.pi)); w.append(wt); prime_of.append(p)
    for p in primes(n0):
        lp = math.log(p); n = p; k = 1
        while n <= n0:
            add(k*lp, lp*math.exp(-s1*k*lp), int(p)); n *= p; k += 1
    # tail n > n0: psi(x) ~ x, sum Lambda(n) g(n) ~ int g(x) dx = int g e^L dL ; Gauss-Legendre in L on [log
     n0, 700]
    xg, wg = np.polynomial.legendre.leggauss(400); a, b = math.log(n0), 700.0
    for x, ww in zip(xg, wg):
        L = (b-a)/2*x+(a+b)/2; add(L, (b-a)/2*ww*math.exp((1-s1)*L), 0)
    return np.array(rows_a), np.array(rows_b), np.array(beta0), np.array(w), np.array(prime_of)


def czeta_pp_data(d, B, PP=300, n0=20000):
    """for primes p <= PP: list of (kvec, wvec, RAk, RBk, b0k) to build max_x f_p(x) on a grid; other n via
     hypot rows."""
    s1 = 0.5+d; groups = []
    for p in primes(PP):
        lp = math.log(p); n = p; k = 1; ks = []; ra = []; rb = []; bz = []
        while n <= n0:
            L = k*lp; w = lp*math.exp(-s1*L)
            ra.append([-w*(L**(j-1)/math.factorial(j-1))/math.pi if kk == 'R' else 0 for (kk, j) in B])
            rb.append([-w*(L**(j-1)/math.factorial(j-1))/math.pi if kk == 'I' else 0 for (kk, j) in B])
            bz.append(w*(math.exp(-d*L)-2)/(L*math.pi)); ks.append(k); n *= p; k += 1
        groups.append((p, np.array(ks), np.array(ra), np.array(rb), np.array(bz)))
    return groups

def elem_data(d, B, Tgrid):
    mp.mp.dps = 25
    s1r = mp.mpf(1)/2+d; s2r = mp.mpf(1)/2+2*d
    base = []; Phi = []
    for T in Tgrid:
        T = mp.mpf(T); s1 = s1r+1j*T; s2 = s2r+1j*T
        Ael = lambda s: mp.im(mp.log(s)+mp.log(s-1))-T/2*mp.log(mp.pi)+mp.im(mp.loggamma(s/2))
        base.append(float((2*Ael(s1)-Ael(s2))/mp.pi-T/(2*mp.pi)*mp.log(T/(2*mp.pi*mp.e))))
        row = []
        for (k, j) in B:
            if j == 1: D = 1/s1+1/(s1-1)-mp.log(mp.pi)/2+mp.psi(0, s1/2)/2
            else: D = s1**(-j)+(s1-1)**(-j)+(-1)**(j-1)/mp.factorial(j-1)/2**j*mp.psi(j-1, s1/2)
            row.append(float(mp.re(D)) if k == 'R' else float(-mp.im(D)))
        Phi.append(row)
    return np.array(base), np.array(Phi)

def optimize(d, mR, mI, Lstar, eta=2e-4, n0=20000, maxit=25, verbose=True):
    B = basis(mR, mI); nB = len(B); i1 = B.index(('R', 1)); i2 = B.index(('R', 2))
    isR = np.array([k == 'R' for k, _ in B])
    bb, tt = [a.ravel() for a in np.meshgrid(np.linspace(0, 0.5, 26), tgrid(300.), indexing='ij')]
    G, Fv = kernel_rows(d, B, bb, tt); Fv = Fv+eta/(1+tt**2)
    RA, RB, b0, w, prime_of = czeta_data(d, B, n0)
    if PPMAX:
        GR = czeta_pp_data(d, B, PPMAX, n0)
```

```
        maskPP = np.array([not (p and p <= PPMAX) for p in prime_of])     # rows of the prime powers p^k with
     p <= PPMAX are replaced by the phase maximum
        assert (~maskPP).sum() == sum(len(g[1]) for g in GR)
    Tg = np.unique(np.concatenate([np.geomspace(1, 1e6, 260), [1.0]]))
    base, Phi = elem_data(d, B, Tg); logT = np.log(Tg)
    sc = np.array([(d-0.5)**j for (_, j) in B])        # scaling for conditioning
    bfine = np.linspace(0, 0.5, 201); tp = np.geomspace(1e-8, 3000., 12000); tfine = np.concatenate([-tp
     [::-1], tp])
    for it in range(maxit):
        y = cp.Variable(nB); z = cp.Variable(); c = cp.multiply(sc, y)
        cons = [G@c >= Fv, 2*d*c[i1]-2*c[i2] >= eta]
        sign = np.where(isR, 1.0, -1.0)
        Fu = base+(Phi@c)/math.pi-c[i1]/(2*math.pi)*logT
        Fl = -base+(Phi@cp.multiply(sign, c))/math.pi-c[i1]/(2*math.pi)*logT
        cons += [z >= Fu, z >= Fl, z >= 7/8-c[i1]/(2*math.pi)*math.log(2*math.pi)]
        if PPMAX:
            xg = np.linspace(0, 2*np.pi, 96, endpoint=False); tp_ = cp.Variable(len(GR))
            for gi, (p, ks, ra, rb, bz) in enumerate(GR):
                Cm = np.cos(np.outer(xg, ks)); Sm = np.sin(np.outer(xg, ks))
                cons.append(tp_[gi] >= Cm@(ra@c)+Sm@(rb@c+bz))
            Cz = cp.sum(tp_)+cp.sum(cp.multiply(w[maskPP], cp.norm(cp.vstack([RA[maskPP]@c, RB[maskPP]@c+b0[
    maskPP]]), 2, axis=0)))
        else:
            Cz = cp.sum(cp.multiply(w, cp.norm(cp.vstack([RA@c, RB@c+b0]), 2, axis=0)))
        obj = c[i1]/(2*math.pi)*Lstar+Cz+z
        prob = cp.Problem(cp.Minimize(obj), cons); prob.solve(solver=cp.CLARABEL)
        assert prob.status in (cp.OPTIMAL, cp.OPTIMAL_INACCURATE), 'solver status: '+str(prob.status)
        cv = sc*y.value
        nb = []; nt = []; worst = -1
        for b in bfine:
            Gr, Fr = kernel_rows(d, B, np.full_like(tfine, b), tfine); v = Fr+eta/(1+tfine**2)-Gr@cv; worst =
     max(worst, v.max())
            idx = np.where((v > 0) & (v >= np.roll(v, 1)) & (v >= np.roll(v, -1)))[0]; nb += [b]*len(idx); nt
     += list(tfine[idx])
        if verbose: print('  it %d obj=%.5f C1=%.5f Cz~%.4f z=%.4f worst=%.2e' % (it, prob.value, cv[i1]/(2*
    math.pi), Cz.value, z.value, worst), flush=True)
        if not nb: break
        G2, F2 = kernel_rows(d, B, np.array(nb), np.array(nt)); G = np.vstack([G, G2]); Fv = np.concatenate([
    Fv, F2+eta/(1+np.array(nt)**2)])
    return dict(d=d, B=B, c=list(map(float, cv)), C1=float(cv[i1]/(2*math.pi)), Cz_est=float(Cz.value), z_est
    =float(z.value),
                A3_est=float(Cz.value+z.value), Lstar=Lstar, eta=eta, PPMAX=PPMAX, discs=[], kappa=[], m=max(
    j for _, j in B))

if __name__ == '__main__':
    d = float(sys.argv[1]); mR = int(sys.argv[2]); mI = int(sys.argv[3]); Ls = float(sys.argv[4]); out = sys.
    argv[5]
    t0 = time.time(); r = optimize(d, mR, mI, Ls)
    print('RESULT d=%.3f mR=%d mI=%d L*=%g  A1=%.5f A3~%.4f  value@L*~%.4f  (%.0fs)' % (d, mR, mI, Ls, r['C1
    '], r['A3_est'], r['C1']*Ls+r['A3_est'], time.time()-t0), flush=True)
    json.dump(r, open(out, 'w'))
```

Appendix C. Outputs of the certification programs

Hybrid: `certify_hybrid.py hyb_m10_P.json 40`

```
d = 9/10  m = 10  discs used: 90  profile e_P (profile_published.py)
C = c_R1/(2pi) + sum kappa_k J_k/pi = 0.074859614271090718755  (ball [0.074860 +/- 3.86e-7] )
kernel part c_R1/(2pi) = 0.06558218278454444
limit of t^2 U at |t| -> oo = 2 d c_R1 - 2 c_R2 = [0.001408437209 +/- 3.24e-13]  positive: True
t in (0,T1]: certified, boxes 2028149
t in [-T1,0): certified, boxes 2294869
|t|>=T1: certified, boxes 277098
```

```
all certified;  C < 0.0748596142711   time 822s
```

Line ℓ_5: kernel inequality

```
d = 1  m = 5  discs used: 0
C = c_R1/(2pi) + sum kappa_k J_k/pi = 0.20175655042691417704  (ball [0.20176 +/- 3.45e-6] )
kernel part c_R1/(2pi) = 0.2017565504269142
limit of t^2 U at |t| -> oo = 2 d c_R1 - 2 c_R2 = [1.133488684 +/- 3.74e-10]  positive: True
t in (0,T1]: certified, boxes 10255
t in [-T1,0): certified, boxes 4207
|t|>=T1: certified, boxes 2
all certified;  C < 0.201756550427   time 1s
```

Line ℓ_5: C_ζ

```
per-prime phase maximum for p <= 1000 : gain [0.13567339 +/- 3.39e-9]
primes 5761455  partial sum (n<=100000000) = [0.700266525818397 +/- 2.62e-16]  psi(N0) = [99998242.7966268
     +/- 1.77e-8]  52s
g monotone on [2000, oo): crude bound [+/- 4.40e-3] < sigma1
tail <= [0.003333488646 +/- 2.02e-13]  (int_{N0}^{e^LB} G = [0.003174276158 +/- 3.32e-14] , int_{e^L1}^{e^LB}
      g = [+/- 3.23e-163] , far tail <= [2.6758e-167 +/- 3.15e-172] )
C_zeta < 0.703600014465    time 52s
```

Line ℓ_5: elementary part, mode N

```
intervals 9215  sup F_u on [1,T0] <= 0.6653643978   sup F_l <= -0.7495433723
T>=T0: F_u <= 0.5045914644  F_l <= -1.245408535  (K = [3.95201 +/- 3.63e-6] )
SUP over T>=1 of max(F_u,F_l) <= 0.665364397774    time 122s
```

Line ℓ_5: elementary part, mode S

```
intervals 9215  sup F_u on [1,T0] <= -0.2236696656   sup F_l <= 0.1394908934
T>=T0: F_u <= -0.3703919570  F_l <= -0.3703919570  (K = [4.11780 +/- 2.69e-8] )
SUP over T>=1 of max(F_u,F_l) <= 0.139490893363    time 124s
```

Line ℓ_{20}: kernel inequality

```
d = 6/5  m = 5  discs used: 0
C = c_R1/(2pi) + sum kappa_k J_k/pi = 0.10822193548420830108  (ball [0.10822 +/- 1.94e-6] )
kernel part c_R1/(2pi) = 0.1082219354842083
limit of t^2 U at |t| -> oo = 2 d c_R1 - 2 c_R2 = [0.02372706155 +/- 2.76e-12]  positive: True
t in (0,T1]: certified, boxes 473575
t in [-T1,0): certified, boxes 279773
|t|>=T1: certified, boxes 1022
all certified;  C < 0.108221935485   time 45s
```

Line ℓ_{20}: C_ζ

```
per-prime phase maximum for p <= 1000 : gain [0.21171880 +/- 3.25e-9]
primes 5761455  partial sum (n<=100000000) = [1.46785649757579 +/- 4.11e-15]  psi(N0) = [99998242.7966268 +/-
     1.77e-8]  53s
g monotone on [2000, oo): crude bound [+/- 0.0145] < sigma1
tail <= [0.0002188390453 +/- 1.36e-15]  (int_{N0}^{e^LB} G = [0.0002060161219 +/- 1.10e-14] , int_{e^L1}^{e^
    LB} g = [+/- 7.27e-234] , far tail <= [7.2489e-238 +/- 1.12e-244] )
C_zeta < 1.46807533663    time 53s
```

Line ℓ_{20}: elementary part, mode N

```
intervals 9215  sup F_u on [1,T0] <= 0.7020707094   sup F_l <= -1.072389149
T>=T0: F_u <= 0.6766231016  F_l <= -1.073376898  (K = [5.21715 +/- 1.54e-6] )
SUP over T>=1 of max(F_u,F_l) <= 0.702070709399    time 123s
```

Line ℓ_{20}: elementary part, mode S

```
intervals 9215  sup F_u on [1,T0] <= -0.1827864943   sup F_l <= -0.1965944846
T>=T0: F_u <= -0.1983603198  F_l <= -0.1983603198  (K = [5.38293 +/- 4.87e-6] )
SUP over T>=1 of max(F_u,F_l) <= -0.182786494342    time 125s
```

Line ℓ_{60}: kernel inequality

```
d = 1  m = 5  discs used: 0
C = c_R1/(2pi) + sum kappa_k J_k/pi = 0.094721574236353990059  (ball [0.094722 +/- 4.26e-7] )
kernel part c_R1/(2pi) = 0.09472157423635399
limit of t^2 U at |t| -> oo = 2 d c_R1 - 2 c_R2 = [0.008043907939 +/- 4.87e-13]  positive: True
t in (0,T1]: certified, boxes 1441685
t in [-T1,0): certified, boxes 483889
|t|>=T1: certified, boxes 4434
all certified;  C < 0.0947215742364   time 111s
```

Line ℓ_{60}: C_ζ

```
per-prime phase maximum for p <= 1000 : gain [0.24105429 +/- 2.19e-10]
primes 5761455  partial sum (n<=100000000) = [2.09743955289942 +/- 3.84e-15]  psi(N0) = [99998242.7966268 +/-
      1.77e-8]  86s
g monotone on [2000, oo): crude bound [+/- 0.0149] < sigma1
tail <= [0.006829934351 +/- 6.03e-14]  (int_{N0}^{e^LB} G = [0.006476069071 +/- 3.90e-13] , int_{e^L1}^{e^LB}
      g = [+/- 1.23e-164] , far tail <= [1.7143e-168 +/- 4.48e-173] )
C_zeta < 2.10426948726    time 87s
```

Line ℓ_{60}: elementary part, mode N

```
intervals 9215  sup F_u on [1,T0] <= 0.7289415578   sup F_l <= -1.047579458
T>=T0: F_u <= 0.7013136727  F_l <= -1.048686327  (K = [4.00282 +/- 3.70e-6] )
SUP over T>=1 of max(F_u,F_l) <= 0.728941557753    time 123s
```

Line ℓ_{60}: elementary part, mode S

```
intervals 9215  sup F_u on [1,T0] <= -0.1584757589   sup F_l <= -0.1717847946
T>=T0: F_u <= -0.1736697487  F_l <= -0.1736697487  (K = [4.16860 +/- 2.71e-6] )
SUP over T>=1 of max(F_u,F_l) <= -0.158475758991    time 125s
```

Line ℓ_{150}: kernel inequality

```
d = 17/20  m = 5  discs used: 0
C = c_R1/(2pi) + sum kappa_k J_k/pi = 0.093480628436037879597  (ball [0.093481 +/- 3.72e-7] )
kernel part c_R1/(2pi) = 0.09348062843603788
limit of t^2 U at |t| -> oo = 2 d c_R1 - 2 c_R2 = [0.007450671016 +/- 2.69e-13]  positive: True
t in (0,T1]: certified, boxes 1658655
t in [-T1,0): certified, boxes 353921
|t|>=T1: certified, boxes 3198
all certified;  C < 0.0934806284361   time 121s
```

Line ℓ_{150}: C_ζ

```
per-prime phase maximum for p <= 1000 : gain [0.25085637 +/- 3.02e-9]
primes 5761455  partial sum (n<=100000000) = [2.21455819652827 +/- 4.70e-15]  psi(N0) = [99998242.7966268 +/-
      1.77e-8]  105s
g monotone on [2000, oo): crude bound [+/- 7.01e-3] < sigma1
tail <= [0.04122525901 +/- 2.13e-12]  (int_{N0}^{e^LB} G = [0.03953310421 +/- 1.61e-12] , int_{e^L1}^{e^LB} g
      = [+/- 2.53e-111] , far tail <= [3.6826e-115 +/- 3.45e-120] )
C_zeta < 2.25578345555    time 106s
```

Line ℓ_{150}: elementary part, mode N

```
intervals 9215  sup F_u on [1,T0] <= 0.7303168470   sup F_l <= -1.045307483
T>=T0: F_u <= 0.7035135754  F_l <= -1.046486424  (K = [3.19479 +/- 4.84e-6] )
SUP over T>=1 of max(F_u,F_l) <= 0.730316846936    time 123s
```

Line ℓ_{150}: elementary part, mode S

```
intervals 9215  sup F_u on [1,T0] <= -0.1575763483   sup F_l <= -0.1695128192
T>=T0: F_u <= -0.1714698460  F_l <= -0.1714698460  (K = [3.36057 +/- 1.57e-6] )
SUP over T>=1 of max(F_u,F_l) <= -0.157576348369    time 125s
```

Line ℓ_{400}: kernel inequality

```
d = 17/20  m = 10  discs used: 0
C = c_R1/(2pi) + sum kappa_k J_k/pi = 0.083986018406625982801  (ball [0.083986 +/- 1.86e-8] )
kernel part c_R1/(2pi) = 0.08398601840662598
limit of t^2 U at |t| -> oo = 2 d c_R1 - 2 c_R2 = [0.0002002739955 +/- 2.69e-14]  positive: True
t in (0,T1]: certified, boxes 11835193
t in [-T1,0): certified, boxes 9801985
|t|>=T1: certified, boxes 4194302
all certified;  C < 0.0839860184067   time 2272s
```

Line ℓ_{400}: C_ζ

```
per-prime phase maximum for p <= 1000 : gain [0.28179774 +/- 3.28e-9]
primes 5761455  partial sum (n<=100000000) = [4.01127229118254 +/- 1.94e-15]  psi(N0) = [99998242.7966268 +/-
      1.77e-8]  163s
g monotone on [5000, oo): crude bound [+/- 0.0114] < sigma1
tail <= [0.4524693305 +/- 1.73e-11]  (int_{N0}^{e^LB} G = [0.4325160730 +/- 4.01e-11] , int_{e^L1}^{e^LB} g =
      [+/- 6.02e-102] , far tail <= [2.8606e-111 +/- 3.11e-116] )
C_zeta < 4.46374162170    time 164s
```

Line ℓ_{400}: elementary part, mode N

```
intervals 9215  sup F_u on [1,T0] <= 0.7375655061   sup F_l <= -1.027807128
T>=T0: F_u <= 0.7209635502  F_l <= -1.029036449  (K = [3.19527 +/- 2.94e-6] )
SUP over T>=1 of max(F_u,F_l) <= 0.737565506071    time 159s
```

Line ℓ_{400}: elementary part, mode S

```
intervals 9215  sup F_u on [1,T0] <= -0.1458267316   sup F_l <= -0.1520124644
T>=T0: F_u <= -0.1540198711  F_l <= -0.1540198711  (K = [3.36106 +/- 6.67e-7] )
SUP over T>=1 of max(F_u,F_l) <= -0.145826731639    time 162s
```

Line ℓ_{6000}: kernel inequality

```
d = 17/20  m = 10  discs used: 0
C = c_R1/(2pi) + sum kappa_k J_k/pi = 0.083814170602092884236  (ball [0.083814 +/- 1.71e-7] )
kernel part c_R1/(2pi) = 0.08381417060209288
limit of t^2 U at |t| -> oo = 2 d c_R1 - 2 c_R2 = [0.0002027460239 +/- 3.01e-14]  positive: True
t in (0,T1]: certified, boxes 14760921
t in [-T1,0): certified, boxes 15866017
|t|>=T1: certified, boxes 4194302
all certified;  C < 0.0838141706021   time 3551s
```

Line ℓ_{6000}: C_ζ

```
per-prime phase maximum for p <= 1000 : gain [0.28466228 +/- 4.66e-10]
primes 5761455  partial sum (n<=100000000) = [4.10087443545128 +/- 2.45e-15]  psi(N0) = [99998242.7966268 +/-
      1.77e-8]  163s
g monotone on [5000, oo): crude bound [+/- 0.0108] < sigma1
tail <= [0.4886359120 +/- 3.99e-12]  (int_{N0}^{e^LB} G = [0.4671146906 +/- 3.15e-11] , int_{e^L1}^{e^LB} g =
      [+/- 6.36e-102] , far tail <= [2.9139e-111 +/- 4.83e-116] )
C_zeta < 4.58951034745    time 164s
```

Line ℓ_{6000}: elementary part, mode N

```
intervals 9215  sup F_u on [1,T0] <= 0.7357375257   sup F_l <= -1.027508246
T>=T0: F_u <= 0.7212794332  F_l <= -1.028720566  (K = [3.19575 +/- 1.13e-6] )
SUP over T>=1 of max(F_u,F_l) <= 0.735737525685    time 159s
```

Line ℓ_{6000}: elementary part, mode S

```
intervals 9215  sup F_u on [1,T0] <= -0.1464103660   sup F_l <= -0.1517135821
T>=T0: F_u <= -0.1537039882  F_l <= -0.1537039882  (K = [3.36154 +/- 2.48e-6] )
SUP over T>=1 of max(F_u,F_l) <= -0.146410366017    time 124s
```

Grid optima of `kernel_only.py` (Remark 2.5; not certified)

```
m=4 d=0.70 all beta C1(grid)=0.08831 worst violation=1.8e-08 it=59  120s
m=4 d=0.80 all beta C1(grid)=0.08894 worst violation=3.0e-13 it=9  20s
m=4 d=0.90 all beta C1(grid)=0.09061 worst violation=1.5e-10 it=3  8s
m=10 d=0.70 all beta C1(grid)=0.08160 worst violation=1.4e-13 it=5  33s
m=10 d=0.80 all beta C1(grid)=0.08071 worst violation=7.4e-08 it=59  233s
m=10 d=0.90 all beta C1(grid)=0.08100 worst violation=8.6e-09 it=59  173s
m=20 d=0.70 all beta C1(grid)=0.07994 worst violation=3.1e-09 it=59  373s
m=20 d=0.80 all beta C1(grid)=0.07983 worst violation=1.6e-09 it=59  378s
m=20 d=0.90 all beta C1(grid)=0.07996 worst violation=1.9e-06 it=59  382s
m=40 d=0.70 all beta C1(grid)=0.07978 worst violation=2.1e-02 it=59  2057s
m=4 d=0.70 constraint at beta=1/2 only C1(grid)=0.04375 worst violation=3.3e-16 it=3  0s
m=4 d=0.80 constraint at beta=1/2 only C1(grid)=0.05000 worst violation=1.6e-12 it=2  0s
m=4 d=0.90 constraint at beta=1/2 only C1(grid)=0.05625 worst violation=2.4e-12 it=2  0s
m=10 d=0.70 constraint at beta=1/2 only C1(grid)=0.01747 worst violation=3.3e-04 it=59  2s
m=10 d=0.80 constraint at beta=1/2 only C1(grid)=0.02002 worst violation=3.5e-14 it=3  0s
m=10 d=0.90 constraint at beta=1/2 only C1(grid)=0.02250 worst violation=1.8e-06 it=59  2s
m=20 d=0.70 constraint at beta=1/2 only C1(grid)=0.01070 worst violation=5.4e-01 it=59  3s
m=20 d=0.80 constraint at beta=1/2 only C1(grid)=0.01257 worst violation=7.6e-02 it=59  4s
m=20 d=0.90 constraint at beta=1/2 only C1(grid)=0.01330 worst violation=1.3e-02 it=59  4s
```

Appendix D. Exact data

The following files hold the coefficients as exact binary floating-point numbers (the decimal strings are the shortest that round-trip), the disc data before rounding to rationals (the certificate rounds c_k, R_k, v_k to denominators $\le 10^4$ and then replaces R_k by $\min(R_k, c_k - \frac{1}{2})$, as in `certify_hybrid.py`), and the parameters of the search. The fields `J_grid` and `C_grid_objective` are the floating-point values of the search programme and are not used by the certificates, which recompute J_k and C_{H} exactly.

`hyb_m10_P.json` (hybrid)

```
{
 "m": 10,
 "d": 0.9,
 "prof": "P",
 "eta": 0.001,
 "B": [
  ["R", 1],
  ["R", 2],
```

```
  ["R", 3],
  ["R", 4],
  ["R", 5],
  ["R", 6],
  ["R", 7],
  ["R", 8],
  ["R", 9],
  ["R", 10],
  ["I", 2],
  ["I", 3],
  ["I", 4],
  ["I", 5],
  ["I", 6],
  ["I", 7],
  ["I", 8],
  ["I", 9],
  ["I", 10]
 ],
 "c": [0.41206500728461565, 0.37015428795149224, 0.2844015811499438, 0.15861857578668956,
  0.06954223006530842, -0.06157764993877496, -0.040570135781007474, 0.07526718292596676,
  -0.03235073058726206, 0.005632605237216712, 0.7334607489350747, -0.13613005947680004,
  0.15136050241561017, -0.11692974368156732, -0.03656894918659249, 0.10601427043647423,
  -0.06227163099672864, 0.016690927091753422, -0.0017941465528435766],
 "discs": [
  [1.03, 0.18000000000000002, -0.5],
  [1.03, 0.18000000000000002, 0.5],
  [1.03, 0.23000000000000004, -0.65],
  [1.03, 0.23000000000000004, 0.65],
  [1.03, 0.28, -0.65],
  [1.03, 0.28, 0.65],
  [1.03, 0.33, -0.65],
  [1.03, 0.33, 0.65],
  [1.03, 0.38, -0.65],
  [1.03, 0.38, 0.65],
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  [1.03, 0.48000000000000004, -1.0],
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  [1.03, 0.53, -1.6],
  [1.03, 0.53, -1.3],
  [1.03, 0.53, -1.0],
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  [1.06, 0.21000000000000005, -0.5],
  [1.06, 0.21000000000000005, 0.5],
  [1.06, 0.26000000000000006, -0.5],
  [1.06, 0.26000000000000006, 0.5],
  [1.06, 0.26000000000000006, 0.65],
  [1.06, 0.41000000000000003, -0.8],
  [1.06, 0.41000000000000003, -0.65],
  [1.06, 0.41000000000000003, 0.65],
  [1.06, 0.41000000000000003, 0.8],
  [1.06, 0.4600000000000001, 0.8],
  [1.06, 0.56, -1.6],
  [1.06, 0.56, -1.3],
  [1.06, 0.56, 1.3],
  [1.1, 0.15000000000000008, -0.35],
  [1.1, 0.2500000000000001, -0.5],
  [1.1, 0.2500000000000001, 0.5],
  [1.1, 0.4000000000000001, 0.8],
  [1.1, 0.45000000000000007, -0.8],
  [1.1, 0.45000000000000007, 0.8],
  [1.1, 0.5000000000000001, -0.8],
  [1.1, 0.5000000000000001, 0.8],
```

```
 [1.1, 0.6000000000000001, -1.0],
 [1.1, 0.6000000000000001, 1.0],
 [1.1, 0.6000000000000001, 1.3],
 [1.0986122886681098, 0.1986122886681098, -0.5],
 [1.0986122886681098, 0.1986122886681098, 0.5],
 [1.0986122886681098, 0.24861228866810978, -0.5],
 [1.0986122886681098, 0.24861228866810978, 0.5],
 [1.0986122886681098, 0.2986122886681098, -0.5],
 [1.0986122886681098, 0.2986122886681098, 0.5],
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 [1.0986122886681098, 0.5986122886681098, -1.3],
 [1.15, 0.1999999999999999, -0.5],
 [1.15, 0.1999999999999999, -0.35],
 [1.15, 0.1999999999999999, 0.35],
 [1.15, 0.4499999999999999, -0.8],
 [1.15, 0.4999999999999999, -0.8],
 [1.15, 0.5499999999999999, -0.8],
 [1.15, 0.5499999999999999, 0.8],
 [1.15, 0.6499999999999999, -1.3],
 [1.15, 0.6499999999999999, -1.0],
 [1.15, 0.6499999999999999, 1.0],
 [1.15, 0.6499999999999999, 1.3],
 [1.2, 0.24999999999999994, -0.5],
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 [1.2, 0.24999999999999994, 0.35],
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 [1.2, 0.7, -1.3],
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 [1.2, 0.7, 1.0],
 [1.2, 0.7, 1.3],
 [1.25, 0.3, -0.35],
 [1.25, 0.3, 0.35],
 [1.25, 0.75, -1.3],
 [1.25, 0.75, -1.0],
 [1.25, 0.75, 1.0],
 [1.25, 0.75, 1.3],
 [1.3, 0.8, -1.0],
 [1.3, 0.8, 1.0],
 [1.4, 0.4499999999999999, -0.35],
 [1.4, 0.4499999999999999, 0.35],
 [1.4, 0.8999999999999999, -1.3],
 [1.4, 0.8999999999999999, 1.3],
 [1.5, 0.55, -0.35],
 [1.5, 0.55, 0.35],
 [1.6094379124341003, 0.6594379124341003, -0.35],
 [1.6094379124341003, 0.6594379124341003, 0.35]
],
"kappa": [0.04645216860607228, 0.04300701194678176, 0.047063043236809085,
 0.14669081272175175, 0.025979954843667673, 0.10379166553442853, 0.0238180047300216,
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  5.25882403645087e-05, 0.0004219252908076343, 0.0018326737860931267,
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 "J_grid": [0.006913876985130073, 0.006913876985130073, 0.010786963484310835,
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 "C_grid_objective": 0.07485961386282106
}
```


line_L5/sol.json (line ℓ_5)


```
{
 "d": 1.0,
 "B": [
  ["R", 1],
  ["R", 2],
  ["R", 3],
  ["R", 4],
  ["R", 5],
  ["I", 2],
  ["I", 3],
  ["I", 4],
  ["I", 5]
 ],
 "c": [1.2676737932696245, 0.7009294514561635, -0.4946330486621655, 0.09751555052498845,
  -0.0005893962931625271, -0.3784073335097401, 0.13473062893139318, -0.01708382778797021,
  0.0001003433684765934],
 "C1": 0.20175655042691418,
 "Cz_est": 0.8391011111542401,
 "z_est": 0.665362725928392,
 "A3_est": 1.504463837082632,
 "Lstar": 5.0,
 "eta": 0.0002,
 "discs": [],
 "kappa": [],
 "m": 5,
 "val": 2.5132465892172027,
```

```
 "PPMAX": 0
}
```


`line_L20/sol.json` (line ℓ_{20})


```
{
 "d": 1.2,
 "B": [
  ["R", 1],
  ["R", 2],
  ["R", 3],
  ["R", 4],
  ["R", 5],
  ["I", 2],
  ["I", 3],
  ["I", 4],
  ["I", 5]
 ],
 "c": [0.6799784749489147, 0.8041106391650762, 0.7774270907421565, -0.008448613496901295,
  2.955972158446658e-05, 1.1843502097112268, -0.576537489831307, 0.00610293149500534,
  -2.1076986306361527e-05],
 "C1": 0.1082219354842083,
 "Cz_est": 1.6797641910534995,
 "z_est": 0.7020688851693405,
 "A3_est": 2.38183307622284,
 "Lstar": 20.0,
 "eta": 0.0002,
 "discs": [],
 "kappa": [],
 "m": 5,
 "val": 4.546271785907006,
 "PPMAX": 0
}
```


`line_L60/sol.json` (line ℓ_{60})


```
{
 "d": 1.0,
 "B": [
  ["R", 1],
  ["R", 2],
  ["R", 3],
  ["R", 4],
  ["R", 5],
  ["I", 2],
  ["I", 3],
  ["I", 4],
  ["I", 5]
 ],
 "c": [0.5951532035147798, 0.5911312495450366, 0.3889216428262093, -0.004325167636924481,
  1.5378902475155497e-05, 0.8079466147098463, -0.3400480079099986, 0.003649167179891613,
  -1.2750467413103112e-05],
 "C1": 0.094721574236354,
 "Cz_est": 2.344884047970574,
 "z_est": 0.7289397780667036,
 "A3_est": 3.0738238260372777,
 "Lstar": 60.0,
 "eta": 0.0002,
 "discs": [],
 "kappa": [],
 "m": 5,
 "val": 8.757118280218517,
 "PPMAX": 0
}
```

line_L150/sol.json (line ℓ_{150})


```
{
 "d": 0.85,
 "B": [
  ["R", 1],
  ["R", 2],
  ["R", 3],
  ["R", 4],
  ["R", 5],
  ["I", 2],
  ["I", 3],
  ["I", 4],
  ["I", 5]
 ],
 "c": [0.5873561110952275, 0.4955273589230774, 0.17191023951649326, -0.04515429156322006,
  0.0002717467554019091 7, 0.5292986070563637, -0.26974254048270235, 0.03950670966367516,
  -0.0002345907376948271 1],
 "C1": 0.09348062843603788,
 "Cz_est": 2.5047342757708364,
 "z_est": 0.7303153300893256,
 "A3_est": 3.235049605860162,
 "Lstar": 150.0,
 "eta": 0.0002,
 "discs": [],
 "kappa": [],
 "m": 5,
 "val": 17.257143871265846,
 "PPMAX": 0
}
```


line_C400/sol.json (line ℓ_{400})


```
{
 "d": 0.85,
 "B": [
  ["R", 1],
  ["R", 2],
  ["R", 3],
  ["R", 4],
  ["R", 5],
  ["R", 6],
  ["R", 7],
  ["R", 8],
  ["R", 9],
  ["R", 10],
  ["I", 2],
  ["I", 3],
  ["I", 4],
  ["I", 5],
  ["I", 6],
  ["I", 7],
  ["I", 8],
  ["I", 9],
  ["I", 10]
 ],
 "c": [0.5276997168610267, 0.44844462233410926, 0.20734915440049123, -0.05170881362706953,
  0.04277439204584691, -0.0021686787194867547, 5.225366348519004e-05,
  -7.194711787780742e-07, 5.548232880956561e-09, -1.890562024872484e-11, 0.57950165380938,
  -0.24304154954646912, 0.09598460587810165, -0.022316290439150017, 0.0010166745274645633,
  -2.326790970420105e-05, 3.096452357377663e-07, -2.3274894920069496e-09,
  7.769753796000765e-12],
 "C1": 0.08398601840662599,
 "Cz_est": 4.723835786462193,
```

```
 "z_est": 0.7375635792618767,
 "A3_est": 5.46139936572407,
 "Lstar": 400.0,
 "eta": 0.0002,
 "discs": [],
 "kappa": [],
 "m": 10,
 "val": 39.05580672837446,
 "PPMAX": 0
}
```


`line_D6000/sol.json` (line ℓ_{6000})


```
{
 "d": 0.85,
 "B": [
  ["R", 1],
  ["R", 2],
  ["R", 3],
  ["R", 4],
  ["R", 5],
  ["R", 6],
  ["R", 7],
  ["R", 8],
  ["R", 9],
  ["R", 10],
  ["I", 2],
  ["I", 3],
  ["I", 4],
  ["I", 5],
  ["I", 6],
  ["I", 7],
  ["I", 8],
  ["I", 9],
  ["I", 10]
 ],
 "c": [0.5266199652605132, 0.44752559745950127, 0.21763335933091524, -0.06632610210813339,
  0.049224751946469816, -0.002454597973796714, 5.8415182530844107e-05,
  -7.96150671490144e-07, 6.08645994127779e-09, -2.0583832928014207e-11, 0.5834246015810649,
  -0.23667212703027896, 0.09567840750457196, -0.01858966229544025, 0.0008313848476117577,
  -1.89384002058713e-05, 2.517567194510178e-07, -1.8927048877024136e-09,
  6.322548809318692e-12],
 "C1": 0.08381417060209288,
 "Cz_est": 4.566001466369842,
 "z_est": 0.7357358598268879,
 "A3_est": 5.30173732619673,
 "Lstar": 6000.0,
 "eta": 0.0002,
 "discs": [],
 "kappa": [],
 "m": 10,
 "val": 508.18676093875405,
 "PPMAX": 300
}
```


Universidad Nacional de Educación a Distancia, Madrid, Spain
Email address: vcasana1@alumno.uned.es